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Supplementary information

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Chromosome-level scaffolding of haplotype-resolved assemblies using Hi-C data without reference genomes

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Table of Contents

Supplementary Methods.....2

1 Simulation of datasets with various adverse factors.....2

2 Scaffolding performance evaluation in simulated datasets.....3

3 Scaffolding performance evaluation in real cases using Hi-C reads.....4

4 Scaffolding performance evaluation in real cases using ground truth.....5

References.....6

Software Commands.....7

1 Simulation of datasets for evaluating scaffolding tools.....7

2 Quality control, mapping, and filtering of Hi-C reads.....11

3 Hi-C-based scaffolding.....12

4 Performance evaluation of scaffolding.....20

5 Genome assembly.....23

6 Comparative genomics analyses.....23

Supplementary Tables.....27

Supplementary Figures.....36

Descriptions of Supplementary Data.....110

Supplementary Methods

1 Simulation of datasets with various adverse factors

In the ALLHiC paper, the authors simulated a haplotype-resolved diploid genome by merging both the genome sequences and Hi-C data of two rice subspecies, *indica* and *japonica*[1]. However, this approach has some limitations. First, in a real diploid, there are background interactions between homologous chromosomes, because they are located in the same nucleus. The artificial merging of data from different subspecies leads to an underestimation of Hi-C links between homologous chromosomes, making it easier to separate them even than those between non-homologous chromosomes (**Supplementary Fig. 2**). Second, the conclusions drawn from experiments on diploids may not apply to polyploids. Third, using only a pair of homologous chromosomes for analysis ignores misassignments between non-homologous chromosomes. Therefore, we constructed a ground truth by manually removing obvious assembly errors based on the published haplotype-resolved autotetraploid genome of *Medicago sativa* XinJiangDaYe[2] (**Supplementary Fig. 1**). We used all 32 chromosomes to construct a ground truth genome of 2.0 Gb.

Using this template, we simulated a series of assemblies with various adverse factors through a pipeline of multiple custom scripts (**Extended Data Fig. 1**). The script "sim_contig.py" fragmented the genome into assemblies with different N50 values (**Supplementary Fig. 3**) and coefficients of variation (CVs) of length (**Supplementary Fig. 5**). Since the contig N50 of the *M. sativa* genome is less than 500 Kb, we retained the gaps represented by Ns in the fragments to simulate higher contiguity. For conciseness, we always refer to these fragments as contigs, regardless of whether there are gaps in them. Based on these assemblies, we used the script "sim_chimeric_contigs.py" to simulate assemblies with different proportions of chimeric contigs (**Supplementary Fig. 9**). Among them, chimeric contigs simulated between homologous chromosomes, non-homologous chromosomes, and within chromosomes were generated at a ratio of 7:2:1, respectively. To simulate different effective sequencing depths, we used "samtools view -s" with 12345 as the seed for random sampling (**Supplementary Fig. 7**). For these assemblies, real Hi-C reads were retrieved from NCBI under the accession numbers SRR9026577 and SRR9026578.

We simulated haplotypes by introducing single nucleotide polymorphisms (SNPs), insertions, and deletions to the first haplotype in the ground truth genome using the script "sim_haplotypes.py". By adjusting the proportion of variations and number of haplotypes generated, we obtained genomes with different sequence divergence levels (**Supplementary Fig. 24**) and ploidies. To simulate reality, we introduced SNPs, insertions, and deletions at a ratio of 10:0.5:0.5, with a 2:1 ratio of transitions to transversions for SNPs (**Supplementary Fig. 23**). Furthermore, these variations were unevenly distributed in bins with a length of 100 Kb and a CV of 0.5. For all assemblies based on these simulated haplotypes, Hi-C reads were generated by sim3C[3], with the

parameter “--trans-rate” dynamically adjusted according to different ploidies to maintain a fixed intensity of chromosome interaction.

Based on these simulated haplotypes and Hi-C reads, we further generated collapsed contigs (**Supplementary Fig. 17**) and switch errors (**Supplementary Fig. 26**). We used the script “sim_collapsed_regions.py” to create two-, three-, and four-haplotype collapsed contigs in a ratio of 7:2:1. Switch errors were introduced by randomly shuffling variations between haplotypes using the script “sim_switch_errors.py”. We also incorporated all the custom scripts above to introduce 5% each of chimeric, collapsed contigs, and switch errors into assemblies with ploidy ranging from 2 to 16. Each contig in simulated assemblies recorded information such as source chromosome, haplotype, position, and error type for subsequent result evaluation.

Three reference genomes of rice (IRGSP-1.0[4]), *Arabidopsis* (TAIR10.1[5]), and human (CHM13v2.0_noY[6]) were used as ground truths to evaluate the accuracy of Hi-C-based scaffolding tools in contig ordering and orientation. Assemblies with varying contig N50 values were also simulated using the script “sim_contigs.py”. HapHiC, ALLHiC, and LACHESIS perform ordering and orientation for individual chromosomes (clusters), differing from the whole-genome scaffolding of SALSA2, 3D-DNA, and YaHS. Thus, contigs in each simulated assembly were partitioned into corresponding chromosomes and ordered and oriented separately, making the evaluation independent of chromosome assignment. Assembly correction of scaffolding tools was disabled for this experiment.

2 Scaffolding performance evaluation in simulated datasets

We initially evaluated the performance of Hi-C-based scaffolding tools in chromosome assignment using metrics including contiguity, anchoring rate, misassignment rates, and the number of scaffolds using a custom script “result_statistics.py”.

In simulated assemblies, all chromosomes were fragmented. Thus, only sequences output by scaffolding tools containing multiple contigs are true scaffolds, while individual contigs are classified as unanchored sequences. The anchoring rate is the ratio of the length of contigs in scaffolds to the total length of all contigs. Ideally, the anchoring rate is 100%, and the number of scaffolds equals the number of chromosomes.

We did not use metrics such as N50 to assess scaffold contiguity because incorrectly separated chromosomes could artificially inflate N50 values. Instead, we designed an indicator with a maximum value of one that is independent of the anchoring rate. First, we calculated the cumulative lengths of contigs based on their source chromosomes for each scaffold and identified the longest source chromosome as the dominant chromosome. Subsequently, we divided this length by the total length of dominant chromosome anchored to all scaffolds to obtain a ratio. A ratio of one signifies that the scaffold entirely comprises contigs from a specific chromosome, and all contigs from this chromosome are distributed only within this scaffold. Finally, we calculated

the average value of this ratio among all scaffolds to obtain the final contiguity. A contiguity of one indicates that all anchored chromosome sequences correspond perfectly to scaffolds in a one-to-one relationship.

For contigs in scaffolds that do not originate from the dominant chromosome, we categorize them as either misassignments between homologous chromosomes or between non-homologous chromosomes, depending on their actual relationship with the dominant chromosome. The misassignment rate is calculated as the proportion of their length to the total length of the anchored genome. If all contigs in the scaffolds originate from the dominant chromosome, the misassignment rate is zero.

Chimeric and collapsed contigs are excluded from statistics because they cannot be considered totally correct when placed in any scaffold. During ALLHiC pruning, the genome of a closely related species, *Medicago truncatula* (MtrunA17r5.0-ANR[7]), was used as a reference. However, chromosomes 4 and 8 of *M. truncatula* have some structural differences compared to those of *M. sativa*[2]. Therefore, we also calculated the contiguity and misassignment rate after excluding these two chromosomes. Since we manually tuned the parameter k, namely the expected number of clusters, for ALLHiC, we calculated Δk as the difference between optimized k and default k.

We also evaluated the accuracy of Hi-C-based scaffolding tools in contig ordering and orientation for assemblies with varying contig N50 values. In preliminary comparisons between HapHiC and ALLHiC, we counted the number of generations for convergence and calculated scores using the "optimize" program of ALLHiC[1]. For a more objective comparison, two metrics were used to evaluate the accuracy of all scaffolding tools. Lin's concordance correlation coefficients[8] (CCCs) were calculated using a custom script "draw_tour_file.py" to measure large-scale consistency between results and reference genomes. "Costs" were also calculated using a modified DERANGE II program[9] (https://github.com/zengxiaofei/Derange2) with the parameters including linear (-L), signed (-S), and a look-ahead value of three. The weights for inversions, transpositions, and transversions were set to one, one, and two, respectively, to simulate the number of steps needed to achieve optimal results in Juicebox[10]. Costs for both the original order and reverse complementary order were calculated for each scaffolding result, and the minimum of the two values were considered the final cost. For SALSA2 and YaHS, which can output multiple scaffolds for each chromosome, we joined these scaffolds as the result of ordering and orientation. However, for 3D-DNA, the intermediate result "0.assembly" was used to ensure contig completeness and result comparability.

3 Scaffolding performance evaluation in real cases using Hi-C reads

In this analysis, we considered the published chromosome-level assemblies as the ground truth. Hi-C reads were first aligned to the assemblies. Subsequently, each assembly was partitioned into 500 Kb bins. We traversed all possible combinations of these bins, counted the number of Hi-C links connecting them, and classified them as intra- and inter-chromosome bin pairs. Similar analysis was also performed for the

scaffolds output by scaffolding tools. To ensure consistency, the scaffolds were globally orientated according to the ground truth. The only difference is that the bin pairs were classified as intra- and inter-scaffold.

We determined both upper and lower bounds for the number of Hi-C links at all bin distances (**Extended Data Fig. 7a**). We identified intra-scaffold bin pairs with Hi-C links beyond these bounds as “discordant”. The cumulative Hi-C links calling into this category formed the discordant intra-scaffold Hi-C links. These two unified metrics allow simultaneous evaluation of chromosome assignment accuracy and contig ordering and orientation. If contigs within a scaffold are incorrectly ordered or oriented, or if contigs from different chromosomes are grouped into the same scaffold, these metrics will increase.

Additionally, we counted the frequencies of inter-chromosome and inter-scaffold bin pairs for each number of Hi-C links (**Extended Data Fig. 7b**). The inter-scaffold bin pairs that outnumbered their inter-chromosome counterparts were identified as “discordant inter-scaffold bin pairs”. Then, the cumulative number of corresponding Hi-C links were defined as the “discordant inter-scaffold Hi-C links”. These two metrics exhibit greater sensitivity to chromosome assignment accuracy.

These analyses were conducted using a script, “get_discordant_HiC_links.py”, which is available at https://github.com/zengxiaofei/HapHiC.

4 Scaffolding performance evaluation in real cases using ground truth

Since the contigs were generated by breaking the N-gaps in the published chromosome-level genomes, the precise positioning and orientation of all contigs within the scaffolds were already known. Based on this information, we designed a set of metrics to evaluate the accuracy of each scaffolding tool in chromosome assignment, contig ordering, and orientation.

Initially, we calculated the lengths of all sources (specifically, the chromosomes in the ground truth) within each scaffold. The longest source within a scaffold was designated as the best. In cases where multiple sources shared the same maximum length, we prioritized them based on their length proportions in the corresponding chromosomes. Notably, a single chromosome could be the best source for multiple scaffolds (a one-to-many relationship). We then selected the scaffold that occupied the largest proportion of the chromosome. Thus, the chromosome was the reciprocal best source of this scaffold (a one-to-one relationship). Subsequently, we defined three types of contigs.

“Unanchored contigs” refer to those that are not part of any scaffold with other contigs and are not derived from the T2T chromosomes. “Newly anchored contigs” are those that were initially unanchored in the ground truth but have been anchored using scaffolding tools. “Translocated contigs” are contigs within the scaffolds that lack reciprocal best sources or are not derived from such sources.

In the subsequent analyses and definitions, we excluded these three types of contigs. The contigs within each scaffold were exclusively derived from the reciprocal best source. By encoding their order and orientation as signed integers, we identified "syntenic contigs" as those within the longest increasing subsequence (LIS) in both forward and reverse strands (**Extended Data Fig. 7c**). Notably, the length of the LIS represents the cumulative length of the contigs, rather than the number of contigs. Using the LIS as a backbone, we further classified the remaining contigs as either inverted, relocated, or both inverted and relocated.

"Inverted contigs" refer to contigs with an orientation opposite to that of the LIS. Additionally, when these contigs are inverted, the LIS lengthens. "Relocated contigs" are those with the same orientation as the LIS but are not part of it. "Inverted and relocated contigs" exhibit an opposite orientation to the LIS. Even if these contigs are inverted, the LIS cannot be extended.

These analyses were performed using a custom script, "evaluate_scaffolds.py", which can be found at https://github.com/zengxiaofei/HapHiC.

References

1. Zhang, X., Zhang, S., Zhao, Q., Ming, R. & Tang, H. Assembly of allele-aware, chromosomal-scale autopolyploid genomes based on Hi-C data. *Nature Plants* **5**, 833-845 (2019).

2. Chen, H. et al. Allele-aware chromosome-level genome assembly and efficient transgene-free genome editing for the autotetraploid cultivated alfalfa. *Nature Communications* **11**, 2494 (2020).

3. DeMaere, M.Z. & Darling, A.E. Sim3C: simulation of Hi-C and Meta3C proximity ligation sequencing technologies. *GigaScience* **7** (2017).

4. Kawahara, Y. et al. Improvement of the *Oryza sativa* Nipponbare reference genome using next generation sequence and optical map data. *Rice* **6**, 4 (2013).

5. Berardini, T.Z. et al. The arabidopsis information resource: Making and mining the "gold standard" annotated reference plant genome. *genesis* **53**, 474-485 (2015).

6. Nurk, S. et al. The complete sequence of a human genome. *Science* **376**, 44-53 (2022).

7. Pecrix, Y. et al. Whole-genome landscape of *Medicago truncatula* symbiotic genes. *Nature Plants* **4**, 1017-1025 (2018).

8. Lawrence, I.K.L. A Concordance Correlation Coefficient to Evaluate Reproducibility. *Biometrics* **45**, 255-268 (1989).

9. Blanchette, M., Kunisawa, T. & Sankoff, D. Parametric genome rearrangement. *Gene* **172**, GC11-GC17 (1996).

10. Durand, N.C. et al. Juicebox provides a visualization system for Hi-C contact maps with unlimited zoom. *Cell systems* **3**, 99-101 (2016).

Software Commands

1 Simulation of datasets for evaluating scaffolding tools

1.1 Custom scripts

All custom Python scripts used in this section are available in the "simulation" directory of the HapHiC GitHub repository (https://github.com/zengxiaofei/HapHiC).

1.2 Simulation of assemblies with different N50 values and coefficients of variation

We used a custom script, "sim_contig.py", to simulate assemblies with different N50 values and coefficients of variation (CVs) of length based on the chromosome-level genome of ground truth (chr_genome.fa):

```
$ python3 sim_contigs.py <chr_genome.fa> <N50> <CV>
```

1.3 Simulation of Hi-C data with different effective sequencing depths

We used SAMtools (version 1.11) for random sampling to simulate Hi-C data with different effective sequencing depths. The parameter "-s" sets the seed and the fraction of reads to be sampled. For example, we used "-s 12345.092" to achieve random sampling of 9.2% of filtered reads with a seed of 12345, resulting in ~1X of effective Hi-C data:

```
$ samtools view <filtered_bam> -s 12345.092 -o <filtered_bam_1X>
```

Similarly, we used a custom script "sample_mnd.py" to randomly sample Hi-C reads from "merged_nodups.txt" file for 3D-DNA:

```
$ python3 sample_mnd.py <merged_nodups_mapq1.txt> <npairs> <fraction> >
<merged_nodups_mapq1_sampled.txt>
```

1.4 Simulation of assemblies with different proportions of chimeric contigs

We used a custom script, "sim_chimeric_contigs.py", to simulate assemblies with different proportions of chimeric contigs based on the contig-level genome assemblies (asm.fa):

```
$ python3 sim_chimeric_contigs.py <asm.fa> --chimeric_ratio <ratio> --
inner_chrom_weight 0.1 --inter_nonhomo_weight 0.2 --inter_homo_weight
0.7
```

where chimeric contigs were generated between homologous chromosomes, non-homologous chromosomes, and within chromosomes at a ratio of 7:2:1, respectively.

1.5 Simulation of datasets with different sequence divergences and ploidies

We used a custom script, “sim_haplotypes.py”, to simulate haplotypes with different sequence divergences and ploidies based on the chromosome-level genome of the first haplotype in the ground truth (first_hap.fa):

```
$ python3 sim_haplotypes.py <first_hap.fa> <div> <ploidy>
```

By default, SNPs, insertions, and deletions were generated at a ratio of 10:0.5:0.5, with a 2:1 ratio of transitions to transversions for SNPs. The command produced a FASTA file of chromosome-level haplotypes (haps.fa) and a file containing allele information (allele_info.txt). Contig-level assemblies can then be simulated based on these haplotypes using “sim_contigs.py”, as described in Section **1.2**.

The corresponding Hi-C reads were generated by sim3C:

```
$ sim3C <haps.fa> <sim.fq> --profile ./mycom.txt -n <npairs> --
efficiency 0.9 --trans-rate <rate> --anti-rate 0 -l 150 -e MboI -m hic
--seed <seed> --machine-profile HiSeqXtruSeqL150
```

where “mycom.txt” is the community profile file indicating that all chromosomes are located in the same cell with equal abundance and copy numbers set to 1. The parameters “-n” and “--trans-rate” were dynamically adjusted according to different ploidies to maintain a fixed intensity of intra- and inter-chromosome Hi-C links, as listed in Supplementary Table 9. To save time, the command was executed using ten parallel jobs with different seeds.

1.6 Simulation of assemblies with different proportions of collapsed contigs

We used a custom script, “sim_collapsed_regions.py”, to simulate different proportions and lengths of collapsed regions. This was performed based on the chromosome-level genome of the first haplotype in the ground truth (first_hap.fa) and the allele information file (allele_info.txt):

```
$ python3 sim_collapsed_regions.py <first_hap.fa> <allele_info.txt> --
collapsed_len <len> --collapsed_ratio <ratio>
```

By default, collapsed regions formed by two, three, and four haplotypes were generated in a ratio of 7:2:1. This command produced a FASTA file as the template and another FASTA file containing simulated collapsed regions. The latter file was subsequently fragmented into contig-level assemblies using “sim_contigs.py”, as described in Section **1.2**.

1.7 Simulation of assemblies with different switch error rates

We used a custom script, “sim_switch_errors.py”, to simulate haplotypes with different switch error rates. This was performed based on the chromosome-level genome of the first haplotype in the ground truth (first_hap.fa) and the allele information file (allele_info.txt):

```
$ python3 sim_switch_errors.py <first_hap.fa> <allele_info.txt> --rate
<rate>
```

This command randomly shuffled variations between haplotypes and generated a new allele information file (new_allele_info.txt). The resulting FASTA files were subsequently fragmented into contig-level assemblies using “sim_contigs.py”, as described in Section **1.2**.

1.8 Simulation of assemblies with different ploidies and three types of assembly errors

We have developed a pipeline that incorporates custom scripts to introduce 5% each of chimeric, collapsed contigs and switch errors in assemblies with different ploidies.

(1) The script “sim_switch_error.py” was used to simulate haplotypes with a 5% switch error rate:

```
# simulate switch errors and generate new_allele_info.txt
$ python3 sim_switch_errors.py <first_hap.fa> <allele_info.txt> --rate
0.05
```

(2) Based on the newly generated allele information file, we simulated 5% of collapsed contigs using “sim_collapsed_regions.py”:

```
# simulate collapsed regions and generate a FASTA file as the template
and a FASTA files (template.fa) containing simulated collapsed regions
(collapsed.fa)
$ python3 sim_collapsed_regions.py <first_hap.fa> <new_allele_info.txt>
# fragment collapsed.fa into contig-level assemblies (ctg_asm.fa)
$ python3 sim_contigs.py <collapsed.fa> 500000 0.2
```

(3) The script “sim_chimeric_contigs.py” was used to generate 5% of chimeric contigs in the fragmented assemblies:

```
# simulate chimeric contigs
$ python3 sim_chimeric_contigs.py <ctg_asm.fa> --chimeric_ratio 0.05 --
inner_chrom_weight 0.1 --inter_nonhomo_weight 0.2 --inter_homo_weight
0.7
```

(4) Corresponding Hi-C reads were simulated by sim3C, based on the template FASTA file generated in step (2):

```
# simulate Hi-C reads based on template.fa
$ sim3C <template.fa> <sim.fq> --profile ./mycom.txt -n <npairs> --
efficiency 0.9 --trans-rate <rate> --anti-rate 0 -l 150 -e MboI -m hic
--seed <seed> --machine-profile HiSeqXtruSeqL150
```

1.9 Simulation of clusters for evaluating the reassignment step in HapHiC

We used a script, “sim_for_reassignment.py”, to simulate a series of clusters with different misassignment rates, anchoring rates, and contiguity levels. The commands are as follows:

```
# (1) simulate misassignments between homologous chromosomes
```

```
$ python3 sim_for_reassignment.py <asm.fa> <rate> --error_type
inter_homo --output_groups

# (2) simulate misassignments between non-homologous chromosomes
$ python3 sim_for_reassignment.py <asm.fa> <rate> --error_type
inter_nonhomo --output_groups

# (3) simulate different contiguity levels
$ python3 sim_for_reassignment.py <asm.fa> <contiguity> --error_type
contiguity --output_groups

# (4) simulate different anchoring rates
$ python3 sim_for_reassignment.py <asm.fa> <rate> --error_type
anchoring_rate --output_groups
```

1.10 Simulation of datasets for evaluating contig ordering and orientation in each scaffolding tool

We performed ordering and orientation on each chromosome in the reference genomes of rice (*Oryza sativa*, IRGSP-1.0), *Arabidopsis* (*Arabidopsis thaliana*, TAIR10.1), and human (*Homo sapiens*, CHM13v2.0_noY). To achieve this, we need to prepare chromosome-specific genome assemblies and mapped Hi-C reads for each scaffolding tool.

These chromosome-level genomes were fragmented into contig-level assemblies (asm.fa) using “sim_contigs.py”, as described in Section **1.2**. Chromosome-specific contigs were extracted based on sequence IDs, which had recorded their source chromosomes.

For 3D-DNA, we split “merged_nodups.txt” files containing mapped Hi-C reads into chromosome-specific subfiles using “split_mnd.py”:

```
$ python3 split_mnd.py <merged_nodups.txt> Chr1 Chr2 Chr3 Chr4 Chr5
[chrn ...]
```

For LACHESIS, SALSA2, and YaHS, we split BAM files containing mapped Hi-C reads into chromosome-specific subfiles using “split_bam.py”:

```
$ python3 split_bam.py <filtered_HiC.bam> Chr1 Chr2 Chr3 Chr4 Chr5
[chrn ...]
```

ALLHiC and HapHiC record mapped Hi-C reads using CLM files. ALLHiC converts all mapped Hi-C reads into a single CLM file, while HapHiC splits the CLM file into chromosome-specific subfiles. We used HapHiC to prepare both the single CLM file and chromosome-specific subfiles for HapHiC and ALLHiC:

```
$ haphic cluster <asm.fa> <filtered_HiC.bam> <nchrs> --skip_clustering
```

In addition, ALLHiC and HapHiC perform ordering and orientation on each cluster (chromosome). Therefore, we simulated chromosome-specific clusters (referred to as “counts_RE.txt” in ALLHiC) using a script “sim_group_files.py” for these two

scaffolding tools:

```
$ python3 sim_group_files.py <asm.fa>
```

1.11 Simulation of contigs for published genomes (real cases)

For published genomes, we generated contig-level assemblies by breaking the “N” gaps and randomly shuffling the ordering and orientation of contigs using custom scripts, namely “fasta_split.py” and “shuffle_fasta.py”. The commands are as follows:

```
# get contig-level assembly (asm.fa) by breaking the "N" gaps in
chromosome-level reference genome (ref.fa)
$ python3 split_fasta.py <ref.fa>

# shuffle the ordering and orientation of contigs in asm.fa
$ python3 shuffle_fasta.py <asm.fa>
```

2 Quality control, mapping, and filtering of Hi-C reads

2.1 Quality control of Hi-C reads

We used fastp (version 0.21.0) to remove adaptors, trim low-quality sequences, and evaluate the quality of raw reads for each Hi-C library:

```
$ fastp -i <raw_1.fq> -I <raw_2.fq> -o <1.fq> -O <2.fq> --threads 28 --
json <libraryID.json> --html <libraryID.html>
```

2.2 General purpose Hi-C read mapping and filtering

We used BWA-MEM (version 0.7.17-r1198-dirty) with the “-5SP” parameter to align Hi-C reads for general purpose. PCR duplicates were identified and marked using samblaster (version 0.1.26). Secondary and supplementary alignments were filtered out using SAMtools (version 1.11) with the flag “3340”. The commands are as follows:

```
$ bwa index <asm.fa>
$ bwa mem -5SP <asm.fa> <1.fq> <2.fq> -t 28 | samblaster | samtools
view -@ 14 -S -h -b -F 3340 -o <HiC.bam>
```

Then, we removed the alignments with a mapping quality of zero (MAPQ = 0) and an edit distance greater than 2 (NM>=3) using a custom script “filter_bam.py”, which is available in the “utils” directory of the HapHiC GitHub repository (https://github.com/zengxiaofei/HapHiC).

```
$ python3 filter_bam.py <HiC.bam> 1 --NM 3 --threads 14 | samtools view
-b -@ 14 -o <filtered_HiC.bam>
```

2.3 Hi-C read mapping and filtering for 3D-DNA scaffolding

We used the Juicer pipeline (https://github.com/aidenlab/juicer, commit: 827fe44) to align and filter Hi-C reads for 3D-DNA as follows:

```
# (1) prepare scripts
```

```
ln -s /path/to/juicer2/CPU scripts

# (2) prepare reads
mkdir fastq && cd fastq
ln -s <1.fq> reads_R1.fastq.gz && ln -s <2.fq> reads_R2.fastq.gz &&
cd ..

# (3) prepare genome
mkdir references && cd references
ln -s <asm.fa> genome.fa && bwa index genome.fa && cd ..

# (4) prepare restriction sites
$ python2 /path/to/juicer2/misc/generate_site_positions.py MboI
genome ../references/genome.fa
$ awk 'BEGIN{OFS="\t"}{print $1, $NF}' genome_MboI.txt >
genome.chrom.sizes

# (5) run Juicer pipeline
$ bash /path/to/juicer2/CPU/juicer.sh -g genome -z references/genome.fa
-p restriction_sites/genome.chrom.sizes -y
restriction_sites/genome_MboI.txt -D `pwd` -s MboI -t 28 --assembly
```

3 Hi-C-based scaffolding

3.1 HapHiC

(1) HapHiC was typically executed with default optional parameters:

```
$ haphic pipeline <asm.fa> <filtered_HiC.bam> <nchrs> --RE <RE>
```

where the parameter “--RE” was set based on the restriction site used in Hi-C library construction. If the restriction site was unclear, the parameter was left as default (GATC). Commands for special cases are listed separately as follows.

(2) We evaluated the performance of HapHiC in scaffolding assemblies with contig N50 values ranging from 2 Mb to 25 Kb. For scaffolding the assembly with an extremely low contig N50 of 25 Kb, we used a lower contig length threshold of N70 for Markov clustering to save time and set lower thresholds for number of Hi-C links and link density to optimize the anchoring rate:

```
$ haphic pipeline <asm.fa> <filtered_HiC.bam> <nchrs> --Nx 70 --
min_link_density 0.00002 --min_links 1
```

(3) For scaffolding assemblies with different effective Hi-C sequencing depths, we used lower thresholds for the number of Hi-C links and link density to optimize the anchoring rate. For effective depths ranging from 1X to 0.02X, the thresholds for number of Hi-C links were set to 20, 10, 4, 2, 1, 1, and 1, while the thresholds for Hi-C link density were set to 0.0001, 0.00005, 0.00002, 0.00001, 0.000005, 0.000003, and 0.000001 (which is 0.1 to 0.002 times the default values):

```
$ haphic pipeline <asm.fa> <filtered_HiC.bam> <nchrs> --
min_link_density <density> --min_links <links>
```

(4) For scaffolding assemblies with simulated chimeric contigs, HapHiC was executed with an additional parameter, "--correct_nrounds", which is required for assembly correction:

```
$ haphic pipeline <asm.fa> <filtered_HiC.bam> <nchrs> --correct_nrounds
2
```

(5) For scaffolding assemblies with simulated switch errors, HapHiC was executed with an additional parameter, "--remove_allelic_links", which is required for removing inter-allele Hi-C links:

```
$ haphic pipeline <asm.fa> <filtered_HiC.bam> <nchrs> --
remove_allelic_links <nhaps>
```

(6) To evaluate the performance of the rank-sum method in identifying chimeric or collapsed contigs, we calculated the rank-sum value of each contig using HapHiC:

```
$ haphic cluster <asm.fa> <filtered_HiC.bam> <nchrs> --Nx 100 --
bin_size 0 --min_RE_sites 0 --density_lower 0 --density_upper 1 --
rank_sum_upper 0 --skip_clustering --verbose 2> <log.txt>
```

(7) To evaluate the performance of the Hi-C link density method in identifying collapsed contigs, we calculated the link density of each contig using HapHiC:

```
$ haphic cluster <asm.fa> <filtered_HiC.bam> <nchrs> --Nx 100 --
bin_size 0 --min_RE_sites 0 --density_lower 0 --density_upper 0 --
rank_sum_upper 1 --skip_clustering --verbose 2> <log.txt>
```

(8) To evaluate the performance of the concordance ratio method in identifying allelic contig pairs, we calculated the concordance ratio of each contig using HapHiC:

```
$ haphic cluster <asm.fa> <filtered_HiC.bam> <nchrs> --Nx 100 --
bin_size 0 --min_RE_sites 0 --density_lower 0 --density_upper 1 --
rank_sum_upper 1 --remove_allelic_links 4 --skip_clustering --verbose
2> <log.txt>
```

(9) To evaluate the performance of reassignment, we executed HapHiC using the simulated clusters as described in Section **1.9**:

```
$ haphic reassign <asm.fa> <full_links.pkl> <simulated_clusters.txt> --
nclusters <nchrs>
```

(10) To evaluate the performance and efficiency of ordering and orientation, we executed HapHiC using the simulated datasets as described in Section **1.10**:

```
$ haphic sort <asm_chr.fa> <HT_links.pkl> <split_clm_dir>
<group_chr.txt>
```

where "asm_chr.fa" and "group_chr.txt" represent the chromosome-specific contigs

and clusters, respectively. “HT_links.pkl” is a pickle file containing Hi-C links between semi-contigs, while “split_clm_dir” is a directory containing chromosome-specific CLM files. Both are generated by “haphic cluster”. This command generates “*.tour.sav” files as the results of fast sorting and “*.tour” files as the final ordering and orientation results after ALLHiC optimization.

To evaluate the efficiency (execution speed and memory usage) of the entire HapHiC pipeline without ALLHiC optimization (fast sorting only):

```
$ haphic pipeline <asm.fa> <filtered_HiC.bam> <nchrs> --skip_allhic
```

(11) Given the extremely low contig contiguity of the *Saccharum spontaneum* AP85-441 genome assembly (contig N50 of ~45 Kb) and the presence of numerous collapsed contigs, we added the parameters “–Nx 70”, “–density_upper 0.9 --rank_sum_upper 0.8”, and “–min_RE_sites 2” to save time, filter out potential collapsed contigs, and optimize the anchoring rate, respectively:

```
$ haphic pipeline <asm.fa> <filtered_HiC.bam> 32 --RE AAGCTT --Nx 70 --
density_upper 0.9 --rank_sum_upper 0.8 --min_RE_sites 2
```

(12) For scaffolding the genome assembly of *M. sativa* XinJiangDaYe, we added the parameters “--min_links 5 --min_link_density 0.00001” to optimize the anchoring rate:

```
$ haphic pipeline <asm.fa> <filtered_HiC.bam> 32 --min_links 5 --
min_link_density 0.00001
```

(13) Given the extremely low contig contiguity of the *M. sativa* Zhongmu-4 genome assembly (contig N50 of ~94 Kb) and the presence of numerous collapsed contigs, we added the parameters “–Nx 70”, “–density_upper 0.8 --rank_sum_upper 0.8”, and “–min_RE_sites 2 --min_links 5” to save time, filter out potential collapsed contigs, and optimize the anchoring rate, respectively:

```
$ haphic pipeline <asm.fa> <filtered_HiC.bam> 32 --RE AAGCTT --Nx 70 --
density_upper 0.8 --rank_sum_upper 0.8 --min_RE_sites 2 --min_links 5
```

(14) Considering the presence of numerous collapsed and chimeric contigs in the genome assembly of *S. spontaneum* Np-X, we added “--correct_nrounds 2” to correct chimeric contigs and “--density_upper 0.9 --rank_sum_upper 0.8” for filtering of collapsed contigs:

```
$ haphic pipeline <asm.fa> <filtered_HiC.bam> 40 --RE AAGCTT --
correct_nrounds 2 --density_upper 0.9 --rank_sum_upper 0.8
```

(15) Considering the presence of large, nearly identical regions in the potato C88 assembly where Hi-C links are not evenly distributed. we added “--normalize_by_nlinks --remove_concentrated_links” to normalize the Hi-C links between contigs:

```
$ haphic pipeline <asm.fa> <filtered_HiC.bam> 48 --RE AAGCTT --
normalize_by_nlinks --remove_concentrated_links
```

For C88.v1 assemblies, an additional parameter –Nx 100 was added since too much contigs were removed by the default filtering criteria of Hi-C link density and rank-sum values:

```
$ haphic pipeline <asm.fa> <filtered_HiC.bam> 48 --RE AAGCTT --
normalize_by_nlinks --remove_concentrated_links --Nx 100
```

(16) For scaffolding the assembly of human diploid genome HG002.v1, we added "—remove_allelic_links 2" to remove the inter-allele Hi-C links and "--density_upper 1 --remove_concentrated_links --normalize_by _nlinks" to deal with the higher link density of sex chromosomes:

```
$ haphic pipeline <asm.fa> <filtered_HiC.bam> 46 --remove_allelic_links
2 --density_upper 1 --remove_concentrated_links --normalize_by_nlinks
```

3.2 ALLHiC

(1) We evaluated the performance of ALLHiC (version 0.9.13) in scaffolding assemblies under various adverse conditions. The chromosome assignment in ALLHiC was typically performed in three different modes with default parameters.

(i) Using a reference genome for pruning and separating homologous groups (referred to as "ALLHiC sep" in figures):

```
# generate Allele.ctg.table for pruning
$ gmap_build -D . -d DB <asm.fa>
# use gmapl instead of gmap for large genomes
$ gmap -D . -d DB -t 28 -f 2 -n <nhaps> <cds.fa> > <gmap.gff3>
$ perl /path/to/ALLHiC/scripts/gmap2AlleleTable.pl <gmap.gff3>

# separate homologous groups
$ samtools sort <filtered_HiC.bam> -o <sorted_filtered_HiC.bam> -@ 28
$ samtools index <sorted_filtered_HiC.bam>
$ perl /path/to/ALLHiC/scripts/partition_gmap.py -r <asm.fa> -g
Allele.ctg.table -b <sorted_filtered_HiC.bam> -t 28

# pruning for each separated homologous group
$ ALLHiC_prune -i <Allele.ctg.table> -b <sorted_filtered_HiC_group.bam>
-r <asm_group.fa>

# partitioning (clustering) for each separated homologous group
$ ALLHiC_partition -b <pruning.bam> -r <asm_group.fa> -e <RE> -k
<nclusters>

# rescue for each separated homologous group
$ ALLHiC_rescue -b <sorted_filtered_HiC_group.bam> -r <asm_group.fa> -c
<clusters_group.txt> -i <counts_RE_group.txt>

# global rescue (when anchoring rate is less than 1)
# generate counts_RE.txt file for all contigs (counts_RE_all.txt)
```

```
$ allhic extract <filtered_HiC_group.bam> <asm.fa> --RE <RE>
$ ALLHiC_rescue -r <asm.fa> -b <filtered_HiC_group.bam> -c
<clusters_all.txt> -i <counts_RE_all.txt>
```

When scaffolding the autotetraploid genome assemblies for *M. sativa* (including XinJiangDaYe, Zhongmu-4, and simulated assemblies derived from XinJiangDaYe), the genome of a closely related species, *M. truncatula* (MtrunA17r5.0-ANR), was selected as the reference based on the species used in their published papers. Considering that chromosomes 4 and 8 of *M. truncatula* have structural differences compared to those in *M. sativa*, the parameter "nclusters" was set starting from 8 to obtain eight primary groups for these chromosomes during the partitioning step. For the rest of the chromosomes, this parameter was set starting from 4.

When scaffolding the autotetraploid genome assemblies for *S. spontaneum* (including both Np-X and AP85-411), the genome of a closely related species, *Sorghum bicolor*, was selected as the reference based on the species used in their published papers. Considering that chromosomes 5 and 8 of *S. bicolor* have structural differences compared to those in *S. spontaneum* AP85-411, the parameter "nclusters" was set starting from 8 to obtain eight primary groups for these chromosomes during the partitioning step. For the rest of the chromosomes in AP85-411 and all chromosomes in Np-X, the parameter "nclusters" was set starting from 4.

When scaffolding the autotetraploid genome assemblies for potato C88, the potato DM8.1 assembly was selected as the reference for the assembly we constructed in this work; while the first haplotype of potato C88.v1 assembly was used as the reference for scaffolding the simulated contigs derived from the C88.v1. The parameter "nclusters" was set to 4.

When scaffolding the diploid genome assembly for *Camellia sinensis* Tieguanyin, the chromosome-level monoploid genome assembly of this sample was selected as the reference based on the method described in their published paper. The parameter "nclusters" was set starting from 2.

When scaffolding the allotriploid genome assembly for *Miscanthus* × *giganteus*, the chromosome-level genome assembly of *Miscanthus sinensis* was used as the reference. The parameter "nclusters" was set starting from 3.

We also used the first haplotype of *M. sativa* XinJiangDaYe as the reference for ALLHiC to validate the effectiveness of the conclusion with an improved reference genome in scaffolding simulated assemblies with varying contig N50 values.

(ii) Using a reference genome for pruning without separating homologous groups (referred to as "ALLHiC" in figures):

```
# generate Allele.ctg.table for pruning
$ gmap_build -D . -d DB <asm.fa>
# use gmapl instead of gmap for large genomes
$ gmap -D . -d DB -t 28 -f 2 -n <nhaps> <cds.fa> > <gmap.gff3>
```

```
$ perl /path/to/ALLHiC/scripts/gmap2AlleleTable.pl <gmap.gff3>

# pruning
$ ALLHiC_prune -i <Allele.ctg.table> -b <filtered_HiC.bam> -r <asm.fa>

# partitioning (clustering)
$ ALLHiC_partition -b <pruning.bam> -r <asm.fa> -e <RE> -k <nclusters>

# rescue (when anchoring rate is less than 1)
$ ALLHiC_rescue -b <filtered_HiC.bam> -r <asm.fa> -c
<pruning_clusters.txt> -i <pruning_counts_RE.txt>
```

(iii) Executing ALLHiC without pruning or separating homologous groups (referred to as "ALLHiC no pruning" in figures):

```
# partitioning (clustering)
$ ALLHiC_partition -b <filtered_HiC.bam> -r <asm.fa> -e <RE> -k
<nclusters>

# rescue (when anchoring rate is less than 1)
$ ALLHiC_rescue -b <filtered_HiC.bam> -r <asm.fa> -c <clusters_all.txt>
-i <counts_RE_all.txt>
```

(2) For the processes of ordering, orientation, and building final scaffolds:

```
# order and orient contigs for each cluster/chromosome
$ allhic optimize <counts_RE_chr.txt> <HiC.clm>

# build final scaffolds/pseudomolecules in the directory containing
*.tour files
$ ALLHiC_build <asm.fa>
```

(3) For scaffolding assemblies with different effective Hi-C sequencing depths, we used lower thresholds for link density to optimize anchoring rate. For effective depths ranging from 1X to 0.02X, the thresholds for Hi-C link density were set to 0.0001, 0.0005, 0.0002, 0.0001, 0.00005, 0.00003, and 0.00002 in the rescue process (which is 0.1 to 0.002 times the default values):

```
# (i) Using a reference genome for pruning and separating homologous
groups
# rescue for each separated homologous group
$ ALLHiC_rescue -r <asm.fa> -b <filtered_HiC_group.bam> -c
<clusters_all.txt> -i <counts_RE_all.txt> -m <density>

# global rescue
$ ALLHiC_rescue -r <asm.fa> -b <filtered_HiC_group.bam> -c
<clusters_all.txt> -i <counts_RE_all.txt> -m <density>

# (ii) Using a reference genome for pruning without separating
homologous groups
$ ALLHiC_rescue -b <filtered_HiC.bam> -r <asm.fa> -c
```

```
<pruning_clusters.txt> -i <pruning_counts_RE.txt> -m <density>

# (iii) Executing ALLHiC without pruning or separating homologous
groups
$ ALLHiC_rescue -b <filtered_HiC.bam> -r <asm.fa> -c <clusters_all.txt>
-i <counts_RE_all.txt> -m <density>
```

(4) When scaffolding assemblies with chimeric contigs, we performed additional assembly correction before the ALLHiC pipeline:

```
$ samtools sort <filtered_HiC.bam> -o <sorted_filtered_HiC.bam> -@ 28
$ samtools index <sorted_filtered_HiC.bam>
$ ALLHiC_corrector -m <sorted_filtered_HiC.bam> -r <asm.fa> -o
<asm_corrected.fa> -t 28
```

Subsequently, Hi-C reads were realigned to “asm_corrected.fa” using the method described in Section **2.2**.

(5) When scaffolding the genome assembly of *M. sativa* XinJiangDaYe, we added “-m 100” in the partitioning step to optimize the results:

```
$ ALLHiC_partition -b <pruning.bam> -r <asm_group.fa> -e <RE> -k
<nclusters> -m 100
```

3.3 LACHESIS

We executed LACHESIS (https://github.com/shendurelab/LACHESIS, commit: 2e27abb) without using a reference genome:

```
$ Lachesis params.ini
```

where the “params.ini” file contained several modifications from the default “test_case.ini” file provided by LACHESIS:

```
# use an arbitrary species name but avoid using string like "human" to
represent a species without a reference genome
SPECIES = alfalfa
OUTPUT_DIR = out
DRAFT_ASSEMBLY_FASTA = <asm.fa>
SAM_DIR = ./bam
SAM_FILES = <filtered_HiC.bam>
RE_SITE_SEQ = <RE>
USE_REFERENCE = 0
CLUSTER_N = <nchrs>
CLUSTER_DRAW_HEATMAP = 0
CLUSTER_DRAW_DOTPLOT = 0
REPORT_DRAW_HEATMAP = 0
```

3.4 SALSA2

SALSA2 (https://github.com/marbl/SALSA, commit: ed76685) was executed without

an assembly graph.

(1) Assembly correction was typically disabled. Given that the BAM file "filtered_HiC.bam" is already sorted by name, the sorting step was skipped:

```
$ bedToBed -i <filtered_HiC.bam> > alignment.bed
$ samtools faidx <asm.fa>
$ python2 run_pipeline.py -a <asm.fa> -l <asm.fa.fai> -b alignment.bed
-e <RE> -m no -o output
```

(2) For scaffolding assemblies with chimeric contigs, assembly correction was enabled:

```
$ python2 run_pipeline.py -a <asm.fa> -l <asm.fa.fai> -b alignment.bed
-e <RE> -m yes -o output
```

3.5 3D-DNA

(1) 3D-DNA (https://github.com/aidenlab/3d-dna, commit: 529ccf4) was typically executed using default parameters without assembly correction:

```
$ bash run-asm-pipeline.sh -r 0 <asm.fa> <merged_nodups.txt>
```

(2) For scaffolding assemblies with different proportions of chimeric contigs, we performed two rounds of assembly correction. The repeat coverage threshold for the misjoin editor was set to 6 to prevent error messages. For scaffolding assemblies with 5% each of chimeric, collapsed contigs, and switch errors, we set the repeat coverage threshold for the misjoin editor to 6 and 10 for diploid and triploid genomes, respectively:

```
$ bash run-asm-pipeline.sh -r 2 --editor-repeat-coverage <cov> <asm.fa>
<merged_nodups.txt>
```

(3) For scaffolding assemblies with low contig N50 values (50 Kb and 25 Kb), and high proportions of collapsed contigs and switch errors (25%), we set a lower coarse stringency value of 30 for the polisher to prevent error messages:

```
$ bash run-asm-pipeline.sh -r 0 --polisher-coarse-stringency 30
<asm.fa> <merged_nodups.txt>
```

3.6 YaHS

(1) YaHS (https://github.com/c-zhou/yahs, commit: 42b8421) was typically executed without assembly correction and memory check:

```
$ samtools faidx <asm.fa>
$ yahs <asm.fa> <filtered_HiC.bam> -e <RE> -q 1 --no-contig-ec --no-
mem-check
```

(2) When encountering difficulties in scaffolding low-contiguity assemblies with YaHS, we specified different ranges of resolutions. For example:

```
$ yahs <asm.fa> <filtered_HiC.bam> -e <RE> -q 1 --no-contig-ec --no-
```

```
mem-check -r
1000,2000,5000,10000,20000,50000,100000,200000,500000,1000000,2000000,5
000000,10000000,20000000,50000000,100000000,200000000,500000000
```

(3) Given that YaHS can only handle up to 45,000 contigs, we used the parameter "-l" to specify the minimum contig length for scaffolding when the number of contigs exceeded this limit. For example:

```
$ yahs <asm.fa> <filtered_HiC.bam> -e <RE> -q 1 --no-contig-ec -l 5000 -
-no-mem-check
```

(4) For scaffolding assemblies with chimeric contigs, assembly correction was enabled:

```
$ yahs <asm.fa> <filtered_HiC.bam> -e <RE> -q 1 --no-mem-check
```

(5) In case of an out-of-memory error, we manually released the cached memory and then executed YaHS without the parameter "--no-mem-check":

```
$ sync && echo 1 > /proc/sys/vm/drop_caches
$ yahs <asm.fa> <filtered_HiC.bam> -e <RE> -q 1 --no-contig-ec
```

3.7 Visualization in Juicebox

To visualize the scaffolding results of HapHiC, ALLHiC, and YaHS in Juicebox, we prepared "*.assembly" and "*.hic" files based on the AGP files (*.agp) output by these scaffolding tools using matlock (https://github.com/phasegenomics/matlock), juicebox scripts (https://github.com/phasegenomics/juicebox_scripts, commit: 4d3d297) and 3D-DNA (https://github.com/aidenlab/3d-dna, commit: 529ccf4):

```
# (1) generate .mnd files (merged nodups)
$ /path/to/matlock bam2 juicer <filtered_HiC.bam> > out.links.mnd
$ sort -k2,2 -k6,6 out.links.mnd > out.sorted.links.mnd

# (2) generate .assembly file
$ /path/to/juicebox_scripts/agp2assembly.py <scaffolds_final.agp>
<scaffolds_final.assembly>

# (3) generate .hic file
$ bash /path/to/3d-dna/visualize/run-assembly-visualizer.sh -p false
<scaffolds_final.assembly> out.sorted.links.mnd
```

Subsequently, the generated "*.hic" and "*.assembly" files were imported into Juicebox (version 1.11.08) for visualization.

4 Performance evaluation of scaffolding

4.1 Custom scripts

All custom Python scripts used in this section are available in the "simulation" directory of the HapHiC GitHub repository (https://github.com/zengxiaofei/HapHiC).

4.2 Performance evaluation of chromosome assignment

HapHiC and ALLHiC provide the results of chromosome assignment in “group*.txt” files (also known as “counts_RE.txt” in ALLHiC). For other scaffolding tools, we converted their results to this format using different custom scripts:

```
# (1) for LACHESIS
$ python3 convert_lachesis_result_to_groups.py <clusters.by_name.txt>
<asm.fa>

# (2) for 3D-DNA
$ python3 convert_assembly_to_groups.py <genome.final.assembly>

# (3) for SALSA2 and YaHS
$ python3 convert_agp_to_groups.py <scaffolds_final.agp>
```

Subsequently, we evaluated the performance of Hi-C-based scaffolding tools in chromosome assignment using metrics such as contiguity, anchoring rate, misassignment rates, and the number of scaffolds. This was accomplished using a custom script named “result_statistics.py”:

```
$ python3 result_statistics.py <asm.fa> <counts_RE_group1.txt>
<counts_RE_group2.txt> [<counts_RE_group3.txt> ...]
```

4.3 Performance evaluation of ordering and orientation

HapHiC and ALLHiC provide the results of ordering and orientation in “*.tour” files. For other scaffolding tools, we converted their results to this format using different custom scripts:

```
# (1) for LACHESIS
$ python3 convert_lachesis_ordering_to_tour.py <asm_chr.fa> <chr_name>
<group0.ordering> <group1.ordering> [<group2.ordering> ...]

# (2) for 3D-DNA
$ python3 convert_assembly_to_tour.py <genome.0.assembly> <chr_name>

# (3) for SALSA2 and YaHS
$ python3 convert_agp_to_tour.py <scaffolds_final.agp> <chr_name>
```

Subsequently, the ordering and orientation results of each scaffolding tools were visualized using a custom script, “draw_tour_file.py”. This script calculates Lin’s concordance correlation coefficients (CCCs) and DERANGE costs for each input “*.tour” file:

```
$ python3 draw_tour_file.py <asm_chr.fa> <chr.tour> <program_name>
<N50> --CCC --derange
```

4.4 Performance evaluation of assembly correction

We evaluated the performance of assembly correction on simulated chimeric contigs for each scaffolding tool. The breakpoints were extracted and analyzed using different

custom scripts:

```
# (1) for HapHiC
$ python3 get_haphic_break_points.py <asm.fa> <asm_corrected.fa> <N50>

# (2) for ALLHiC
$ python3 get_allhic_break_points.py <asm.fa> <asm_corrected.fa> <N50>

# (3) for 3D-DNA
$ python3 get_3d_dna_break_points.py <asm.fa> <genome.FINAL.assembly>
<N50>

# (4) for SALSA2
$ python3 get_salsa_break_points.py <asm.fa> <scaffolds_final.agp>

# (5) for YaHS
$ python3 get_yahs_break_points.py <asm.fa> <scaffolds_final.agp>
```

We also compared the performance of assembly correction between HapHiC and ALLHiC in the genome assembly of *S. spontaneum* Np-X:

```
$ python3 correction_analysis.py <asm.fa> <filtered_HiC.bam>
<ctg_annoation_list.txt> <scaffolds_HapHiC.agp>
<scaffolds_ALLHiC.agp> > correction_statistics.txt
```

where “ctg_annoation_list.txt” is a list file that contains manually annotated chimeric and non-chimeric contigs.

4.5 Performance evaluation of the rank-sum method in identifying chimeric and collapsed contigs

We evaluated the performance of the rank-sum method in identifying chimeric and collapsed contigs using the “log.txt” files generated in Section **3.1** (6). The custom scripts “chimeric_contig_statistics.py” and “collapsed_contig_statistics.py” were utilized for this purpose:

```
# (1) for chimeric contigs
$ python3 chimeric_contig_statistics.py <asm.fa> <log.txt>
<program_name> <N50>

# (2) for collapsed contigs
$ python3 collapsed_contig_statistics.py <asm.fa> <log.txt>
<program_name> <N50> --method rank_sum
```

4.6 Performance evaluation of the Hi-C link density method in identifying collapsed contigs

We evaluated the performance of the Hi-C link density method in identifying collapsed contigs using the “log.txt” files generated in Section **3.1** (7). The custom script “collapsed_contig_statistics.py” was utilized for this purpose:

```
$ python3 collapsed_contig_statistics.py <asm.fa> <log.txt>
```

```
<program_name> <N50> --method link_density
```

4.7 Performance evaluation of the concordance ratio method and the pruning method in identifying allelic contig pairs

We evaluated the performance of the concordance ratio method and pruning method in identifying allelic contig pairs using the "log.txt" files generated in Section **3.1** (8). The custom script "allelic_contig_statistics.py" was utilized for this purpose:

```
$ python3 allelic_contig_statistics.py <log.txt> <Allele.ctg.table>
<N50>
```

5 Genome assembly

5.1 Genome assembly of *M.* × *giganteus*

The genome of *M.* × *giganteus* was assembled using hifiasm (version 0.13-r308). The parameter "-l0" was employed to generate primary unitigs (p_utg):

```
$ hifiasm -o Mgi -l0 -t 28 <hifi.fq>
```

In addition, the genome of *M.* × *giganteus* was assembled using HiCanu (version 2.1.1) for comparison with the hifiasm assembly, in order to explain the source of diagonally distributed inter-allele Hi-C links:

```
$ canu -p Mgi -d Mgi_minOverlapLength_200 \
    genomeSize=2g \
    -pacbio-hifi <hifi.fq> \
    minOverlapLength=200
```

5.2 Genome assembly of potato C88

The genome of potato (*Solanum tuberosum*) C88 was assembled using hifiasm (version 0.19.0-r534) to generate primary unitigs (p_utg):

```
$ hifiasm <hifi.fq> -l 0 --primary -t 64
```

6 Comparative genomics analyses

6.1 Custom scripts

All custom Python scripts used in this section are available in the "simulation" directory of the HapHiC GitHub repository (https://github.com/zengxiaofei/HapHiC).

6.2 *K*-mer analysis

We performed a *k*-mer analysis on each potato C88 scaffold using a custom script, "haplotype_kmers.py", based on the published reference genome (C88.v1):

```
$ python3 haplotype_kmers.py <C88.v1.fa> <scaffold_*.fa>
```

where "scaffold_*.fa" represents the FASTA file of each scaffold and the default *k*-mer

size was set to 201 bp.

6.3 Gene synteny analysis

Initially, we simply identified coding sequences in each subgenome of *M.* × *giganteus* through GMAP (version 2019-12-01) mapping, using the coding sequences of *M. sinensis* as queries:

```
$ gmap_build -D . -d DB <subgenome.fa>
$ gmap -D . -d DB -t 28 -f 2 -n 1 ref.cds > <subgenome.gff>
# extract coding sequences using gffread from cufflinks (version 2.2.1)
$ gffread Msisub.gff -g genome.fa -x <subgenome.cds>
```

Subsequently, we performed gene synteny comparisons between the subgenomes of *Miscanthus* species and generated a karyotype plot using MCscan in JCVI utility libraries (version 1.1.18):

```
# prepare BED file for subgenomes
$ python3 -m jcvi.formats.gff bed --type=<type> --key=<key>
<subgenome.gff> -o <subgenome.bed>

# perform synteny analysis between subgenome pairs
$ python3 -m jcvi.compara.catalog ortholog <subgenomeA> <subgenomeB> --
no_strip_names --cscore 0.99
$ python3 -m jcvi.compara.synteny screen --minspan=30 --simple
subgenomeA.subgenomeB.anchors subgenomeA.subgenomeB.anchors.new

# retain synteny blocks between orthologous chromosomes
$ python3 filter_homologous_anchors.py
subgenomeA.subgenomeB.anchors.simple <subgenomeA.bed> <subgenomeB.bed>
"Chr01A,Chr02A,Chr03A,..." "Chr01B,Chr02B,Chr03B,..." >
subgenomeA.subgenomeB.anchors.simple.ortho

# draw karyotype plot
$ python3 -m jcvi.graphics.karyotype seqids layout
```

6.4 Genome alignment and structural variation identification

(1) We performed a sequence alignment between potato C88 scaffolds and each haplotype of C88.v1 using unimap (https://github.com/lh3/unimap, version 0.1-r41):

```
# sequence alignment for each haplotype of C88.v1
$ unimap -d <hap*.umi> <C88.v1.hap*.fa> -t 28
$ unimap -c <hap*.umi> <scaffolds.fa> -x asm5 --cs -N 50 --secondary=no
-t 28 > <scaffolds_hap*.paf>

# concatenate all PAF files into a single file
$ cat scaffolds_hap1.paf scaffolds_hap2.paf scaffolds_hap3.paf
scaffolds_hap4.paf > scaffolds_allhaps.paf
```

Subsequently, we extracted the corresponding alignments from the PAF files for each

chromosome and recalculate the number of matching bases (Column 10) based on the "de" tag (Column 21). For example:

```
$ grep "chr7_" ${aln} | egrep "group25|group28|group29|group30" | cut -
f 1-12,21 | sed 's/de:f://g' | awk '{print
$1"\t"$2"\t"$3"\t"$4"\t"$5"\t"$6"\t"$7"\t"$8"\t"$9"\t"(1-
$13)*$11"\t"$11"\t"$12}' > haphic_chr7.paf
```

Finally, we visualized the output PAF files using a modified version of paf2dotplot (https://github.com/zengxiaofei/paf2dotplot):

```
$ paf2dotplot.r -s -c 0.99 -p 4 <chr*.paf>
```

(2) We performed a sequence alignment and identified structural variations between chromosome 2 of *M.* × *giganteus* and *M. sinensis* using Minimap2 (version 2.26-r1175) and SyRI (version 1.6.3):

```
# sequence alignment
$ minimap2 -ax asm5 --eqx Msiref_Chr02.fa Msisub_Chr02.fa -t 28 >
Msisub_vs_Msiref_Chr02.sam

# structural variation identification
$ syri -c Msisub_vs_Msiref_Chr02.sam -r Msiref_Chr02.fa -q
Msisub_Chr02.fa -k -F S

# extract SVs identified by JCVI
$ python3 extract_SVs_from_simple.py Msi.Msisub.anchors.simple.ortho
Msi.bed

# visualization
$ plotsr --sr syri.out --genomes genomes.txt -s 100000 --tracks
tracks.txt -o plotsr_Msiref_ref.pdf -W 6 -H 3 -S 0.5
```

6.5 Analysis of genetic maps

Initially, we aligned the genetic markers of each map to the A subgenomes of *M.* × *giganteus* (MgiA) and the *M. sinensis* genome (MsiA) using BWA-ALN and BWA-SAMSE (version 0.7.17-r1198-dirty) with default parameters:

```
$ bwa index <genome*.fa>
$ bwa aln <genome*.fa> <map*.fa> > <map*_genome*.sai> -t 28
$ bwa samse <genome*.fa> <map*_genome*.sai> <map*.fa> >
<map*_genome*.sam>
$ samtools view -b <map*_genome*.sam> -o <map*_genome*.bam>
```

After alignment, each pair of BAM files generated using the same genetic map were converted to a pair of CSV files, retaining only shared genetic markers mapped to the same linkage groups:

```
$ python3 bam_to_csv_shared.py --sam_LG_only <map*_genome_MgiA.bam>
<map*_genome_MsiA.bam>
```

Subsequently, we visualized the agreements between the genetic maps and each genome using ALLMAPS in JCVI utility libraries:

```
$ python3 -m jcvi.assembly.allmaps merge MapA_M_genome*.csv
MapA_P_genome*.csv MapB_M_genome*.csv MapB_P_genome*.csv
MapC_M_genome*.csv MapC_P_genome*.csv -o combined.bed
$ python3 mock_agp_file.py <genome*.fa> > combined.chr.agp
$ cp combined.chr.agp combined.agp
$ cp combined.bed combined.lifted.bed
$ python3 -m jcvi.assembly.allmaps plotall combined.bed
```

where the custom script “mock_agp_file.py” is available in the “utils” directory of the HapHiC GitHub repository at https://github.com/zengxiaofei/HapHiC.

Supplementary Tables

Supplementary Table 1 | Statistics of simulated assemblies with different N50 values

Simulation (N50/CV)	Total length (bp)	Number of contigs	Max length (bp)	Avg length (bp)	Actual N50 (bp)
2 Mb/0.2	2,044,810,672	1,028	3,571,011	1,989,115	2,064,317
1 Mb/0.2	2,044,810,672	2,049	1,785,505	997,955	1,037,798
800 Kb/0.2	2,044,810,672	2,558	1,350,484	799,379	832,577
500 Kb/0.2	2,044,810,672	4,084	892,752	500,688	520,506
300 Kb/0.2	2,044,810,672	6,826	535,651	299,562	311,687
200 Kb/0.2	2,044,810,672	10,256	357,101	199,377	207,685
100 Kb/0.2	2,044,810,672	20,470	178,550	99,893	103,997
50 Kb/0.2	2,044,810,672	40,928	89,617	49,961	52,025
25 Kb/0.2	2,044,810,672	81,856	45,829	24,981	25,978

Supplementary Table 2 | Statistics of simulated assemblies with different coefficients of variation (CVs)

Simulation (N50/CV)	N10 (bp)	N30 (bp)	N50 (bp)	N70 (bp)	N90 (bp)
500 Kb/0.2	647,202	573,137	520,506	466,757	397,161
500 Kb/0.3	670,526	572,006	534,413	429,894	333,633
500 Kb/0.4	716,123	595,095	507,578	417,377	301,438
500 Kb/0.6	760,920	611,098	500,829	397,033	255,800
500 Kb/0.8	819,645	637,888	512,337	398,322	241,392
500 Kb/1	821,080	630,147	499,628	381,094	224,603
500 Kb/1.5	848,800	636,684	501,865	373,211	213,587
500 Kb/3	868,450	641,236	498,918	362,597	206,077

Supplementary Table 3 | Parameters for simulating different effective Hi-C sequencing depths

Sampling method	Simulation (depth)	Fraction	Number of read pairs	Number of read bases	Calculated depth
sample_mnd.py (For 3D-DNA)	All	NA	97,500,525	19,692,884,346	9.6X
	1X	0.104	10,140,054	2,048,066,632	1.0X
	0.5X	0.0519	5,060,277	1,022,089,650	0.5X
	0.2X	0.0208	2,028,010	409,614,990	0.2X
	0.1X	0.0104	1,014,005	204,807,554	0.1X
	0.05X	0.00519	506,027	102,209,804	0.05X
	0.03X	0.00312	304,201	61,441,704	0.03X
	0.02X	0.00208	202,801	40,962,706	0.02X
SAMtools (For other scaffolding tools)	All	NA	110,443,496	22,324,670,268	10.9X
	1X	0.092	10,161,564	2,054,060,928	1.0X
	0.5X	0. 0458	5,062,902	1,023,406,738	0.5X
	0.2X	0. 0183	2,023,015	408,930,034	0.2X
	0.1X	0. 00916	1,012,324	204,631,670	0.1X
	0.05X	0. 00458	506,602	102,407,568	0.05X
	0.03X	0. 00275	304,633	61,578,334	0.03X
	0.02X	0. 00183	202,516	40,937,902	0.02X

Supplementary Table 4 | Statistics of simulated assemblies with different lengths and proportions of chimeric contigs

Simulation (proportion/length)	Total length (bp)	Total length of chimeric contigs (bp)	Number of chimeric contigs	Avg length of chimeric contigs (Kb)	Calculated proportion
20%/2 Mb	2,044,810,672	400,417,814	204	1,962.8	19.6%
20%/1 Mb	2,044,810,672	403,812,563	407	992.2	19.7%
20%/800 Kb	2,044,810,672	408,663,191	510	801.3	20.0%
20%/500 Kb	2,044,810,672	406,041,196	815	498.2	19.9%
20%/300 Kb	2,044,810,672	407,366,972	1,364	298.7	19.9%
20%/200 Kb	2,044,810,672	410,395,141	2,050	200.2	20.1%
20%/100 Kb	2,044,810,672	409,065,751	4,092	100.0	20.0%
20%/50 Kb	2,044,810,672	408,112,589	8,184	49.9	20.0%
5%/500 Kb	2,044,810,672	101,078,754	202	500.4	4.9%
10%/500 Kb	2,044,810,672	201,539,322	406	496.4	9.9%
15%/500 Kb	2,044,810,672	304,899,629	611	499.0	14.9%
20%/500 Kb	2,044,810,672	406,041,196	815	498.2	19.9%
25%/500 Kb	2,044,810,672	507,655,379	1,020	497.7	24.8%
30%/500 Kb	2,044,810,672	608,293,353	1,224	497.0	29.7%
35%/500 Kb	2,044,810,672	710,821,147	1,427	498.1	34.8%
40%/500 Kb	2,044,810,672	815,148,811	1,632	499.5	39.9%

Supplementary Table 5 | Statistics of simulated assemblies with different proportions of collapsed contigs

Simulation (proportion/length)	Total length (bp)	Total length of collapsed contigs (bp)	Number of collapsed contigs	Avg length of collapsed contigs (Kb)	Calculated proportion
20%/2 Mb	1,618,461,908	326,000,509	163	2,000.0	20.1%
20%/1 Mb	1,636,461,918	327,000,255	328	997.0	20.0%
20%/800 Kb	1,639,262,554	327,200,663	410	798.1	20.0%
20%/500 Kb	1,636,461,582	327,500,111	658	497.7	20.0%
20%/300 Kb	1,639,162,248	327,600,641	1,095	299.2	20.0%
20%/200 Kb	1,642,462,159	327,799,923	1,640	199.9	20.0%
20%/100 Kb	1,642,161,390	327,800,153	3,285	100.0	20.0%
20%/50 Kb	1,641,511,968	327,850,417	6,570	49.9	20.0%
5%/500 Kb	1,957,462,001	97,999,972	198	495.0	5.0%
10%/500 Kb	1,837,461,910	184,000,064	370	497.3	10.0%
15%/500 Kb	1,733,962,036	260,000,265	523	497.1	15.0%
17.5%/500 Kb	1,685,961,813	294,499,854	589	500.0	17.5%
20%/500 Kb	1,636,461,582	327,500,111	658	497.7	20.0%
22.5%/500 Kb	1,594,462,108	359,000,004	718	500.0	22.5%
25%/500 Kb	1,553,461,670	388,500,153	777	500.0	25.0%

Supplementary Table 6 | Statistics of simulated haplotypes at different sequence divergence levels

Simulation (sequence divergence)	Calculated sequence divergence between haplotypes (number of variations divided by average length of two haplotypes)					
	Hap1 vs Hap2	Hap1 vs Hap3	Hap1 vs Hap4	Hap2 vs Hap3	Hap2 vs Hap4	Hap3 vs Hap4
2%	2.00%	1.99%	2.00%	1.99%	2.01%	1.99%
1.5%	1.50%	1.49%	1.50%	1.50%	1.51%	1.50%
1%	1.00%	1.00%	1.00%	1.00%	1.01%	1.00%
0.5%	0.50%	0.50%	0.50%	0.50%	0.50%	0.50%
0.4%	0.40%	0.40%	0.40%	0.40%	0.40%	0.40%
0.3%	0.30%	0.30%	0.30%	0.30%	0.30%	0.30%
0.2%	0.20%	0.20%	0.20%	0.20%	0.20%	0.20%
0.1%	0.10%	0.10%	0.10%	0.10%	0.10%	0.10%

Supplementary Table 7 | Statistics of simulated haplotypes with different switch error rates

Simulation (switch error rate)	Calculated switch error rate between haplotypes (number of switched variations divided by variations between two haplotypes)						
	Hap1 vs Hap2	Hap1 vs Hap3	Hap1 vs Hap4	Hap2 vs Hap3	Hap2 vs Hap4	Hap3 vs Hap4	Total
2.5%	0.42%	0.42%	0.42%	0.42%	0.41%	0.41%	2.50%
5%	0.84%	0.83%	0.84%	0.83%	0.83%	0.83%	4.99%
7.5%	1.26%	1.25%	1.25%	1.25%	1.24%	1.24%	7.49%
10%	1.67%	1.66%	1.67%	1.66%	1.66%	1.66%	9.99%
15%	2.52%	2.50%	2.50%	2.49%	2.49%	2.48%	14.98%
20%	3.36%	3.33%	3.34%	3.32%	3.33%	3.31%	19.98%
25%	4.19%	4.16%	4.17%	4.15%	4.17%	4.14%	24.97%

Supplementary Table 8 | Statistics of simulated assemblies at different ploidy levels

Simulation (ploidy)	Total length (bp)	Number of contigs	Max length (bp)	Avg length (bp)	N50 (bp)
1	524,615,222	1,051	892,752	499,158	515,805
2	1,049,230,598	2,099	892,752	499,872	519,610
3	1,573,846,741	3,144	892,752	500,587	522,203
4	2,098,461,825	4,194	892,752	500,349	520,439
6	3,147,691,937	6,304	892,752	499,317	519,824
8	4,196,923,741	8,414	920,319	498,802	520,056
10	5,246,154,432	10,530	920,345	498,210	519,137
12	6,295,384,748	12,626	957,377	498,605	519,888
16	8,393,845,927	16,833	957,294	498,654	519,914

Supplementary Table 9 | Sim3C parameters for simulating Hi-C reads of genomes at different ploidy levels

Simulation (ploidy)	Number of read pairs	Trans rate
1	87,500,000	0.104203152
2	175,000,000	0.199530516
3	262,500,000	0.276520509
4	350,000,000	0.34
6	525,000,000	0.438529089
8	700,000,000	0.511461318
10	875,000,000	0.567624683
12	1,050,000,000	0.612206217
16	1,400,000,000	0.678504085

Supplementary Table 10 | Statistics of simulated assemblies with different ploidies and 5% each of chimeric, collapsed, and switch errors

Simulation (ploidy)	Total length (bp)	Number of contigs	Max length (bp)	Avg length (bp)	N50 (bp)
2	999,730,885	2,121	917,743	471,349	506,006
3	1,481,346,483	3,168	892,752	467,597	500,023
4	1,957,461,928	4,247	952,262	460,905	500,008
6	2,938,691,961	6,584	946,643	446,338	500,002
8	3,919,423,846	9,071	913,634	432,083	499,999
10	4,902,653,840	12,067	944,770	406,286	499,996
12	5,883,384,546	15,449	1,005,068	380,826	499,995
16	7,852,345,231	24,867	579,692	315,773	499,991

Supplementary Table 11 | Statistics of simulated assemblies for rice (IRGSP-1.0), *Arabidopsis* (TAIR10.1), and human (CHM13v2.0_noY) with varying contig N50 values

Simulation (ploidy)	N50	Avg contigs per chromosome
Rice	2 Mb	15.6
	500 Kb	62.5
	100 Kb	310.8
	25 Kb	1,246.4
Arabidopsis	2 Mb	12
	500 Kb	48
	100 Kb	239.2
	25 Kb	952.4
Human	8 Mb	16.7
	2 Mb	66.7
	500 Kb	265.9
	100 Kb	1,329.7

Supplementary Table 12 | Comparison between corrected and uncorrected genome assemblies of *S. spontaneum* Np-X for HapHiC and ALLHiC

Contig length	Published assembly (before correction)	Assembly corrected by HapHiC	Assembly corrected by ALLHiC
Max length	3,761,197	3,761,197 (-0%)	1,779,478 (-52.67%)
N10	998,139	978,242 (-1.99%)	314,000 (-68.54%)
N20	737,538	686,238 (-6.96%)	241,636 (-67.24%)
N30	582,565	545,211 (-6.41%)	195,608 (-66.42%)
N40	476,428	438,675 (-7.92%)	161,904 (-66.02%)
N50	381,898	348,775 (-8.67%)	138,999 (-63.60%)
N60	296,647	272,959 (-7.99%)	118,448 (-60.07%)
N70	223,057	206,651 (-7.36%)	98,999 (-55.62%)
N80	152,457	143,688 (-5.75%)	80,999 (-46.87%)
N90	80,422	77,553 (-3.57%)	55,709 (-30.73%)

Supplementary Table 13 | Comparison of genome assemblies between *M.* × *giganteus* and other published *Miscanthus* species

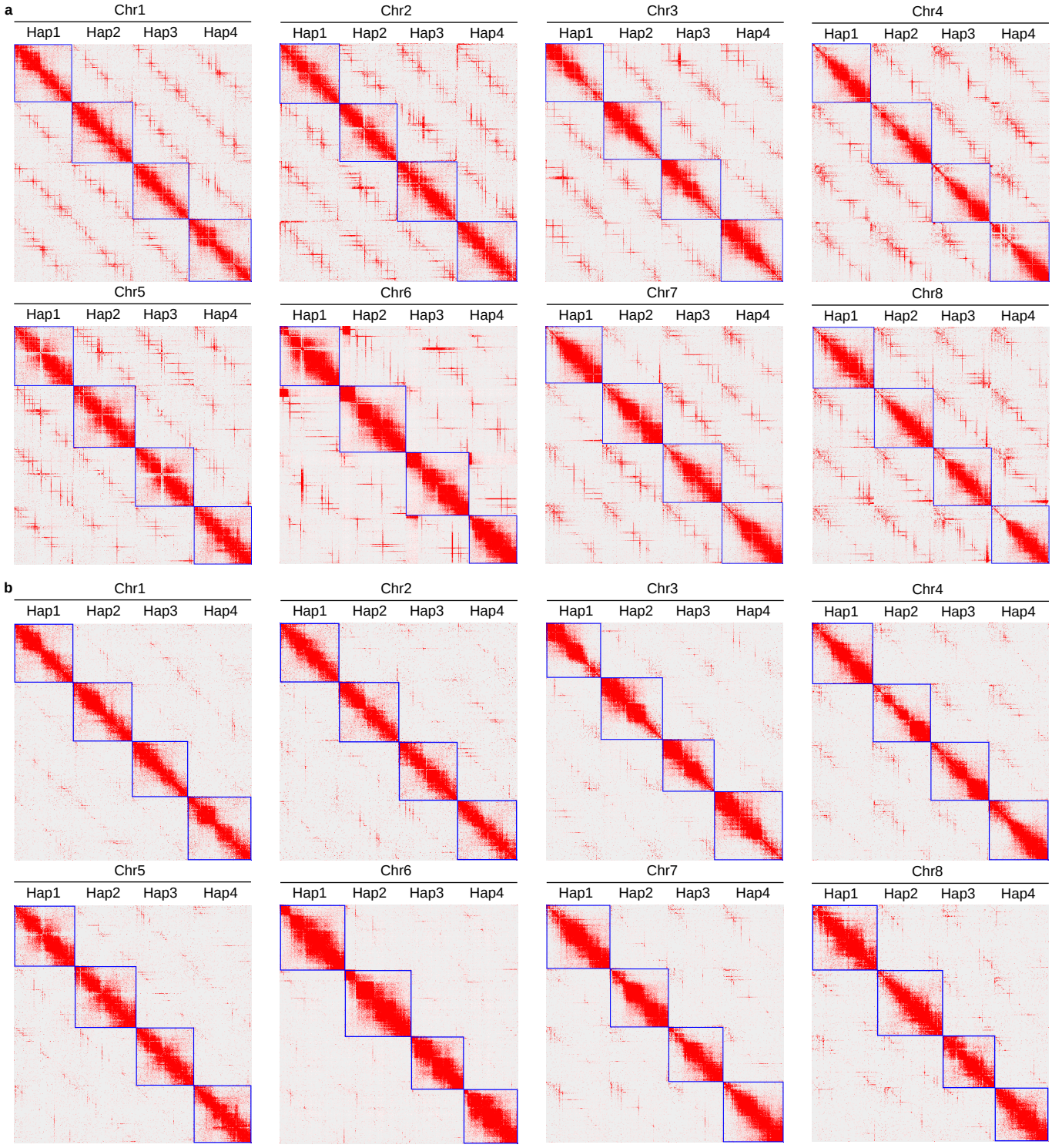
Species	Total length of assembly (Gb)	Number of contigs	Contig N50 (bp)	Is chromosome-level	Is haplotype-resolved
M. × *giganteus*	6.11	5,823	2,176,999	Yes	Yes
M. sinensis	1.85	170,175	33,106	Yes	No
M. sacchariflorus	1.24	1,189,594	2,242	No	No
M. lutarioriparius	2.08	2,621	1,588,283	Yes	No
M. floridulus	2.68	7,351	673,655	Yes	No

Supplementary Table 14 | Pseudocode for optimal recommendation algorithm

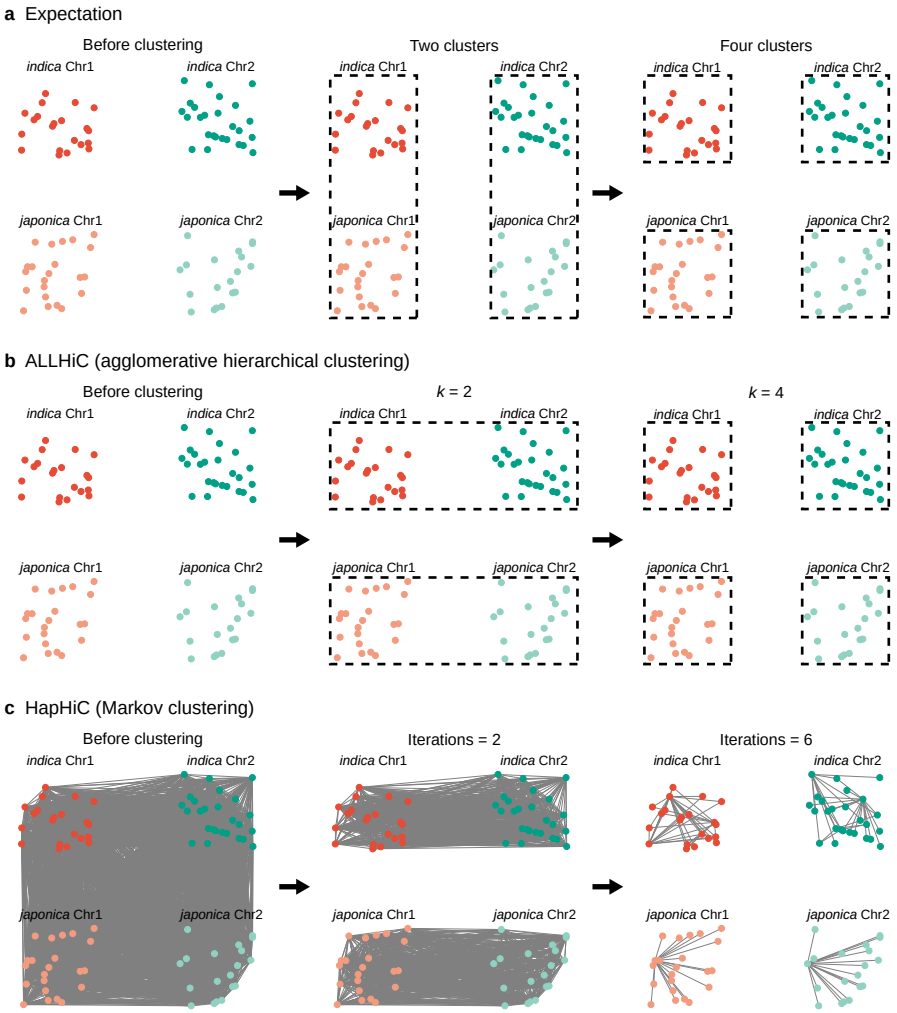
Optimal inflation recommendation

n: the number of Markov clusters for a given inflation value
k: the known number of chromosomes
r: the length ratio threshold of adjacent clusters
L: a reversely sorted list to store Markov cluster length for a given inflation value
D: a dictionary with inflation values as keys and the number of clusters meeting the threshold r as values
for float $i \in$ closed interval [min_inflation, max_inflation] **do**
    if $n < k$ **then**
        continue
    for integer $j \in$ closed interval [1, $n - 1$] **do**
        if $L[j] / L[j-1] < r$ **then**
            if $j \geq k$ **then**
                $D[i] = j$
            break
    $D[i] = n$
return min(D, key=lambda $i : D[i]$)

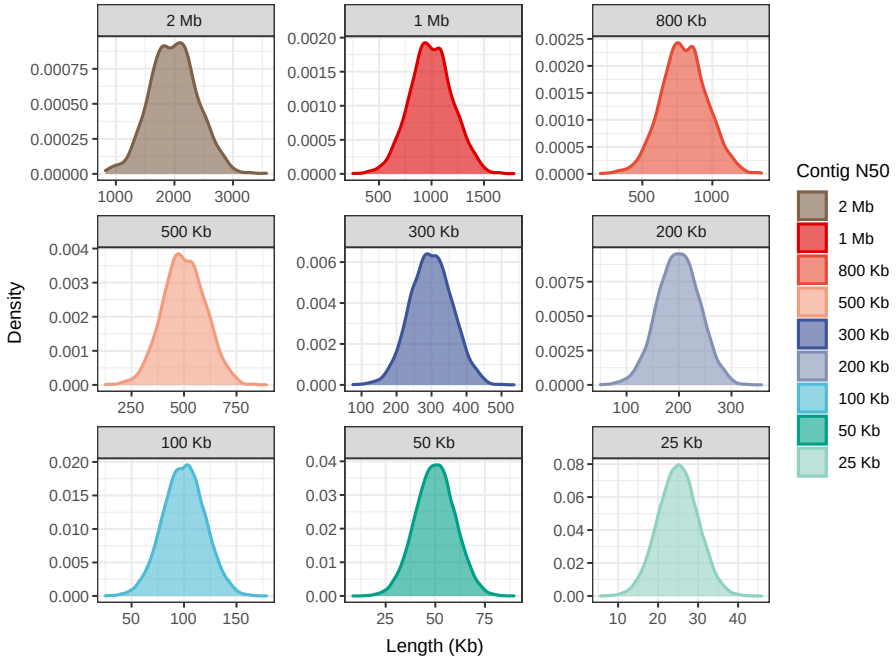
Supplementary Figures



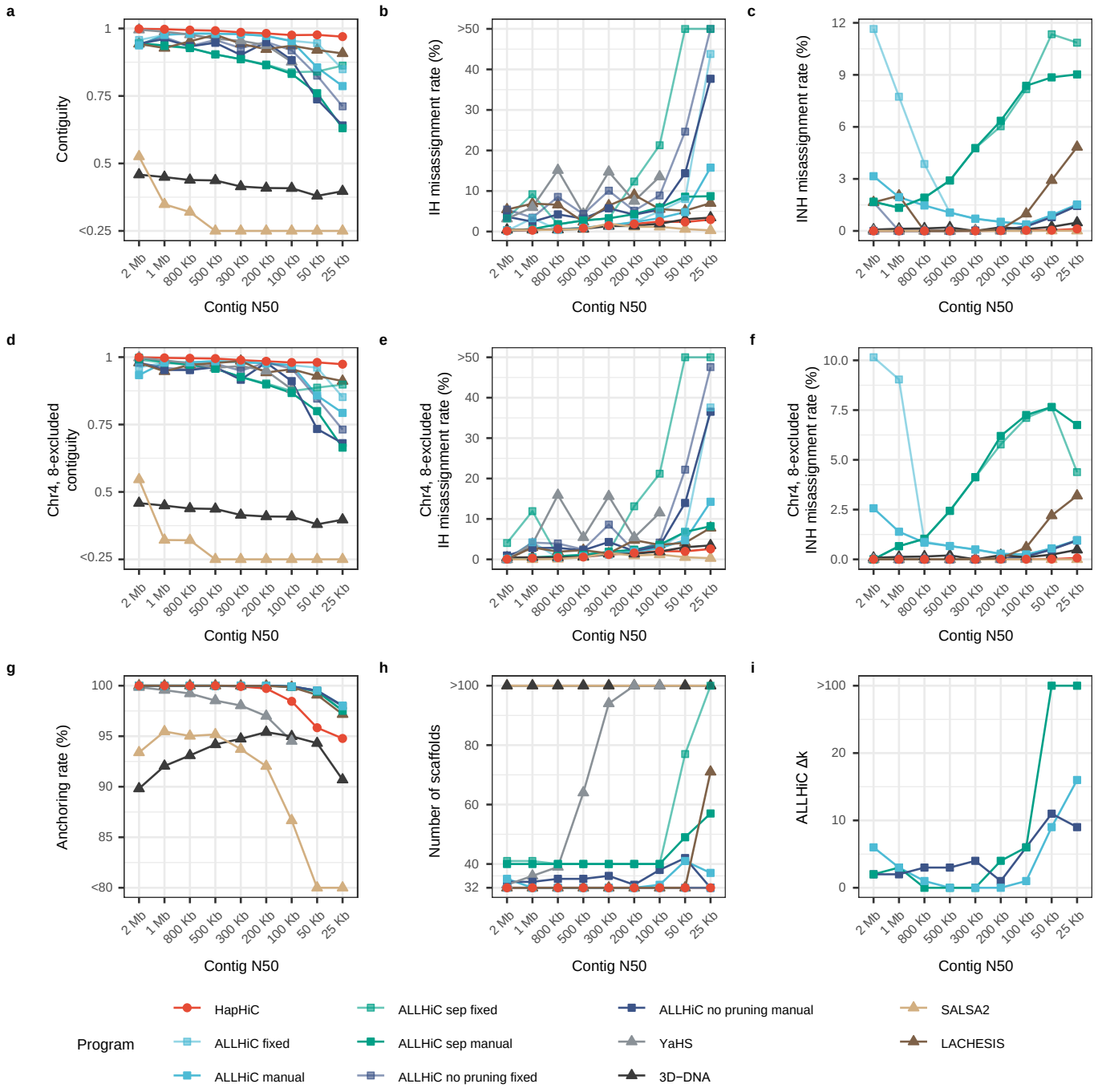
Supplementary Fig. 1 | Generation of the ground truth for evaluating scaffolding tools using the published haplotype-resolved autotetraploid genome of *M. sativa* XinJiangDaYe. **a**, The Hi-C contact maps of published *M. sativa* XinJiangDaYe genome. **b**, The Hi-C contact maps of the ground truth, generated after the manual removal of obvious assembly errors.



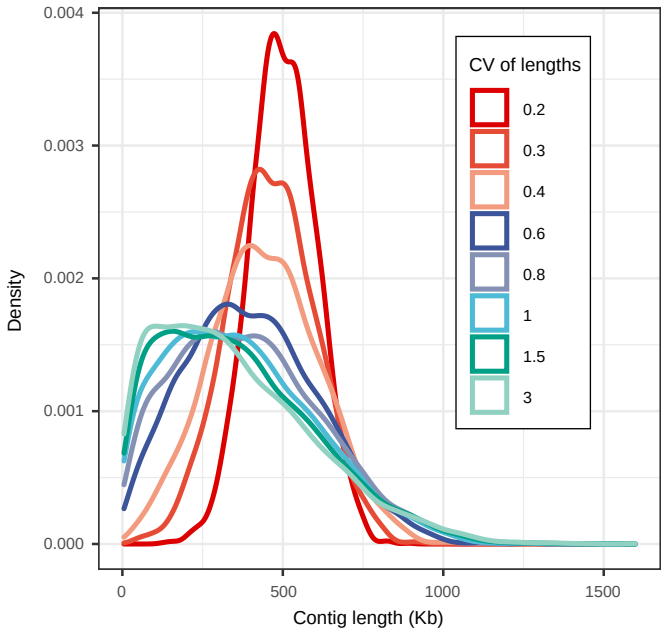
Supplementary Fig. 2 | Artificial merging of data from two rice subspecies, *indica* and *japonica*, leads to an underestimation of Hi-C links between homologous chromosomes. **a**, Ideally, the contigs between homologous chromosomes should be more difficult to separate than those between non-homologous chromosomes. **b**, However, the artificial merging of data from *indica* and *japonica* makes the contigs between homologous chromosomes easier to separate than those between non-homologous chromosomes in ALLHiC. **c**, This phenomenon can also be observed in HapHiC.



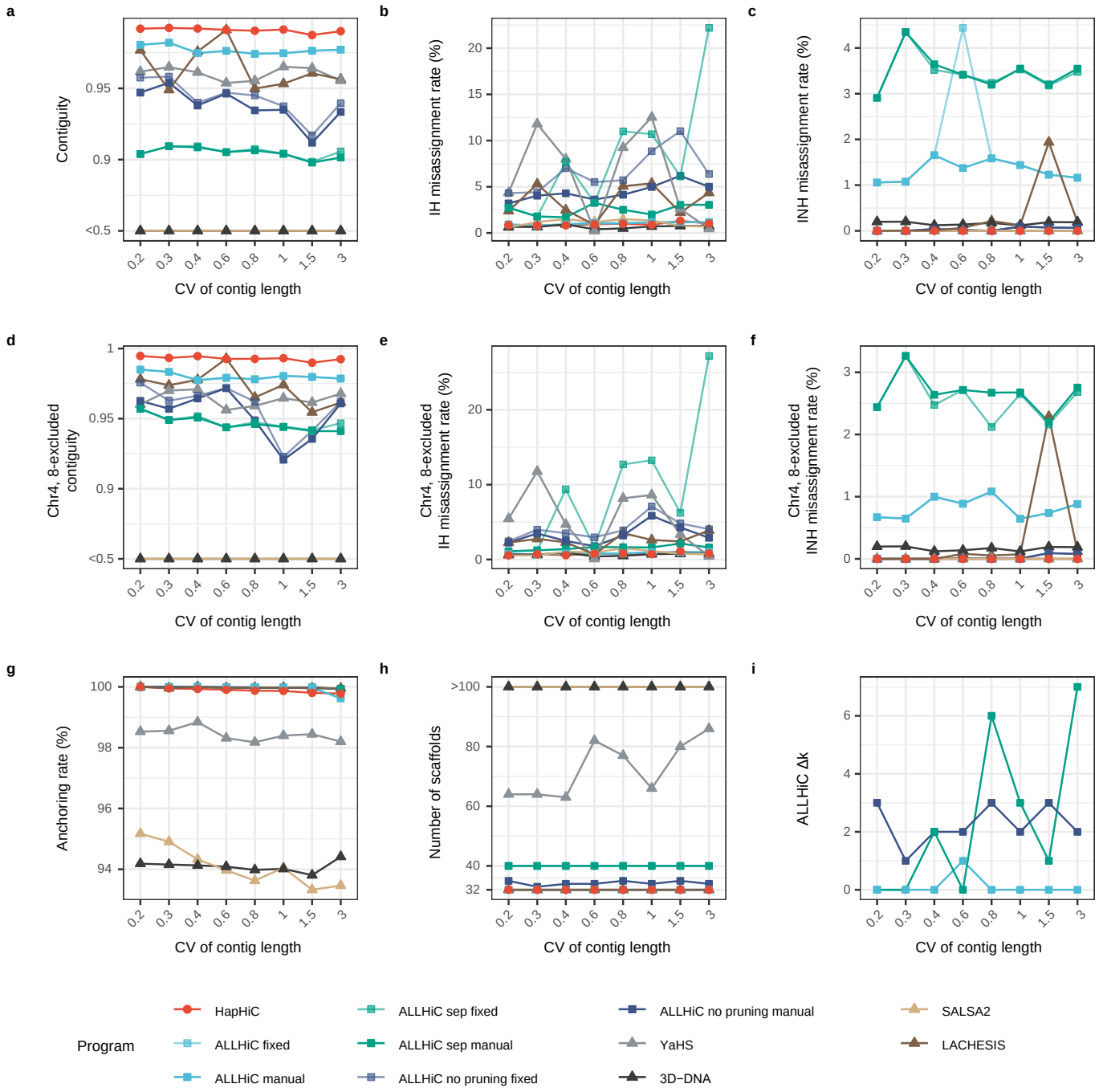
Supplementary Fig. 3 | Distribution of contig lengths in simulated assemblies, with the contig N50 values ranging from 2 Mb to 25 Kb.



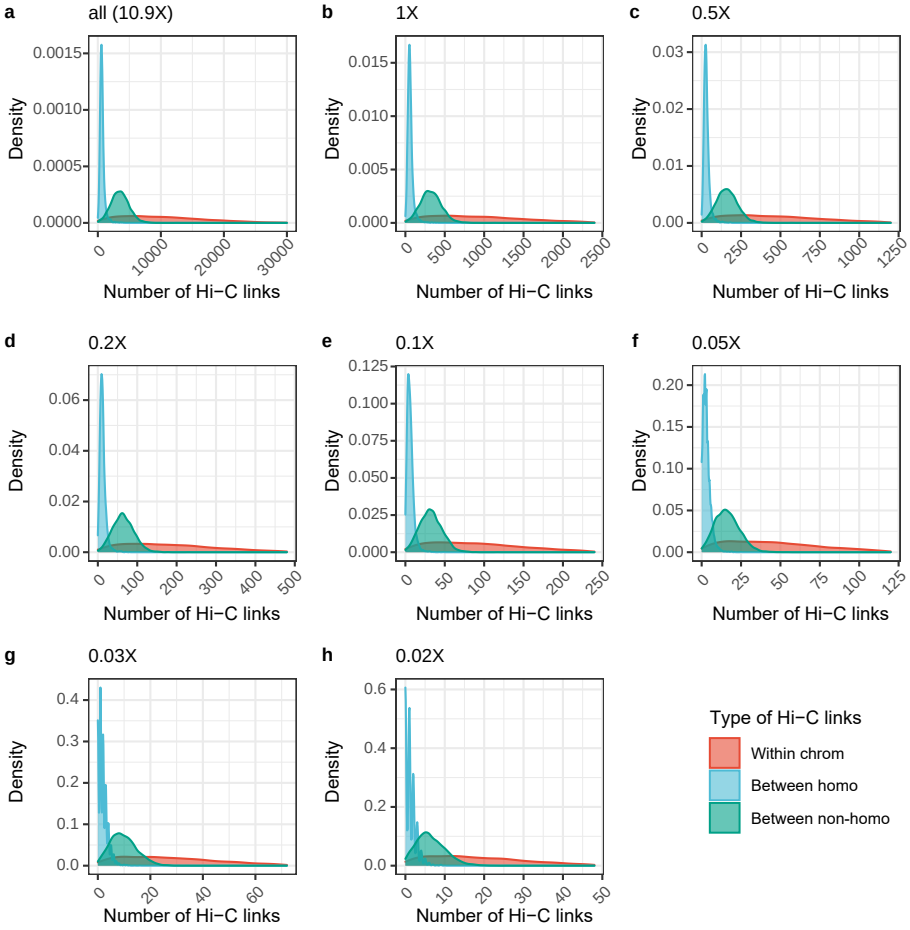
Supplementary Fig. 4 | Performance evaluation of Hi-C-based scaffolding tools in chromosome assignment on assemblies with varying contig N50 values. The evaluation metrics include contiguity (**a**), misassignment rate between homologous chromosomes (**b**), misassignment rate between non-homologous chromosomes (**c**), anchoring rate (**g**), number of scaffolds (**h**), and Δ*k* of manual parameter tuning for ALLHiC (**i**). During ALLHiC pruning, the genome of a closely related species, *M. truncatula* (MtrunA17r5.0-ANR), was used as a reference. However, chromosomes 4 and 8 of *M. truncatula* have some structural differences compared to those of *M. sativa*. Therefore, we also calculated the contiguity and misassignment rate after excluding these two chromosomes (**d-f**). The parameters for these scaffolding tools are provided in Section **3 Hi-C-based scaffolding** of **Software Commands**.



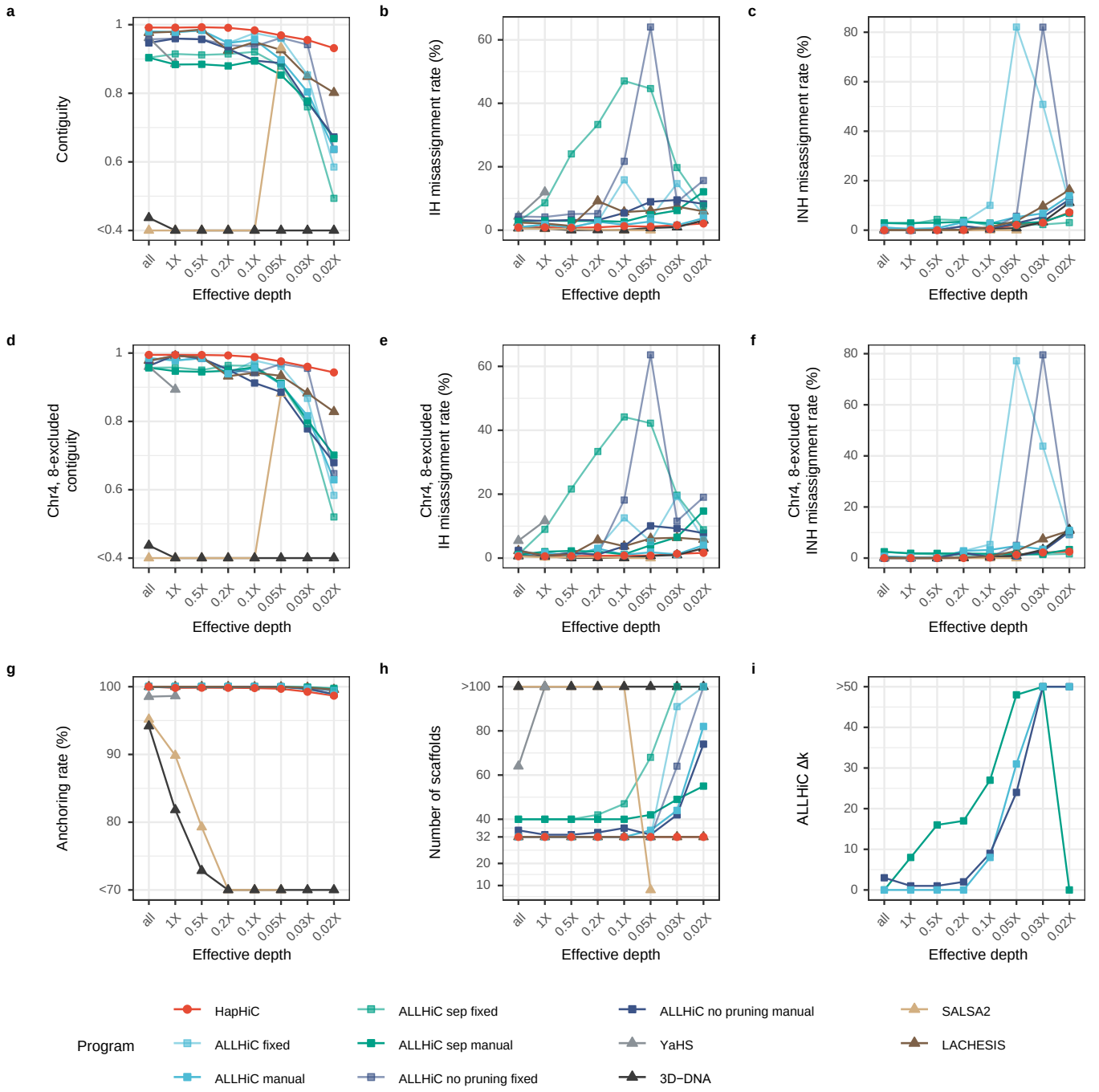
Supplementary Fig. 5 | Distribution of contig lengths in simulated assemblies, with the coefficients of variation (CVs) of contig length ranging from 0.2 to 3.



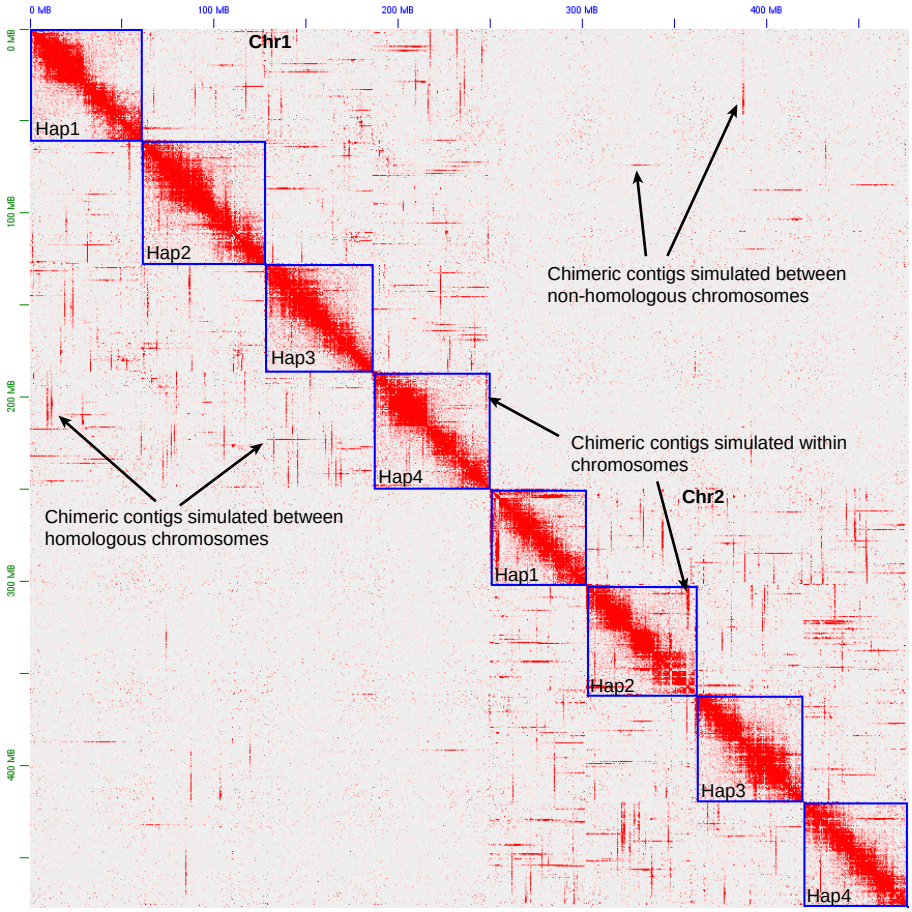
Supplementary Fig. 6 | Performance evaluation of Hi-C-based scaffolding tools in chromosome assignment on assemblies with varying coefficients of variation (CVs) of contig length. The evaluation metrics include contiguity (**a**), misassignment rate between homologous chromosomes (**b**), misassignment rate between non-homologous chromosomes (**c**), anchoring rate (**g**), number of scaffolds (**h**), and Δ*k* of manual parameter tuning for ALLHiC (**i**). During ALLHiC pruning, the genome of a closely related species, *M. truncatula* (MtrunA17r5.0-ANR), was used as a reference. However, chromosomes 4 and 8 of *M. truncatula* have some structural differences compared to those of *M. sativa*. Therefore, we also calculated the contiguity and misassignment rate after excluding these two chromosomes (**d-f**). The parameters for these scaffolding tools are provided in Section **3 Hi-C-based scaffolding** of **Software Commands**.



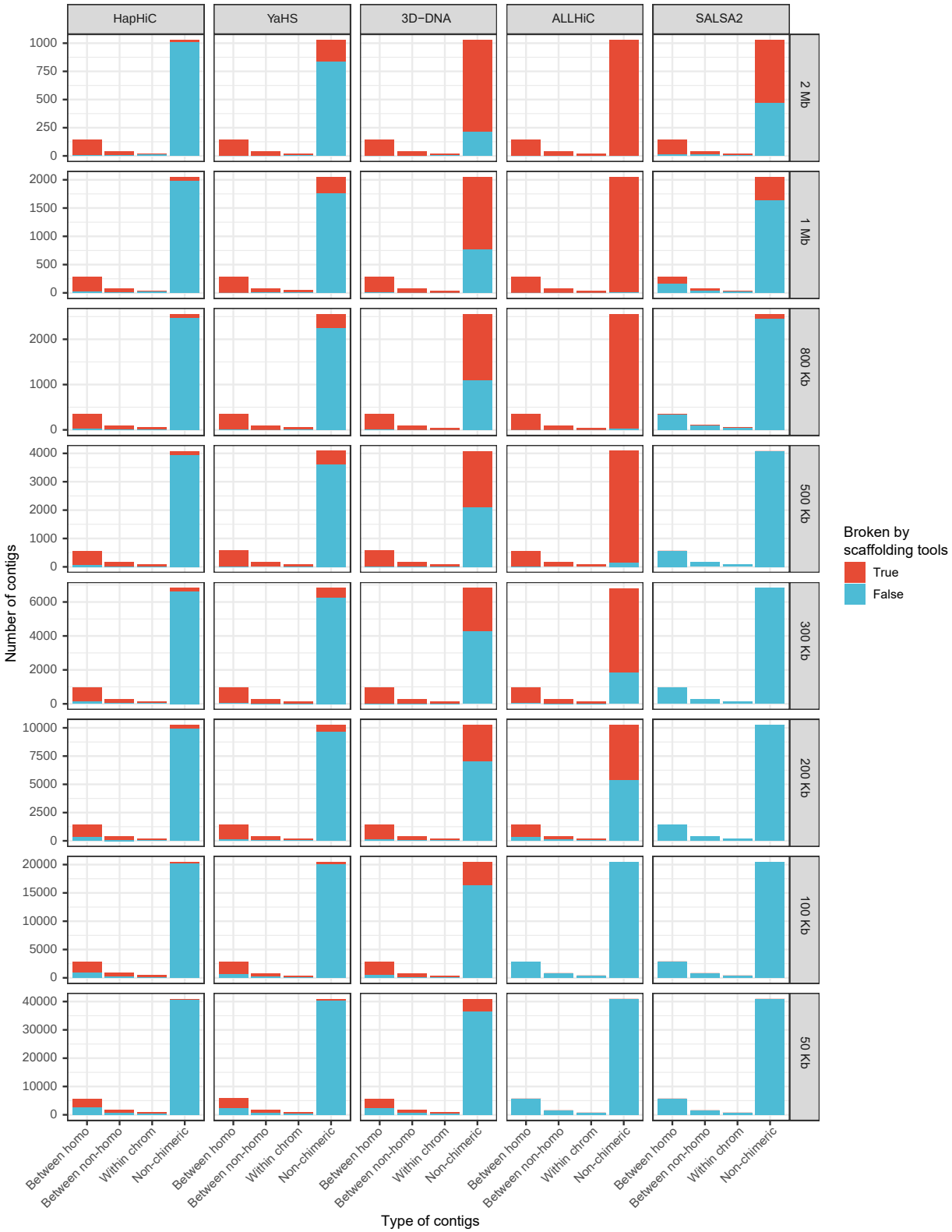
Supplementary Fig. 7 | Distribution of the number of Hi-C links for contigs at varying effective sequencing depths ranging from 10.9X to 0.02X (**a**-**h**).



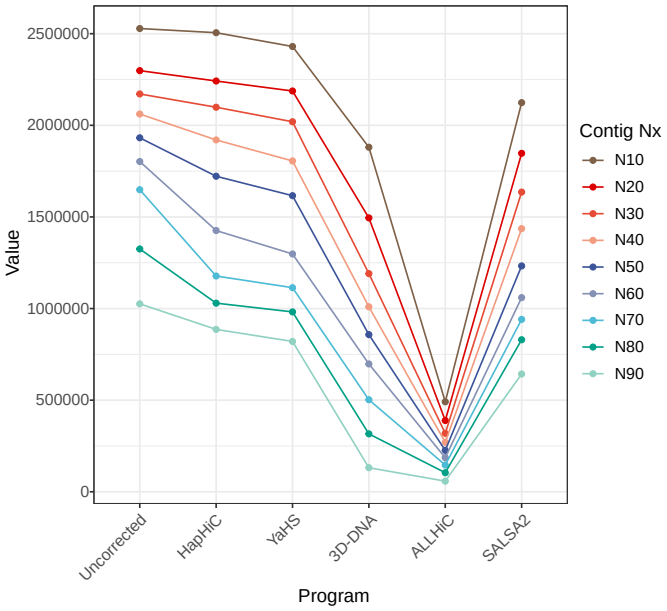
Supplementary Fig. 8 | Performance evaluation of Hi-C-based scaffolding tools in chromosome assignment on assemblies with varying effective Hi-C sequencing depths. The evaluation metrics include contiguity (**a**), misassignment rate between homologous chromosomes (**b**), misassignment rate between non-homologous chromosomes (**c**), anchoring rate (**g**), number of scaffolds (**h**), and Δ*k* of manual parameter tuning for ALLHiC (**i**). During ALLHiC pruning, the genome of a closely related species, *M. truncatula* (MtrunA17r5.0-ANR), was used as a reference. However, chromosomes 4 and 8 of *M. truncatula* have some structural differences compared to those of *M. sativa*. Therefore, we also calculated the contiguity and misassignment rate after excluding these two chromosomes (**d-f**). The parameters for these scaffolding tools are provided in Section **3 Hi-C-based scaffolding** of **Software Commands**.



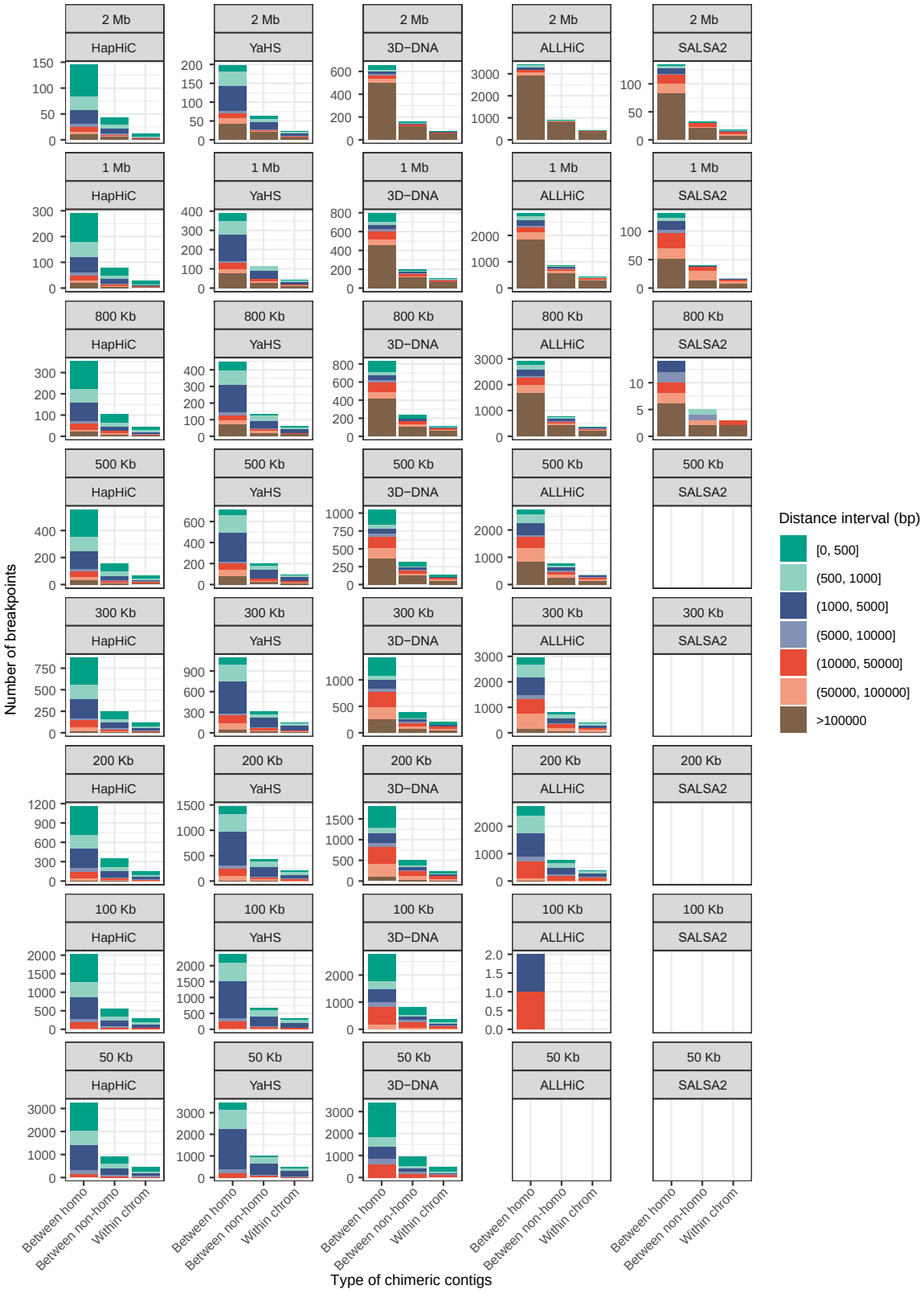
Supplementary Fig. 9 | Hi-C contact map of simulated chimeric contigs. Arrows indicate examples of simulated chimeric contigs.



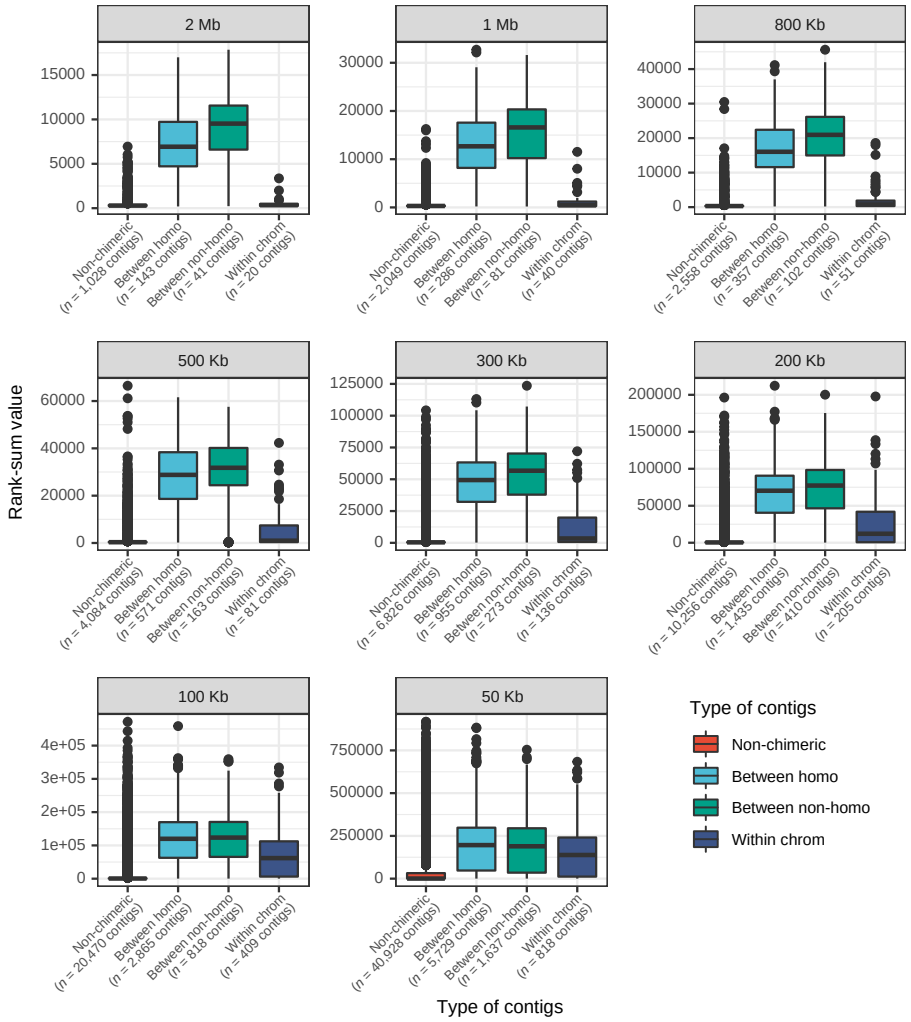
Supplementary Fig. 10 | Performance evaluation of Hi-C-based scaffolding tools in identifying chimeric contigs. The chimeric contigs are classified based on their simulated misjoins: between homologous chromosomes (between homo), between non-homologous chromosomes (between non-homo), and within the same chromosome (within chrom). Ideally, all simulated chimeric contigs should be identified and broken by Hi-C-based scaffolding tools (indicated in red), while non-chimeric contigs should remain intact (indicated in blue).



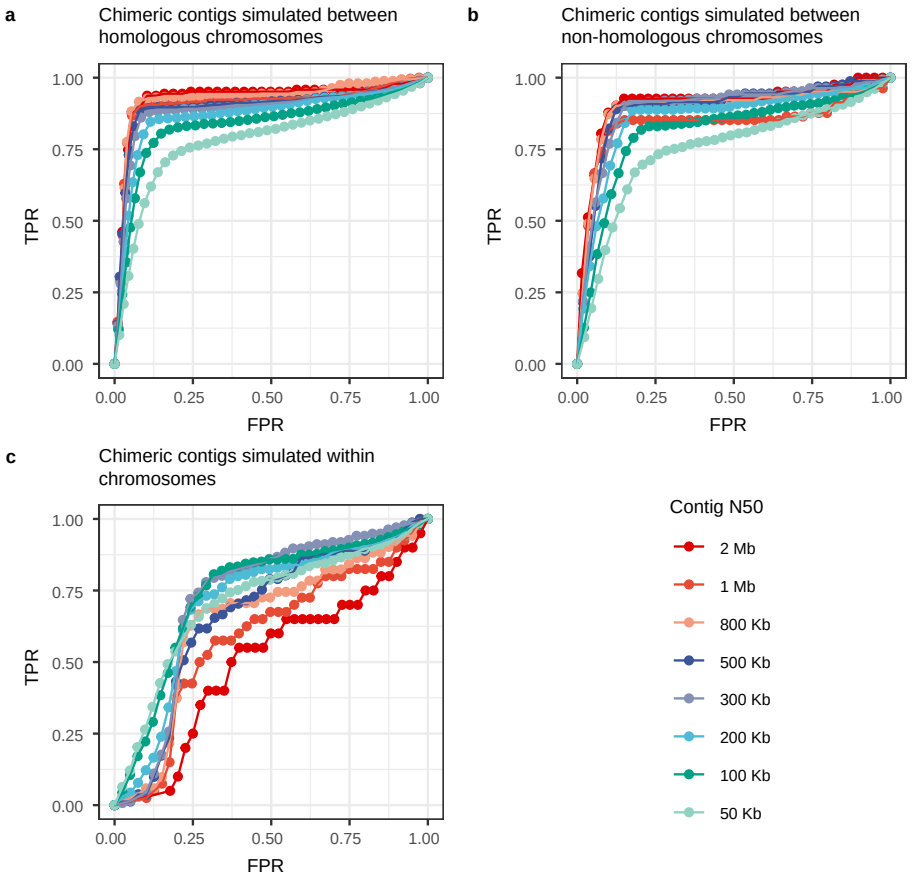
Supplementary Fig. 11 | Comparison of Contig N10 to N90 values of assemblies before and after assembly correction by each Hi-C-based scaffolding tool.



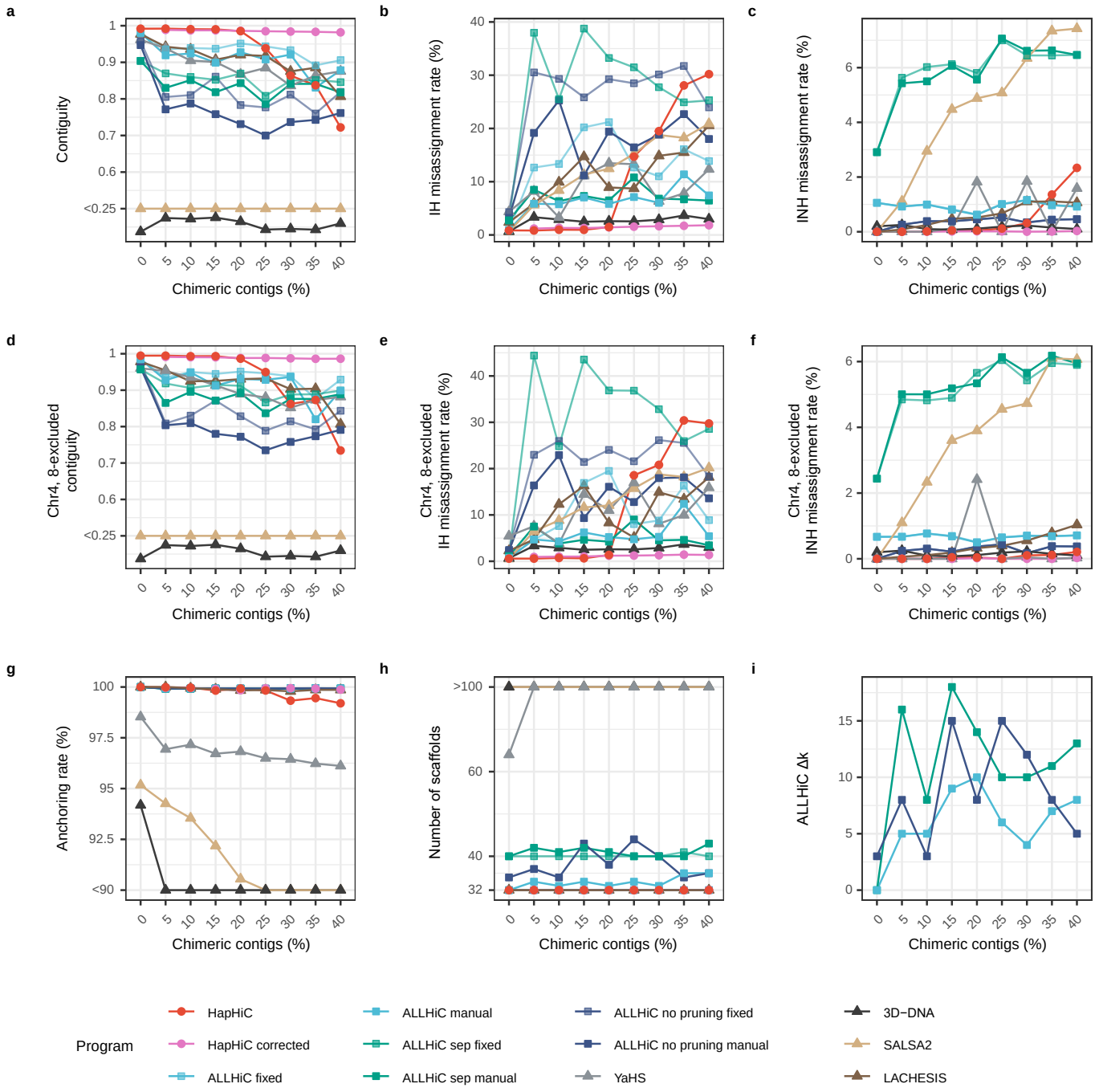
Supplementary Fig. 12 | Performance evaluation of Hi-C-based scaffolding tools in determining breakpoints for chimeric contigs. The chimeric contigs are classified based on their simulated misjoins: between homologous chromosomes (between homo), between non-homologous chromosomes (between non-homo), and within the same chromosome (within chrom). A closer distance between the determined breakpoint and the actual simulated misjoin indicates better results.



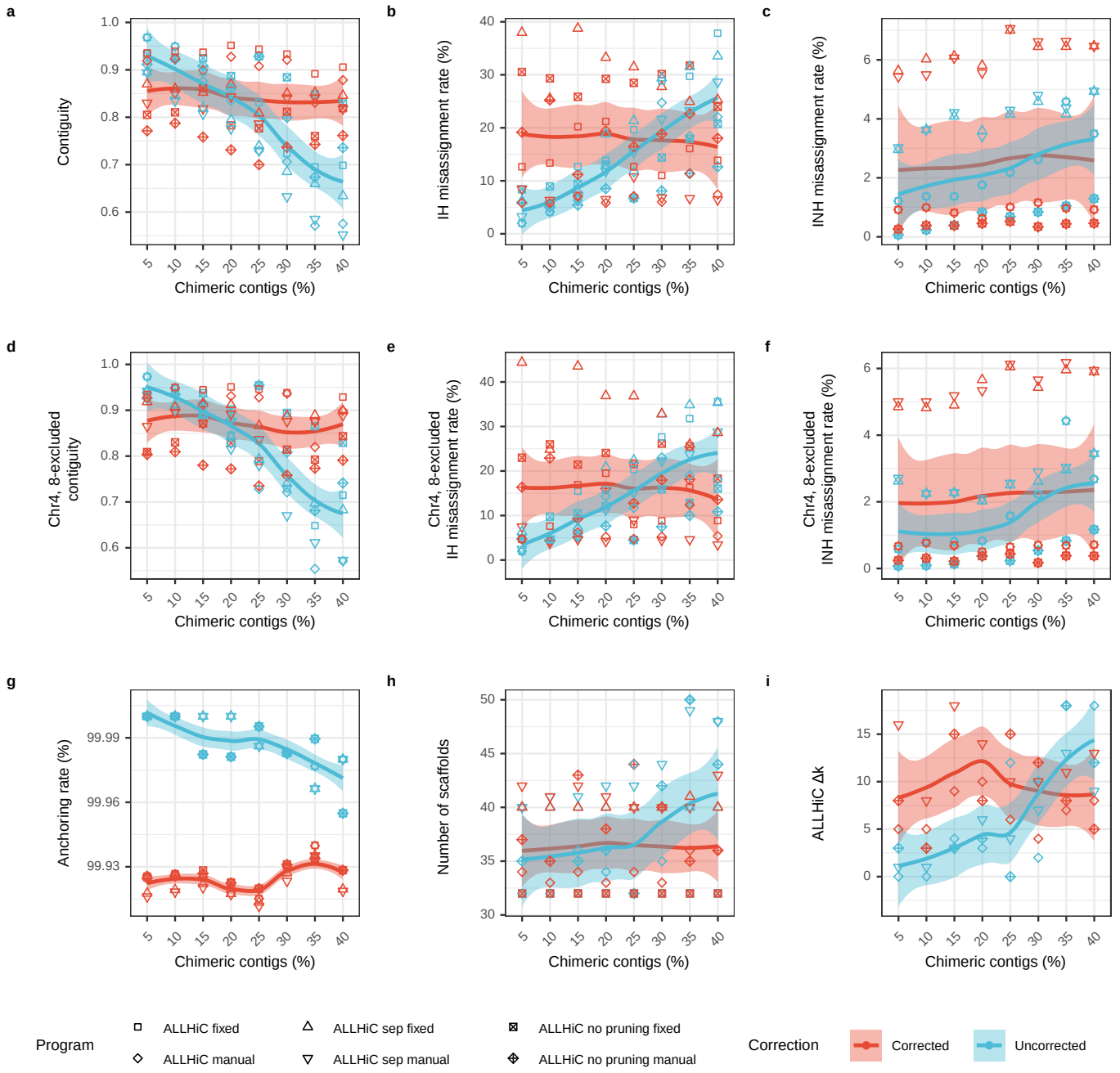
Supplementary Fig. 13 | Rank-sum values of chimeric and non-chimeric contigs of varying lengths, ranging from 2 Mb to 50 Kb. The chimeric contigs are classified based on their simulated misjoins: between homologous chromosomes (between homo), between non-homologous chromosomes (between non-homo), and within the same chromosome (within chrom). Box plots: the box depicts the interquartile range; the line represents the median value; and the whiskers represent the minimum and maximum values (excluding outliers).



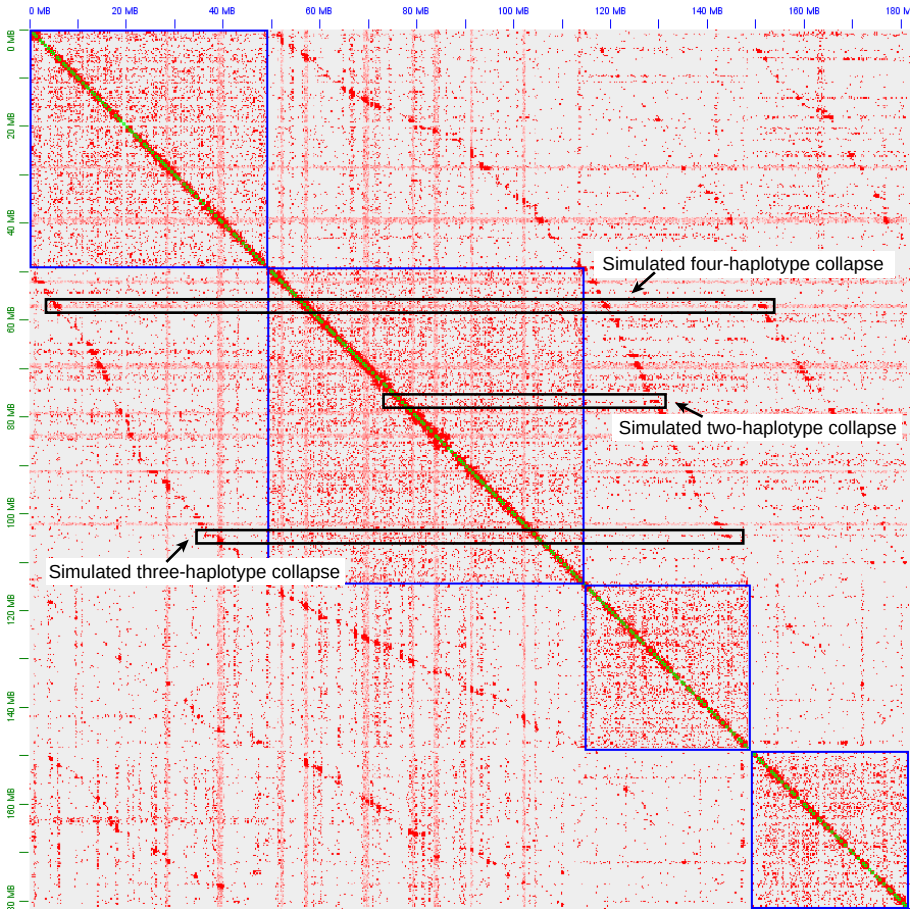
Supplementary Fig. 14 | Receiver operating characteristics (ROC) curves demonstrate the performance of the rank-sum algorithm in identifying chimeric contigs of varying lengths. The chimeric contigs are classified based on their simulated misjoins: between homologous chromosomes (**a**), between non-homologous chromosomes (**b**), and within the same chromosome (**c**).



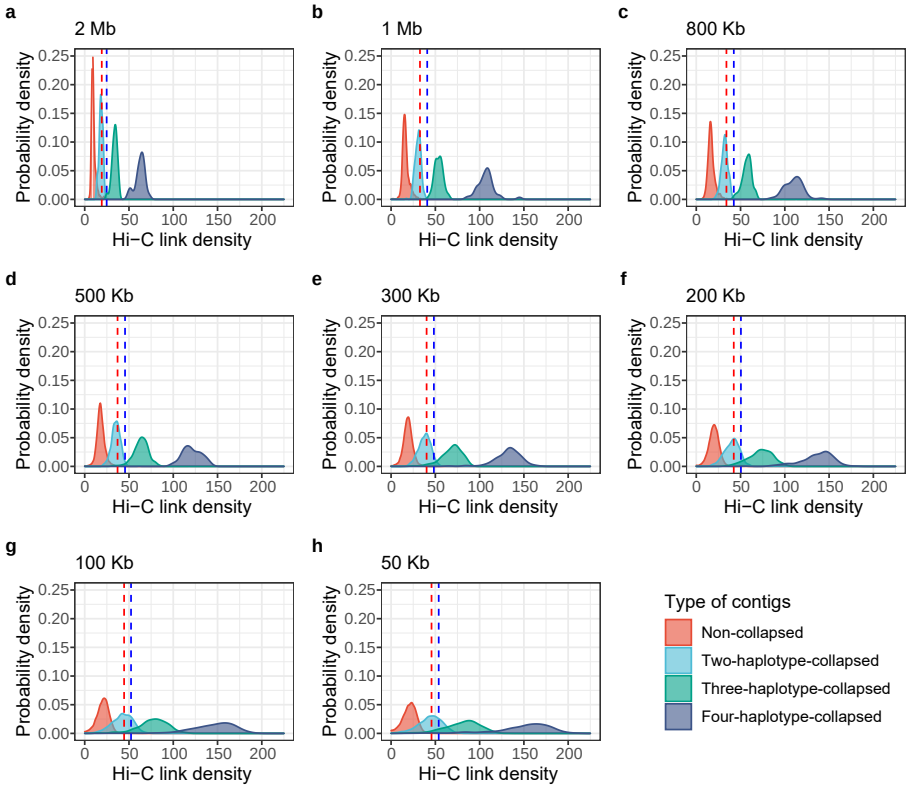
Supplementary Fig. 15 | Performance evaluation of Hi-C-based scaffolding tools in chromosome assignment on assemblies with varying proportions of chimeric contigs. The evaluation metrics include contiguity (**a**), misassignment rate between homologous chromosomes (**b**), misassignment rate between non-homologous chromosomes (**c**), anchoring rate (**g**), number of scaffolds (**h**), and Δ*k* of manual parameter tuning for ALLHiC (**i**). During ALLHiC pruning, the genome of a closely related species, *M. truncatula* (MtrunA17r5.0-ANR), was used as a reference. However, chromosomes 4 and 8 of *M. truncatula* have some structural differences compared to those of *M. sativa*. Therefore, we also calculated the contiguity and misassignment rate after excluding these two chromosomes (**d-f**). The parameters for these scaffolding tools are provided in Section **3 Hi-C-based scaffolding** of **Software Commands**.



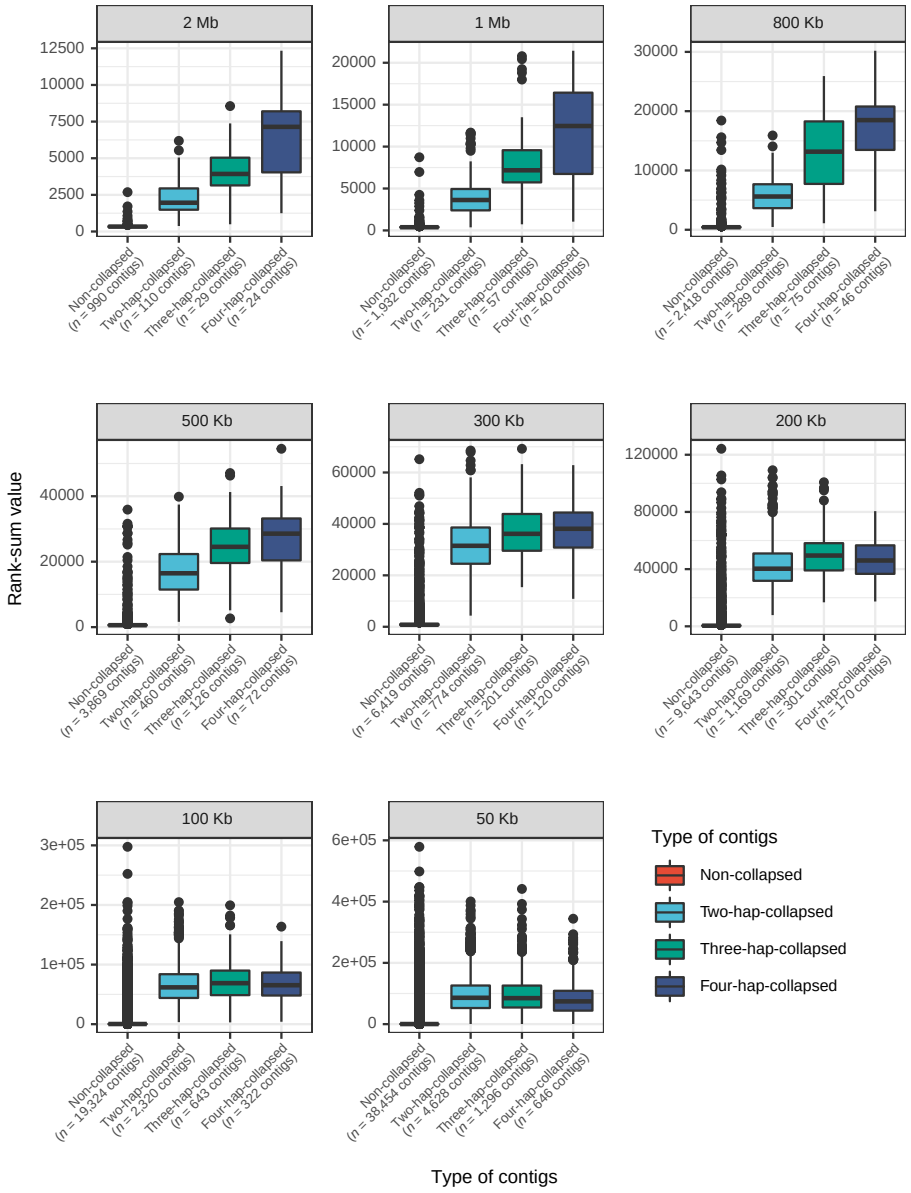
Supplementary Fig. 16 | Performance evaluation of ALLHiC with and without assembly correction in chromosome assignment on assemblies with varying proportions of chimeric contigs. The evaluation metrics include contiguity (**a**), misassignment rate between homologous chromosomes (**b**), misassignment rate between non-homologous chromosomes (**c**), anchoring rate (**g**), number of scaffolds (**h**), and Δ*k* of manual parameter tuning for ALLHiC (**i**). During ALLHiC pruning, the genome of a closely related species, *M. truncatula* (MtrunA17r5.0-ANR), was used as a reference. However, chromosomes 4 and 8 of *M. truncatula* have some structural differences compared to those of *M. sativa*. Therefore, we also calculated the contiguity and misassignment rate after excluding these two chromosomes (**d-f**). Curves: LOESS curve fitting; shaded areas: 95% confidence interval. The parameters for these scaffolding tools are provided in Section **3 Hi-C-based scaffolding** of **Software Commands**.



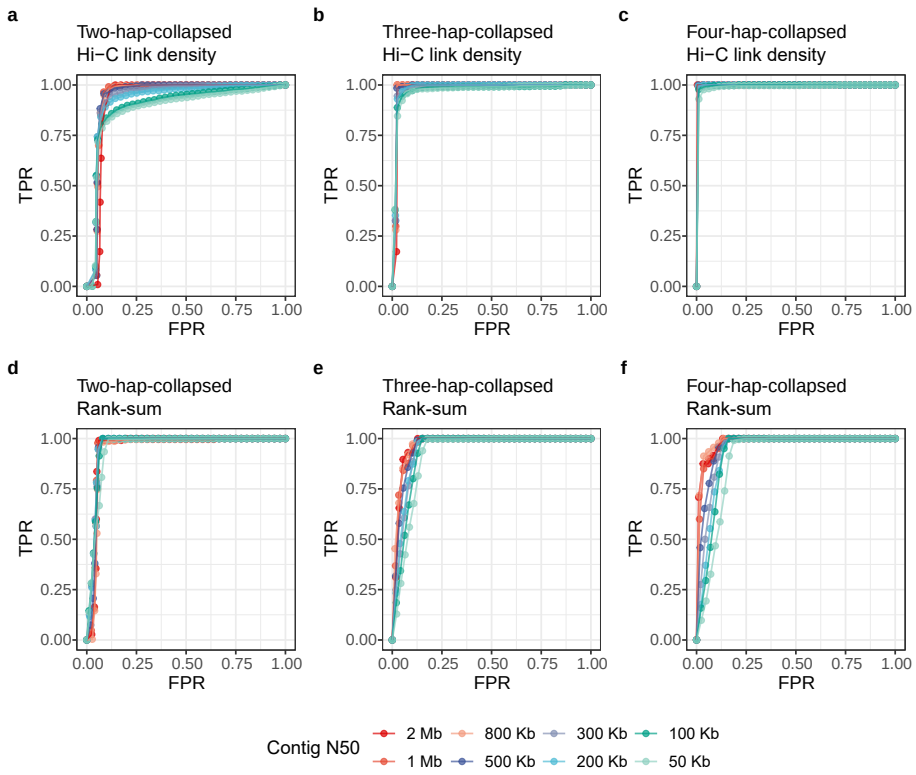
Supplementary Fig. 17 | Hi-C contact map of simulated collapsed contigs. Arrows indicate examples of collapsed chimeric contigs.



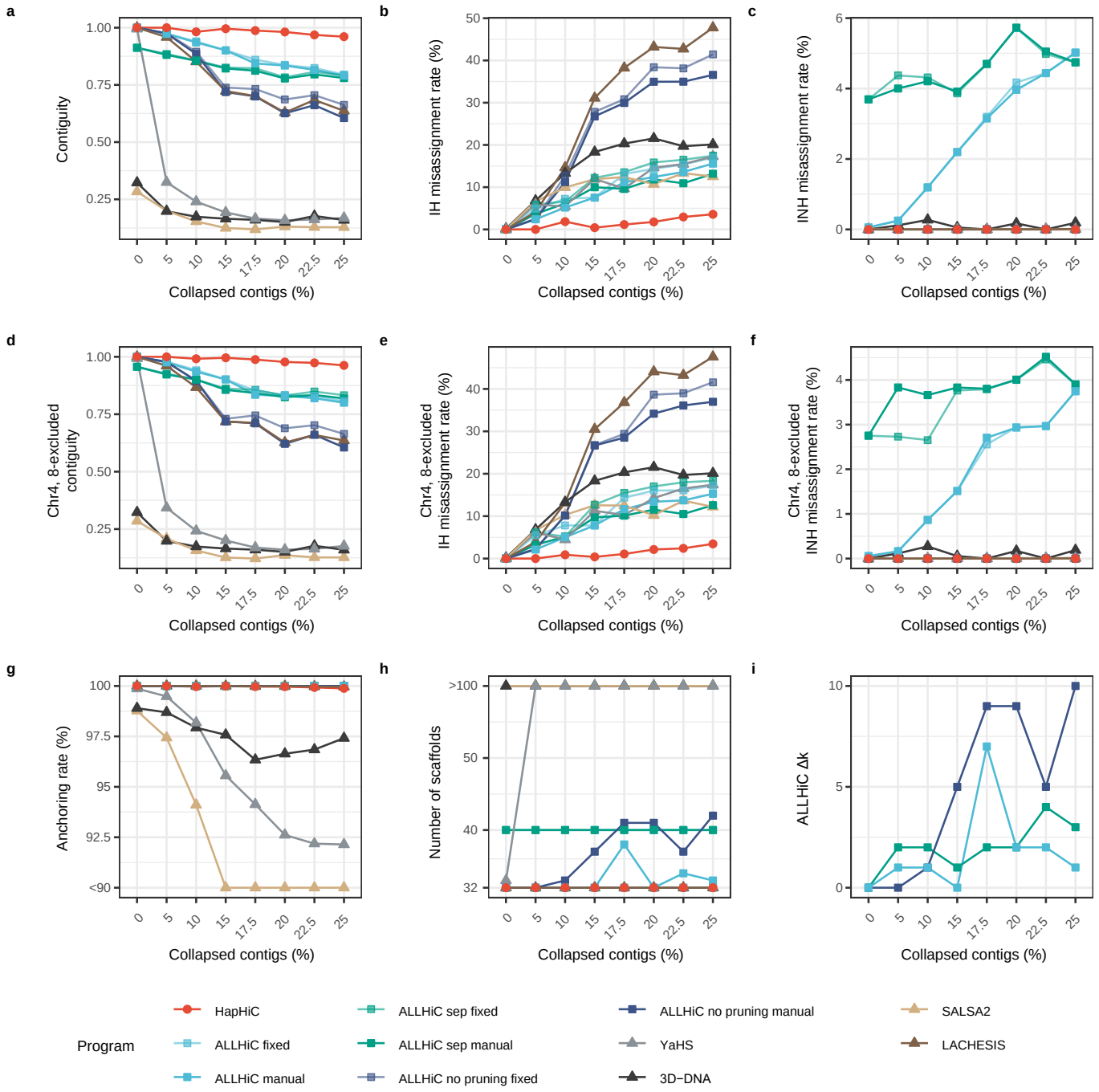
Supplementary Fig. 18 | Distribution of Hi-C link density for collapsed and non-collapsed contigs, with the contig lengths ranging from 2 Mb to 50 Kb (**a**-**h**). The collapsed contigs are classified based on the number of haplotypes contributing to the collapse: two haplotypes (two-haplotype-collapsed), three haplotypes (three-haplotype-collapsed), and four haplotypes (four-haplotype-collapsed).



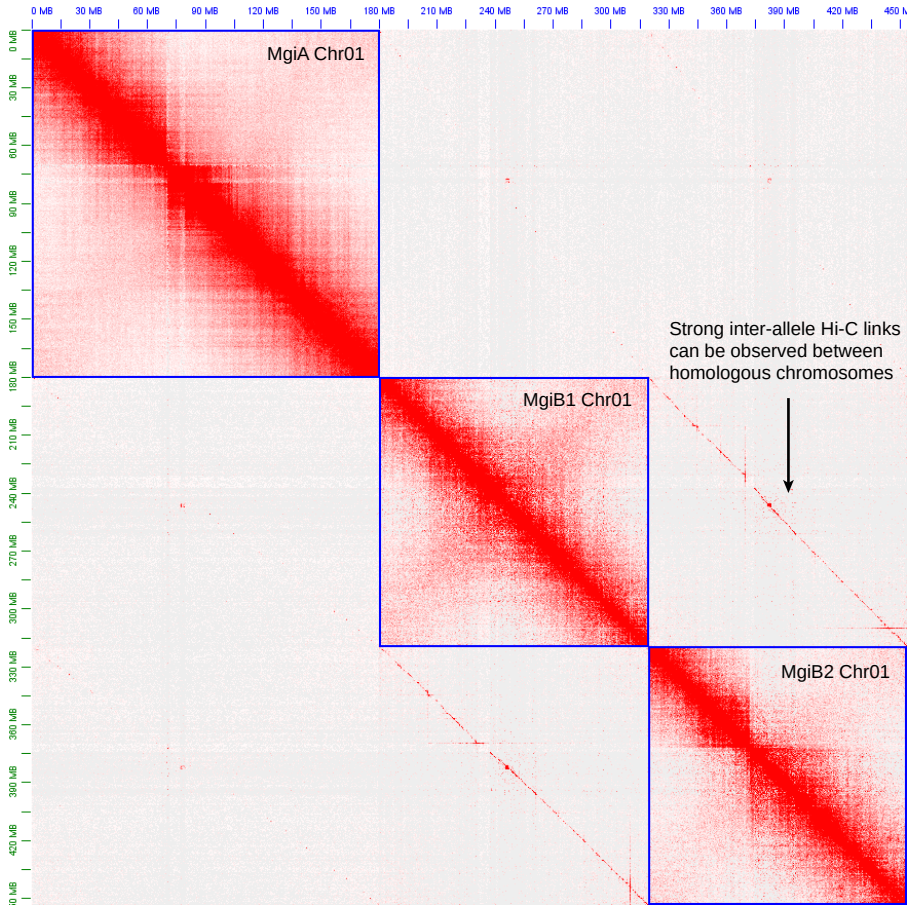
Supplementary Fig. 19 | Rank-sum values of collapsed and non-collapsed contigs of varying lengths, ranging from 2 Mb to 50 Kb. The collapsed contigs are classified based on the number of haplotypes contributing to the collapse: two haplotypes (two-haplotype-collapsed), three haplotypes (three-haplotype-collapsed), and four haplotypes (four-haplotype-collapsed). Box plots: the box depicts the interquartile range; the line represents the median value; and the whiskers represent the minimum and maximum values (excluding outliers).



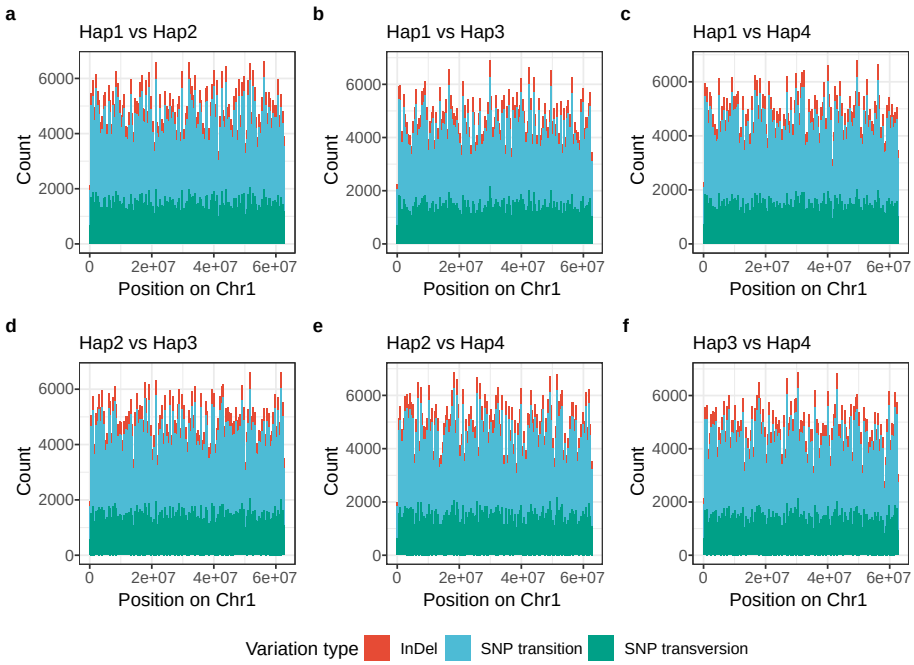
Supplementary Fig. 20 | Receiver operating characteristics (ROC) curves demonstrate the performance of link density and rank-sum algorithm in identifying chimeric contigs of varying contig lengths. The collapsed contigs are classified based on the number of haplotypes contributing to the collapse: two haplotypes (two-haplotype-collapsed, **a** and **d**), three haplotypes (three-haplotype-collapsed, **b** and **e**), and four haplotypes (four-haplotype-collapsed, **c** and **f**).



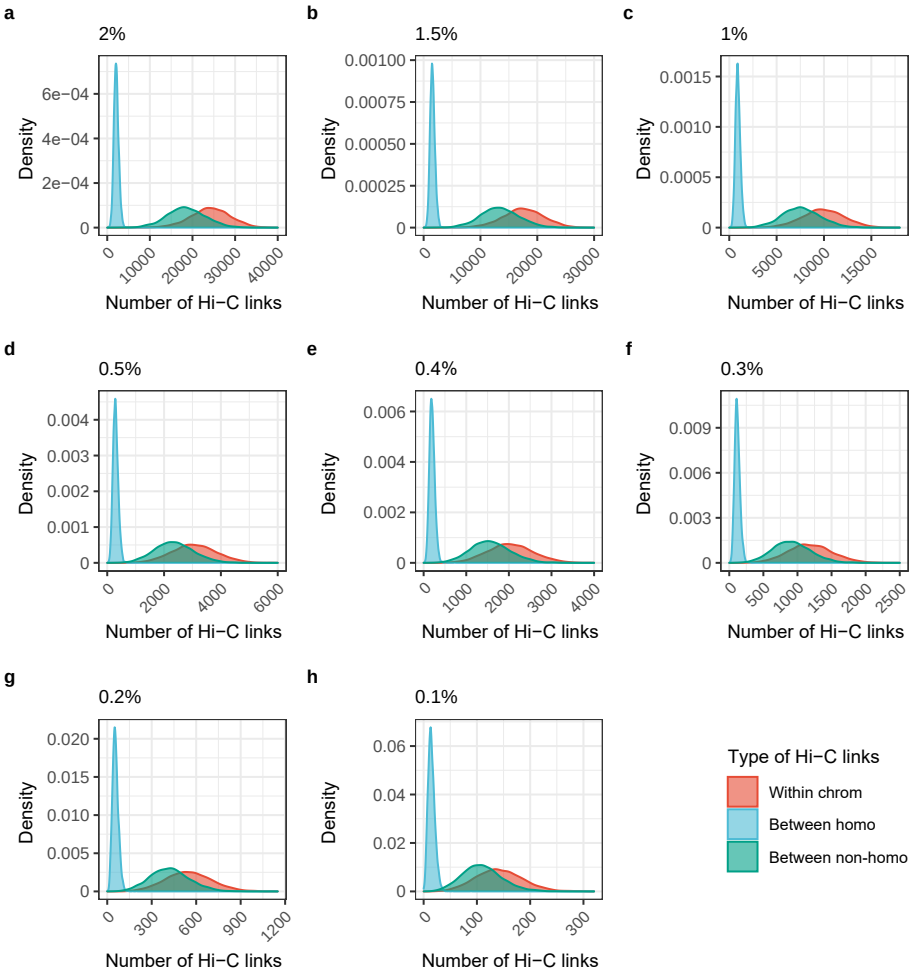
Supplementary Fig. 21 | Performance evaluation of Hi-C-based scaffolding tools in chromosome assignment on assemblies with varying proportions of collapsed contigs. The evaluation metrics include contiguity (**a**), misassignment rate between homologous chromosomes (**b**), misassignment rate between non-homologous chromosomes (**c**), anchoring rate (**g**), number of scaffolds (**h**), and Δ*k* of manual parameter tuning for ALLHiC (**i**). During ALLHiC pruning, the genome of a closely related species, *M. truncatula* (MtrunA17r5.0-ANR), was used as a reference. However, chromosomes 4 and 8 of *M. truncatula* have some structural differences compared to those of *M. sativa*. Therefore, we also calculated the contiguity and misassignment rate after excluding these two chromosomes (**d-f**). The parameters for these scaffolding tools are provided in Section **3 Hi-C-based scaffolding** of **Software Commands**.



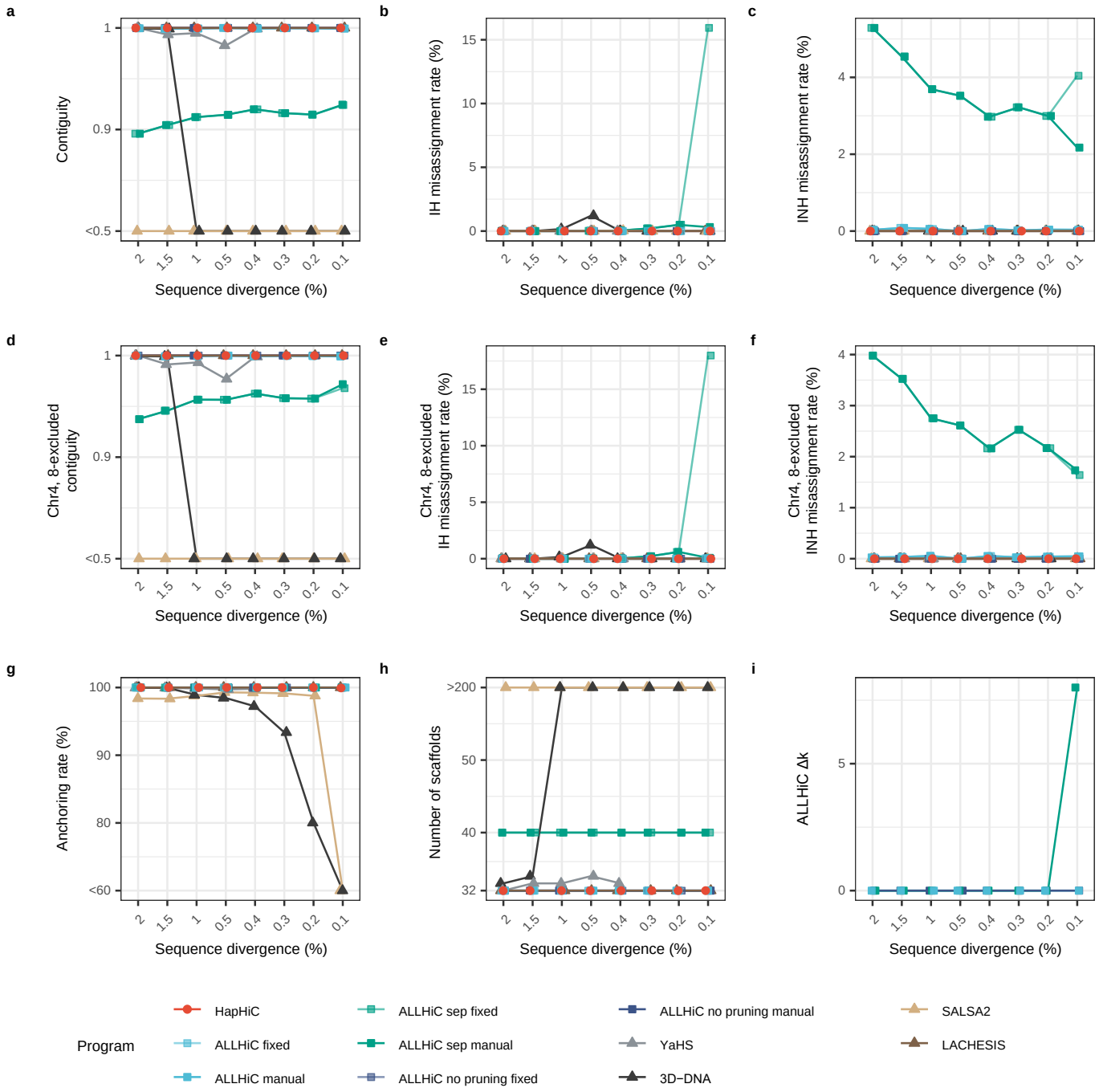
Supplementary Fig. 22 | Hi-C contact map for the three haplotypes of chromosome 1 in the *M.* × *giganteus* genome illustrates the strong inter-allele Hi-C links between homologous chromosomes.



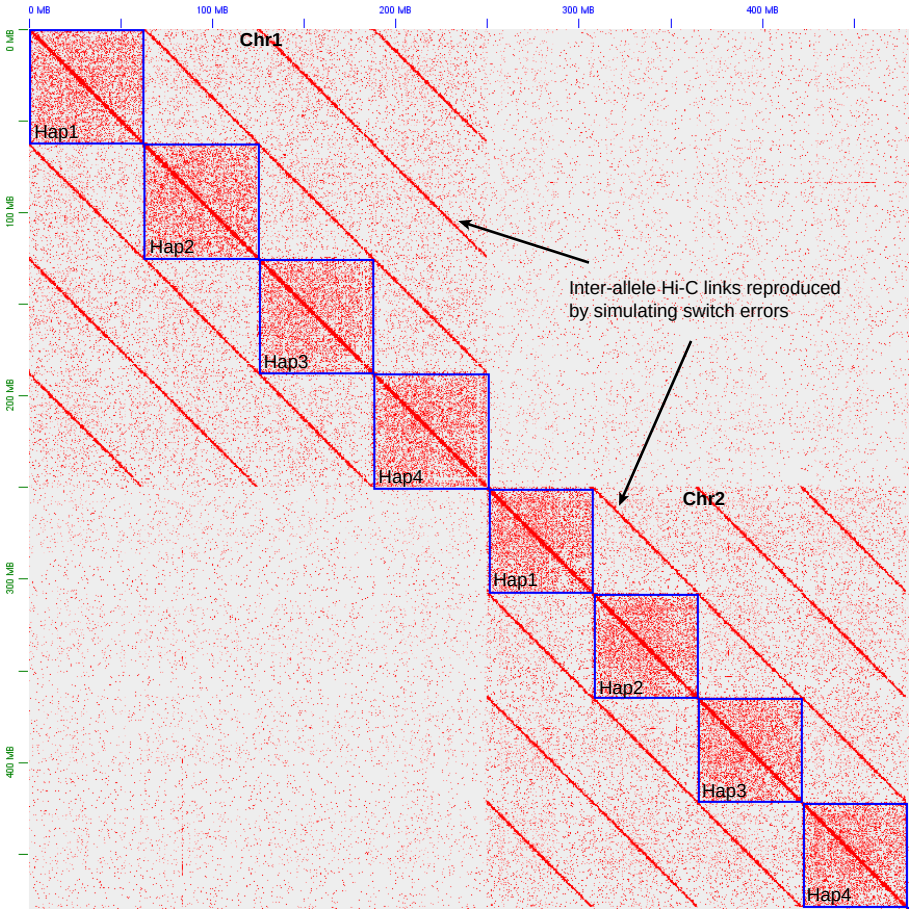
Supplementary Fig. 23 | Distribution of simulated variations between haplotypes along chromosome 1. The comparisons include: haplotype 1 vs haplotype 2 (**a**), haplotype 1 vs haplotype 3 (**b**), haplotype 1 vs haplotype 4 (**c**), haplotype 2 vs haplotype 3 (**d**), haplotype 2 vs haplotype 4 (**e**), haplotype 3 vs haplotype 4 (**f**).



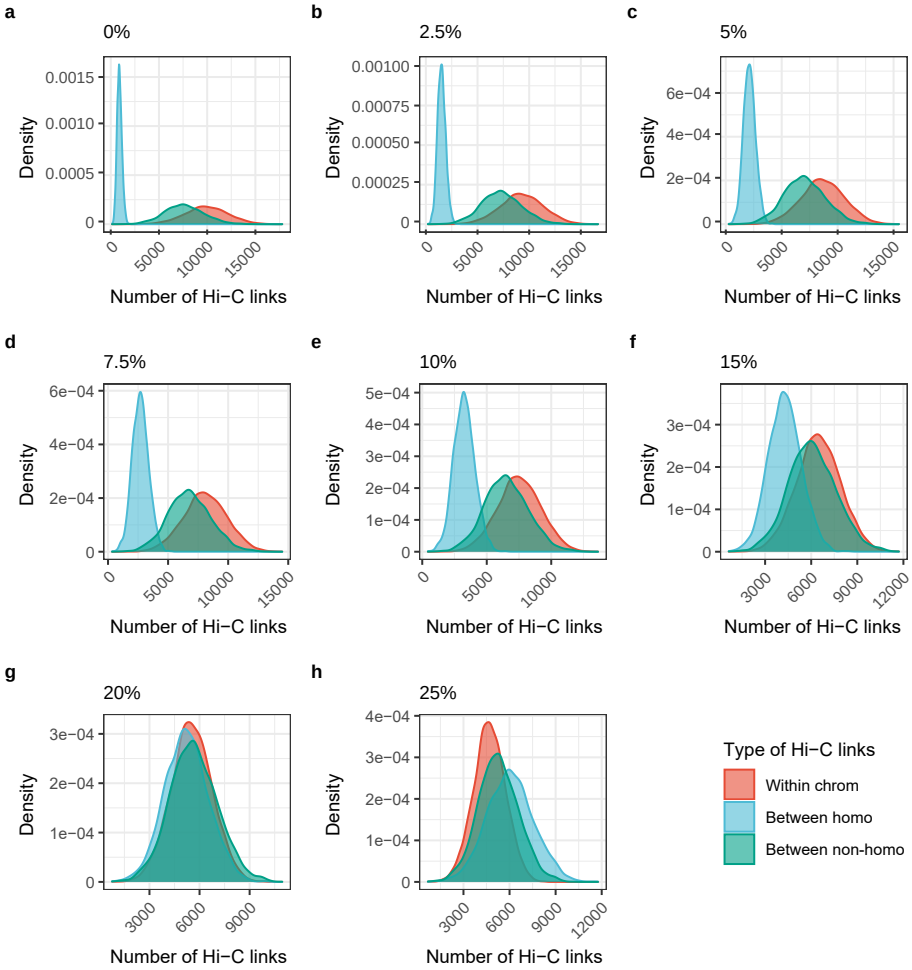
Supplementary Fig. 24 | Distribution of the number of Hi-C links for contigs, with the sequence divergence levels ranging from 2% to 0.1% (**a**-**h**).



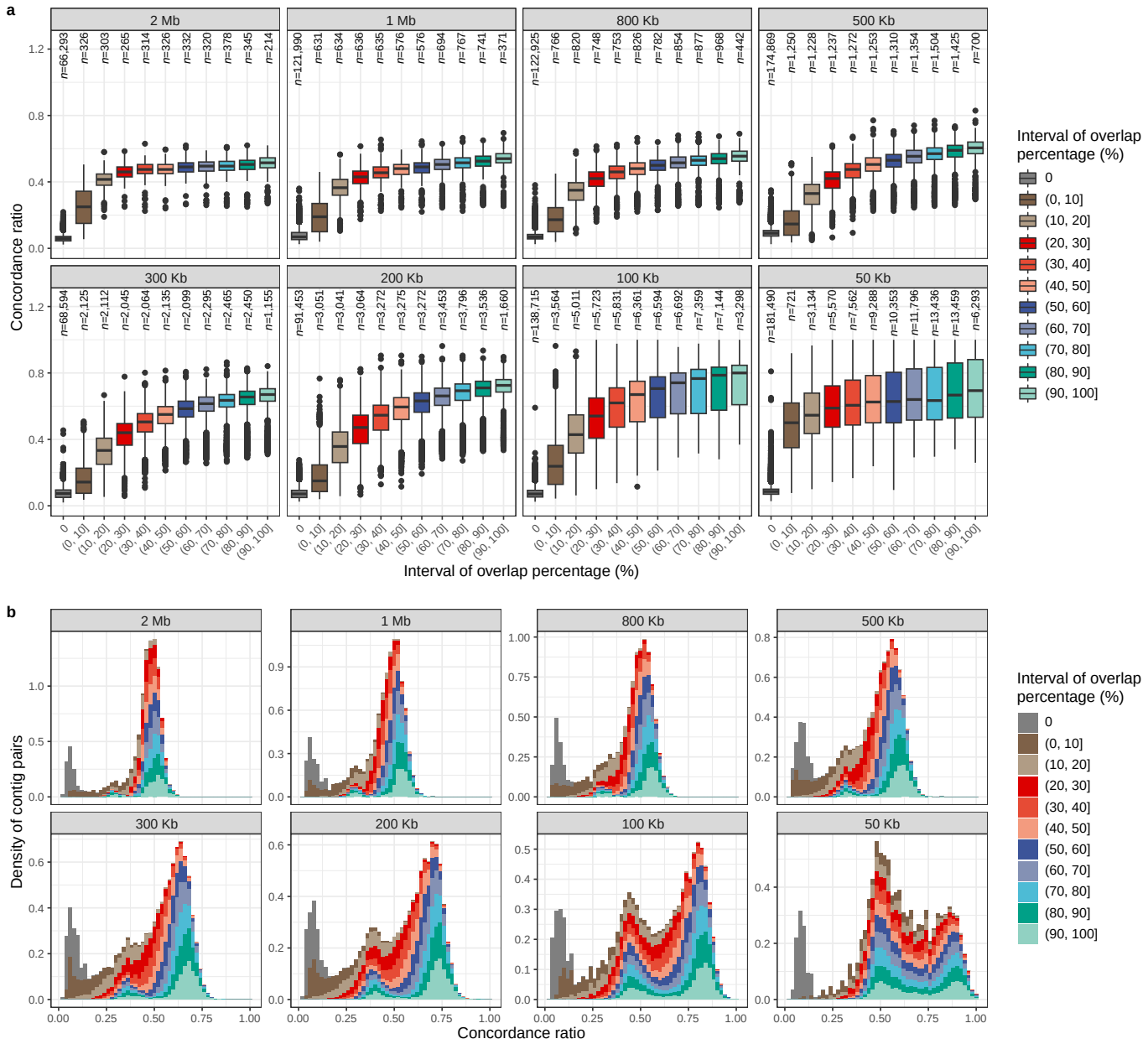
Supplementary Fig. 25 | Performance evaluation of Hi-C-based scaffolding tools in chromosome assignment on assemblies with varying sequence divergence levels. The evaluation metrics include contiguity (**a**), misassignment rate between homologous chromosomes (**b**), misassignment rate between non-homologous chromosomes (**c**), anchoring rate (**g**), number of scaffolds (**h**), and Δ*k* of manual parameter tuning for ALLHiC (**i**). During ALLHiC pruning, the genome of a closely related species, *M. truncatula* (MtrunA17r5.0-ANR), was used as a reference. However, chromosomes 4 and 8 of *M. truncatula* have some structural differences compared to those of *M. sativa*. Therefore, we also calculated the contiguity and misassignment rate after excluding these two chromosomes (**d-f**). The parameters for these scaffolding tools are provided in Section **3 Hi-C-based scaffolding** of **Software Commands**.



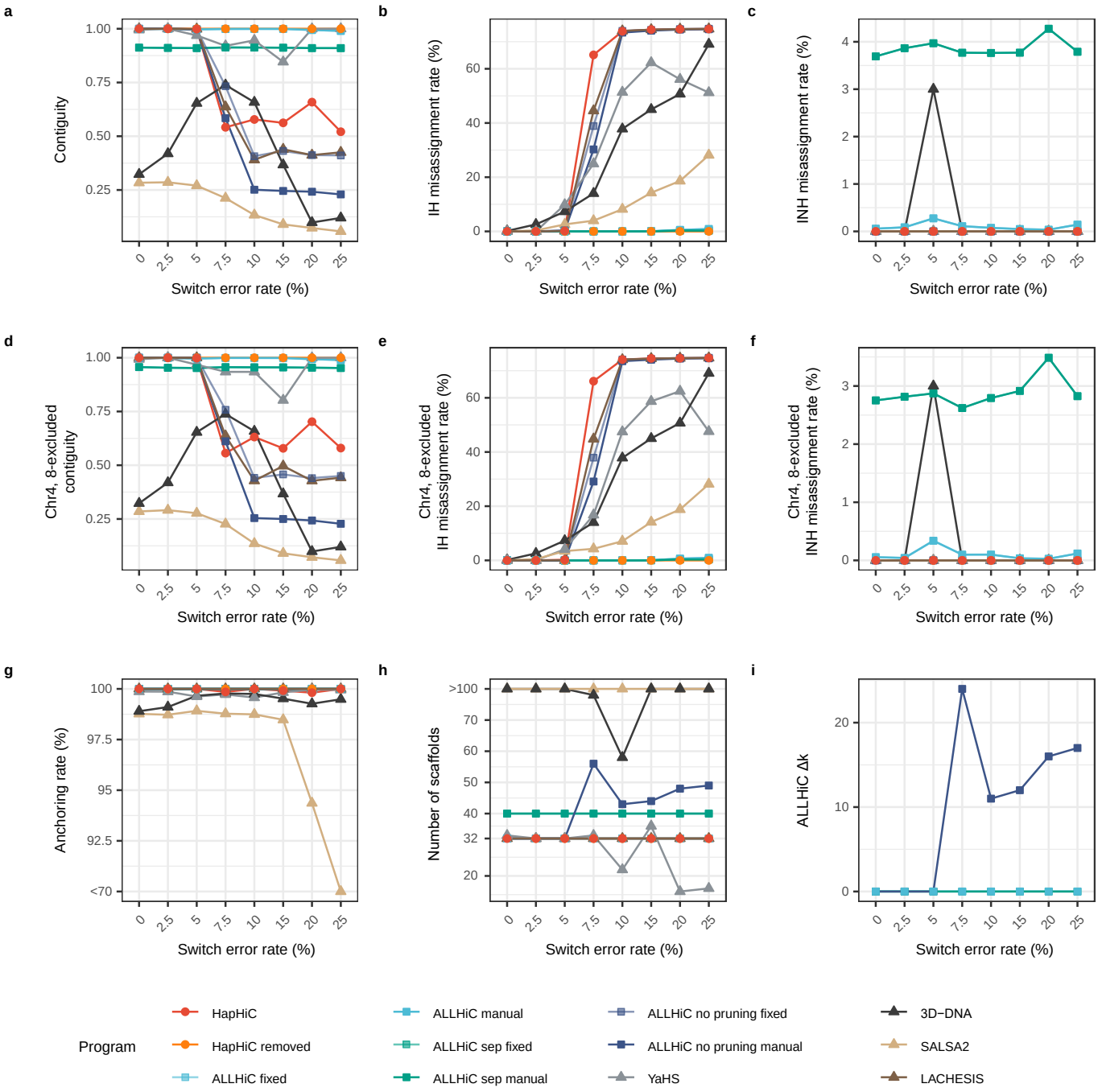
Supplementary Fig. 26 | Hi-C contact map of haplotypes with simulated switch errors. Arrows indicate inter-allele Hi-C links observed between haplotypes.



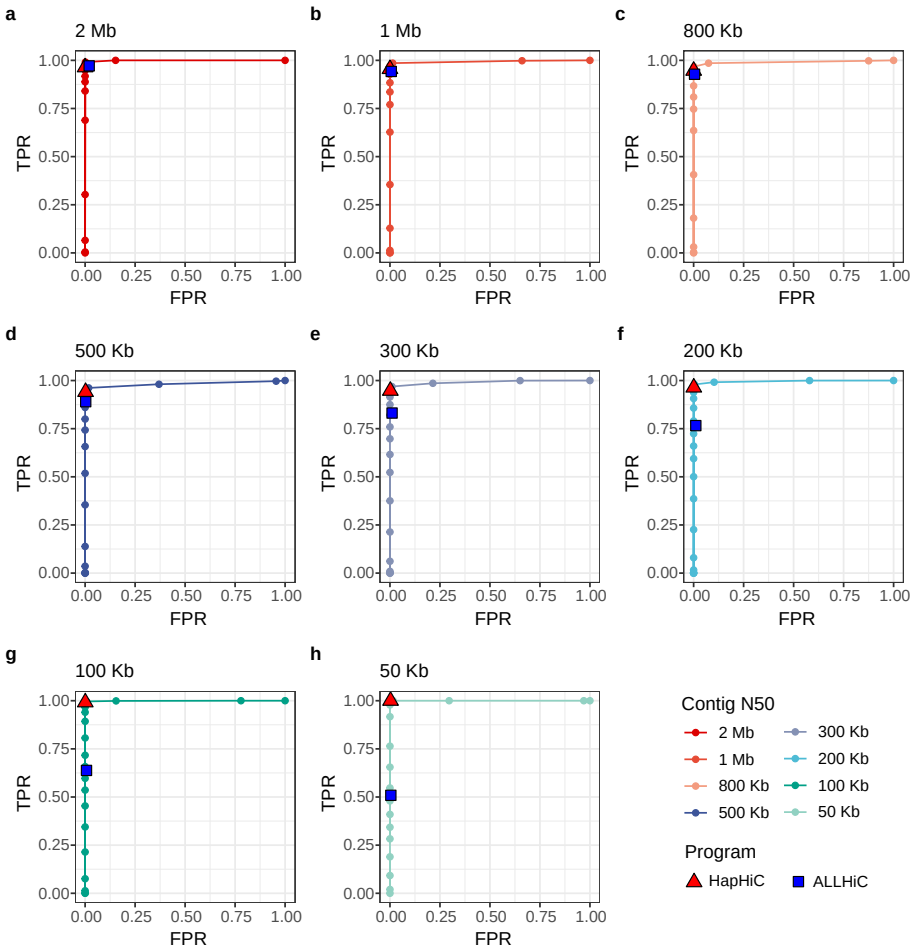
Supplementary Fig. 27 | Distribution of the number of Hi-C links for contigs with varying switch error rate ranging from 0 to 25% (**a**-**h**).



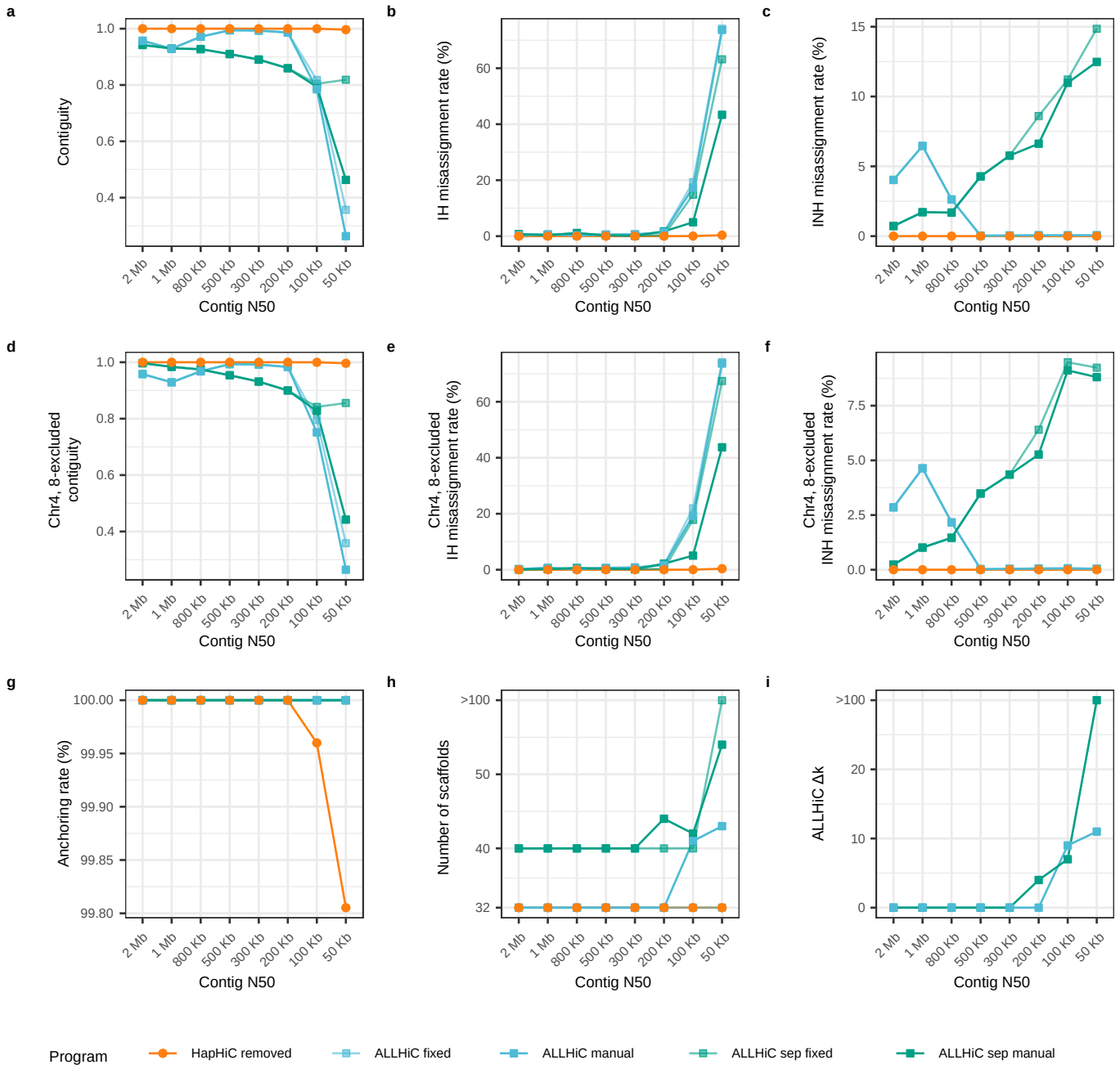
Supplementary Fig. 28 | Evaluation of the concordance ratio method in identifying allelic contig pairs. **a**, The concordance ratios increase with the percentage of overlap. "*n*" represents the number of contig pairs. **b**, The threshold demonstrates stable performance in detecting allelic contig pairs across varying contig lengths. Box plots: the box depicts the interquartile range; the line represents the median value; and the whiskers represent the minimum and maximum values (excluding outliers).



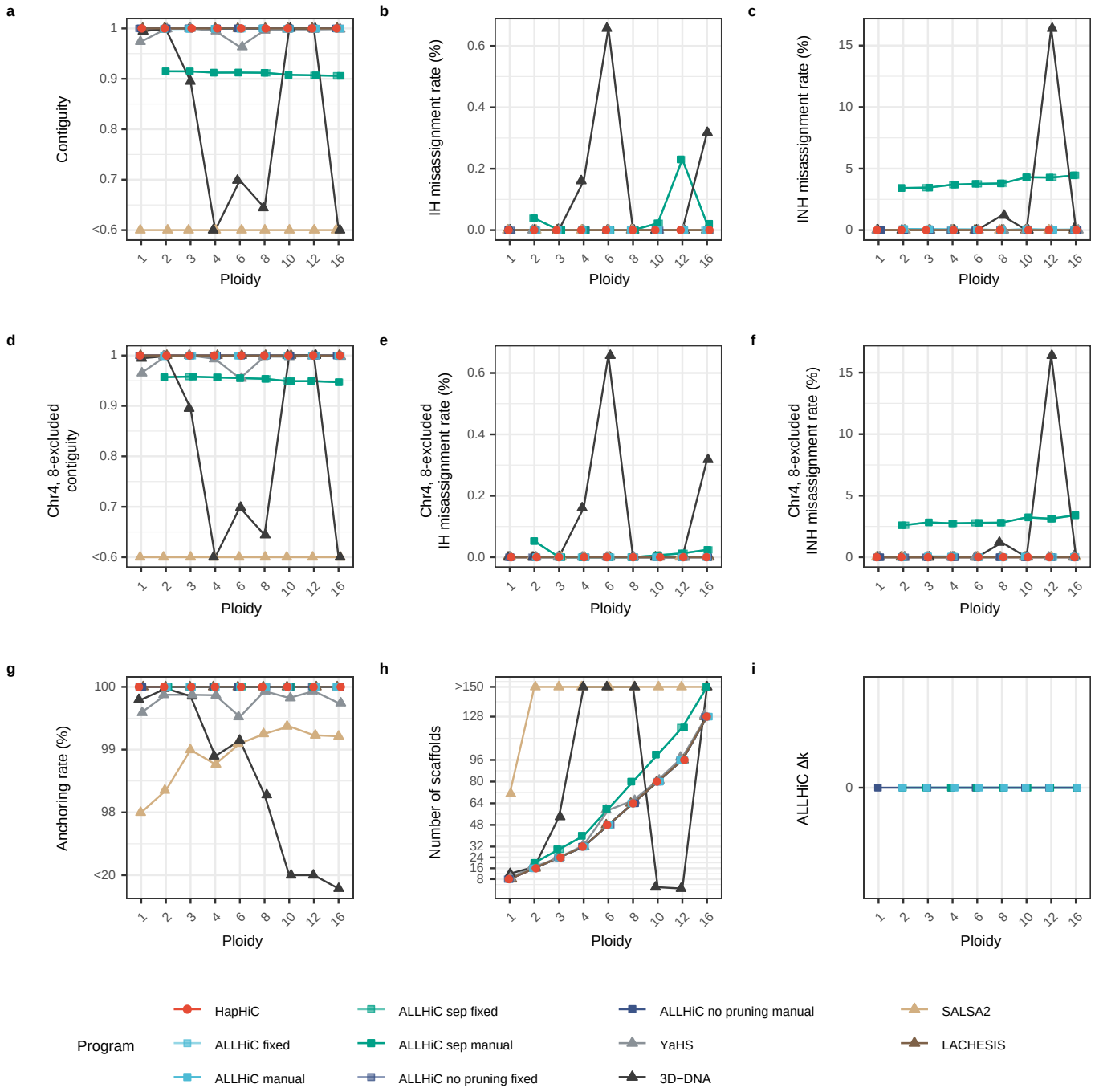
Supplementary Fig. 29 | Performance evaluation of Hi-C-based scaffolding tools in chromosome assignment on assemblies with varying switch error rates. The evaluation metrics include contiguity (**a**), misassignment rate between homologous chromosomes (**b**), misassignment rate between non-homologous chromosomes (**c**), anchoring rate (**g**), number of scaffolds (**h**), and Δk of manual parameter tuning for ALLHiC (**i**). During ALLHiC pruning, the genome of a closely related species, *M. truncatula* (MtrunA17r5.0-ANR), was used as a reference. However, chromosomes 4 and 8 of *M. truncatula* have some structural differences compared to those of *M. sativa*. Therefore, we also calculated the contiguity and misassignment rate after excluding these two chromosomes (**d-f**). The parameters for these scaffolding tools are provided in Section **3 Hi-C-based scaffolding** of **Software Commands**.



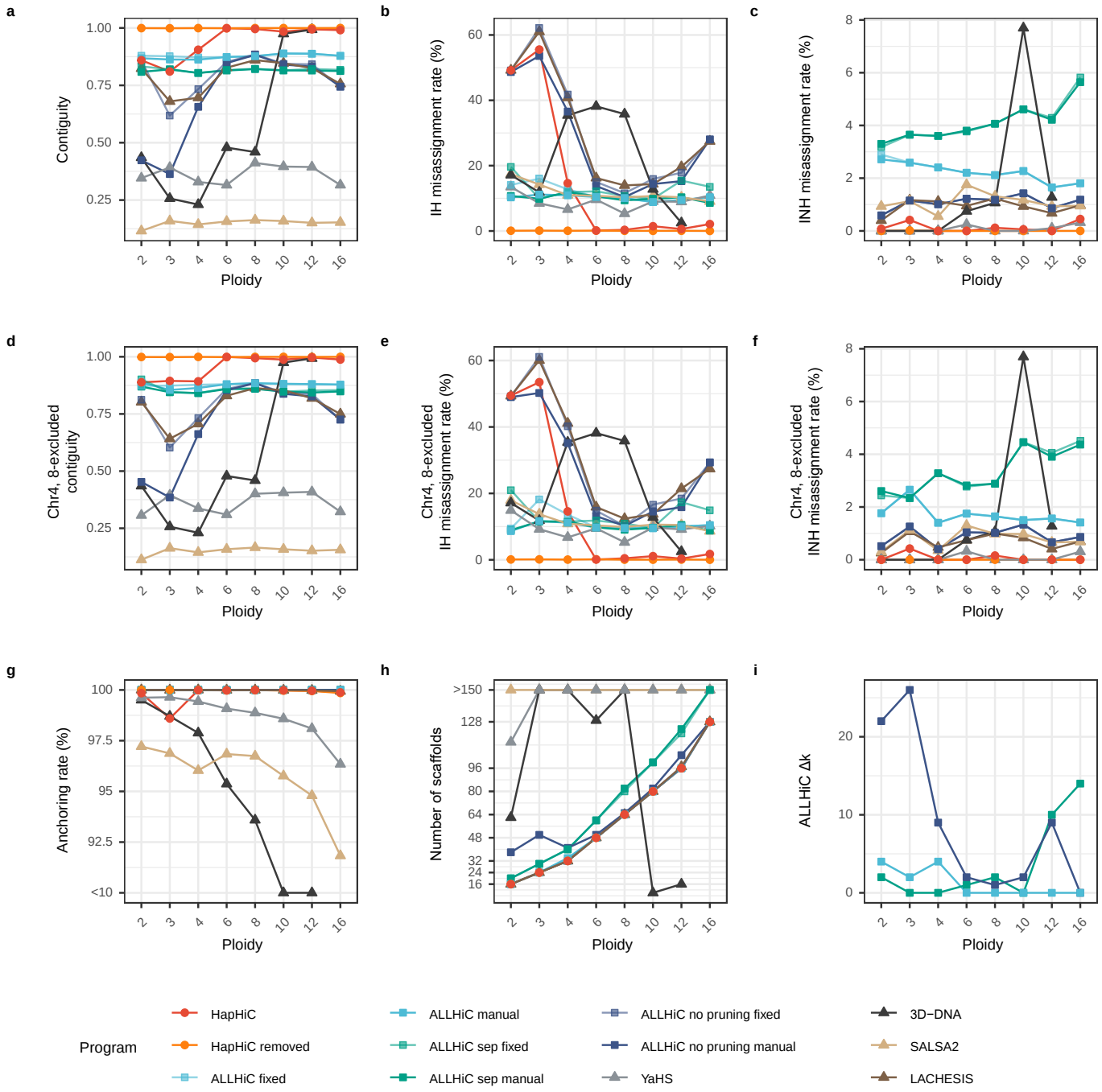
Supplementary Fig. 30 | Receiver operating characteristics (ROC) curves demonstrate the performance of the concordance ratio method in identifying allelic contig pairs of varying lengths ranging from 2 Mb to 50 Kb (**a**-**h**). Red triangles represent the performance of the default concordance ratio (0.2), while blue rectangles denote the performance of the reference-dependent method implemented in ALLHiC.



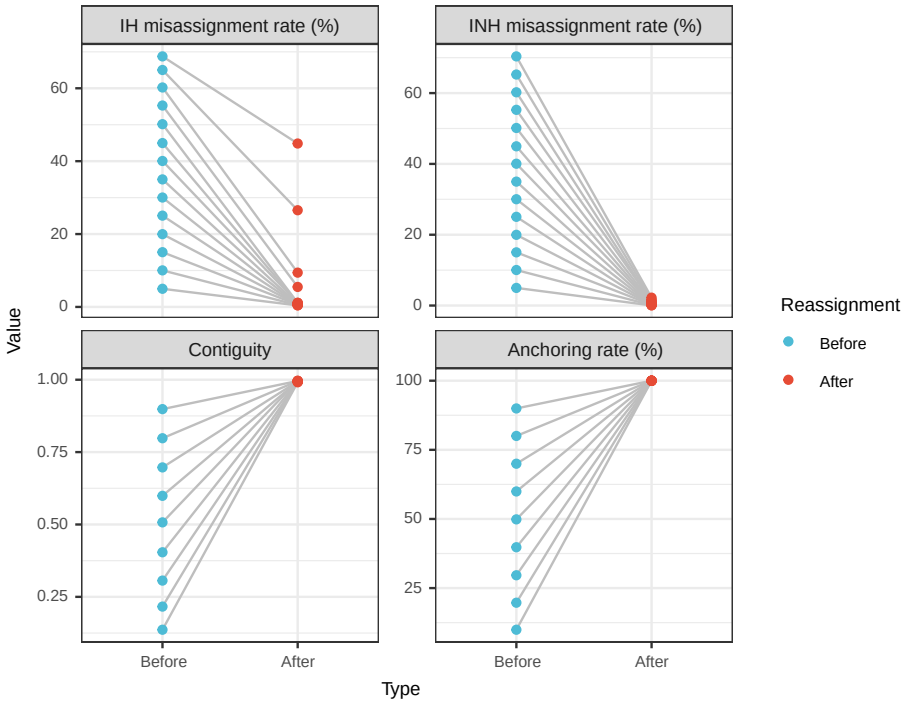
Supplementary Fig. 31 | Performance evaluation of Hi-C-based scaffolding tools in chromosome assignment on assemblies with identical switch error rate (20%) but varying contig N50 values. The evaluation metrics include contiguity (**a**), misassignment rate between homologous chromosomes (**b**), misassignment rate between non-homologous chromosomes (**c**), anchoring rate (**g**), number of scaffolds (**h**), and Δ*k* of manual parameter tuning for ALLHiC (**i**). During ALLHiC pruning, the genome of a closely related species, *M. truncatula* (MtrunA17r5.0-ANR), was used as a reference. However, chromosomes 4 and 8 of *M. truncatula* have some structural differences compared to those of *M. sativa*. Therefore, we also calculated the contiguity and misassignment rate after excluding these two chromosomes (**d-f**). The parameters for these scaffolding tools are provided in Section **3 Hi-C-based scaffolding** of **Software Commands**.



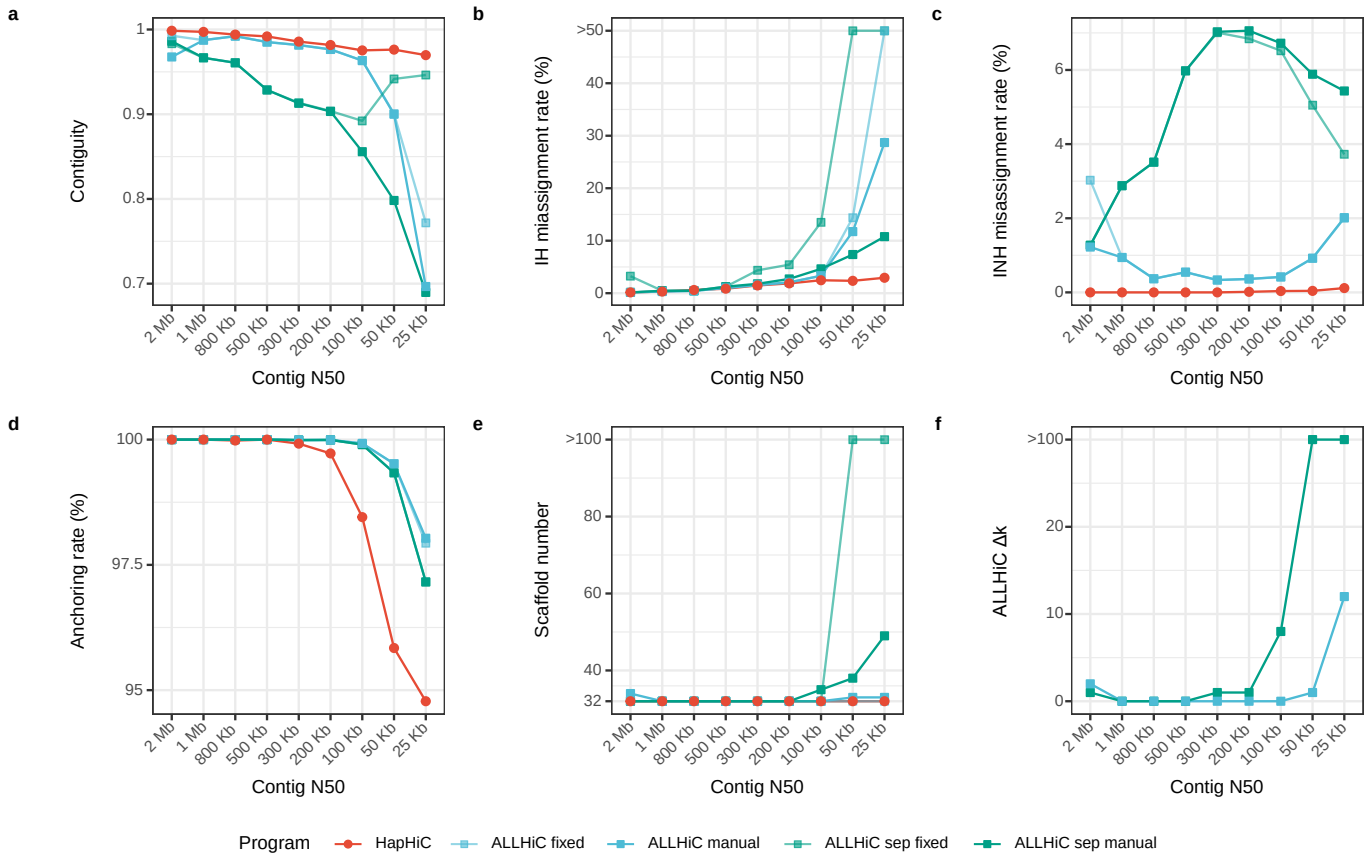
Supplementary Fig. 32 | Performance evaluation of Hi-C-based scaffolding tools in chromosome assignment on assemblies with varying ploidies. The evaluation metrics include contiguity (**a**), misassignment rate between homologous chromosomes (**b**), misassignment rate between non-homologous chromosomes (**c**), anchoring rate (**g**), number of scaffolds (**h**), and Δ*k* of manual parameter tuning for ALLHiC (**i**). During ALLHiC pruning, the genome of a closely related species, *M. truncatula* (MtrunA17r5.0-ANR), was used as a reference. However, chromosomes 4 and 8 of *M. truncatula* have some structural differences compared to those of *M. sativa*. Therefore, we also calculated the contiguity and misassignment rate after excluding these two chromosomes (**d-f**). The parameters for these scaffolding tools are provided in Section **3 Hi-C-based scaffolding** of **Software Commands**.



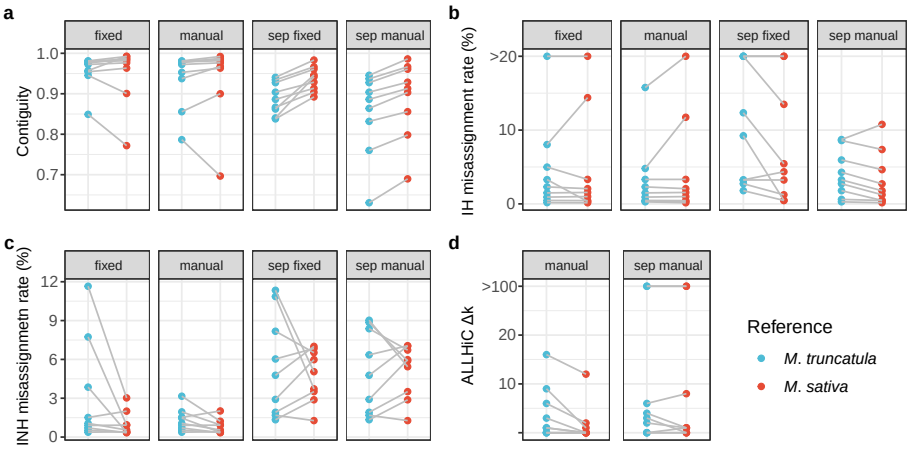
Supplementary Fig. 33 | Performance evaluation of Hi-C-based scaffolding tools in chromosome assignment on assemblies with varying ploidies and 5% each of chimeric, collapsed, and switch errors. The evaluation metrics include contiguity (**a**), misassignment rate between homologous chromosomes (**b**), misassignment rate between non-homologous chromosomes (**c**), anchoring rate (**g**), number of scaffolds (**h**), and Δ*k* of manual parameter tuning for ALLHiC (**i**). During ALLHiC pruning, the genome of a closely related species, *M. truncatula* (MtrunA17r5.0-ANR), was used as a reference. However, chromosomes 4 and 8 of *M. truncatula* have some structural differences compared to those of *M. sativa*. Therefore, we also calculated the contiguity and misassignment rate after excluding these two chromosomes (**d-f**). The parameters for these scaffolding tools are provided in Section **3 Hi-C-based scaffolding** of **Software Commands**.



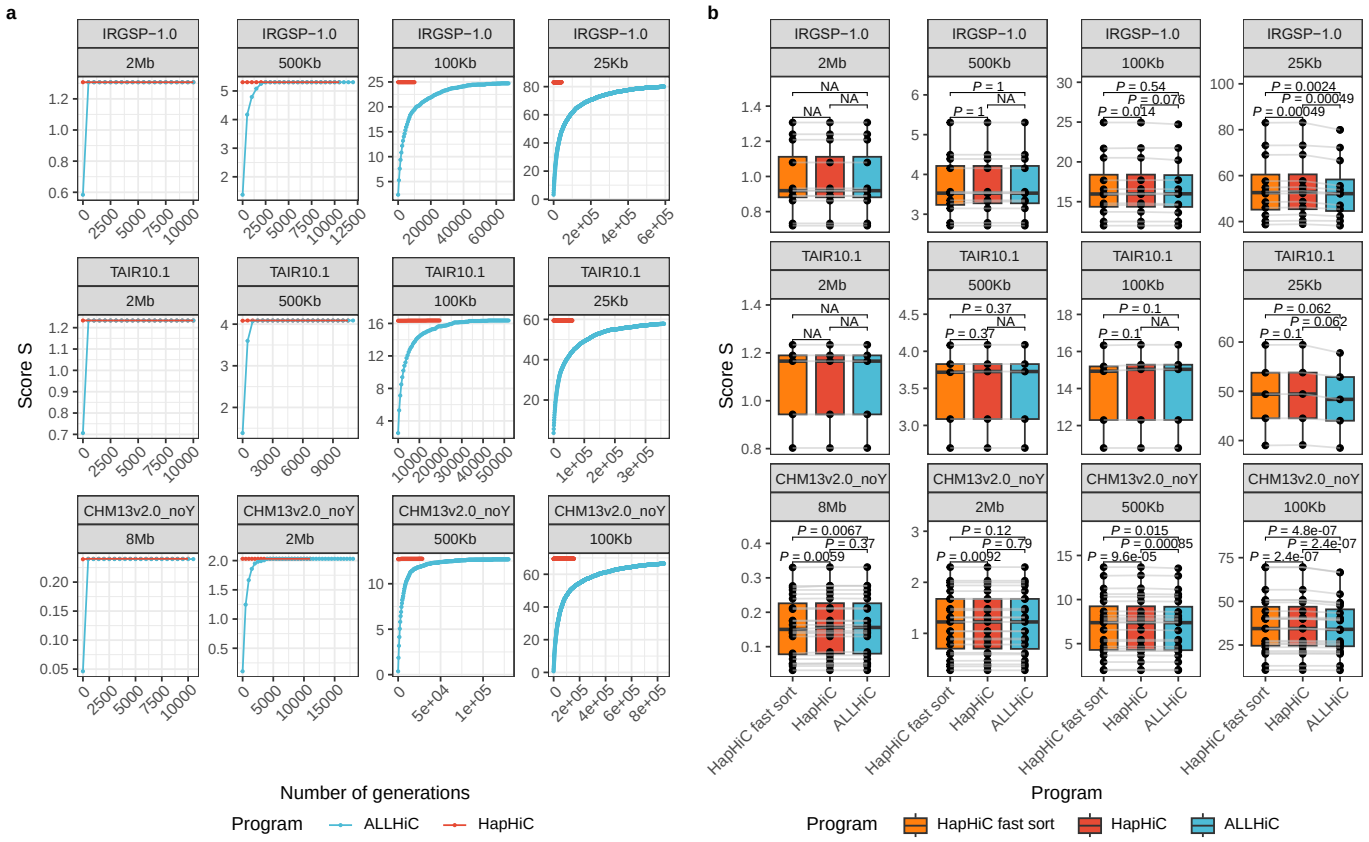
Supplementary Fig. 34 | Performance evaluation of the reassignment process in HapHiC. The evaluation metrics include misassignment rate between homologous chromosomes (IH misassignment rate), misassignment rate between non-homologous chromosomes (INH misassignment rate), contiguity, and anchoring rate.



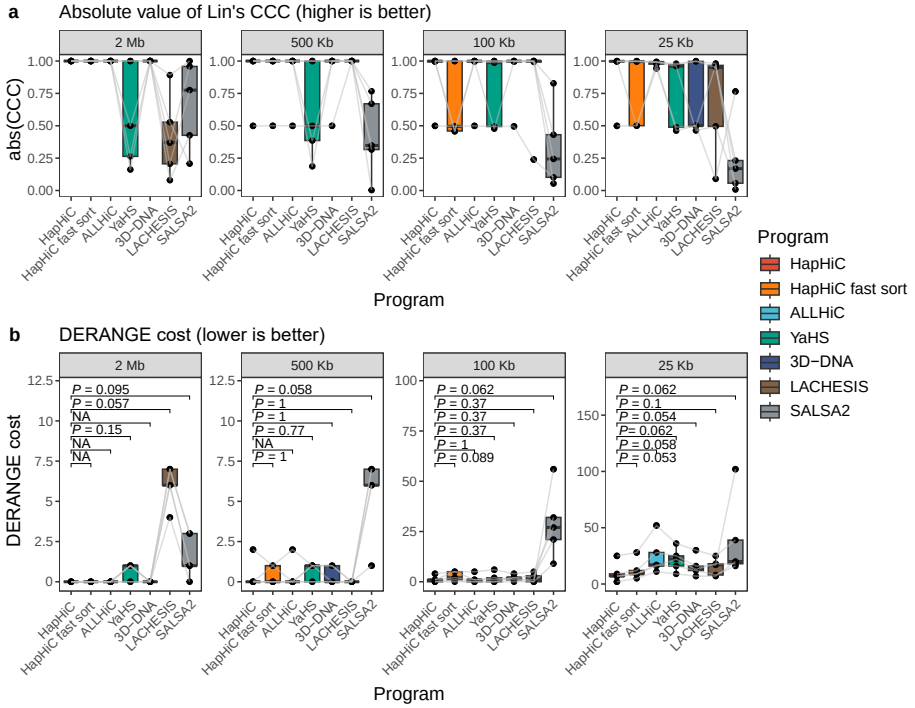
Supplementary Fig. 35 | Performance evaluation of ALLHiC using the first haplotype of *M. sativa* XinJiangDaYe as the reference genome. The evaluation metrics include contiguity (**a**), misassignment rate between homologous chromosomes (**b**), misassignment rate between non-homologous chromosomes (**c**), anchoring rate (**d**), number of scaffolds (**e**), and Δ*k* of manual parameter tuning for ALLHiC (**f**).



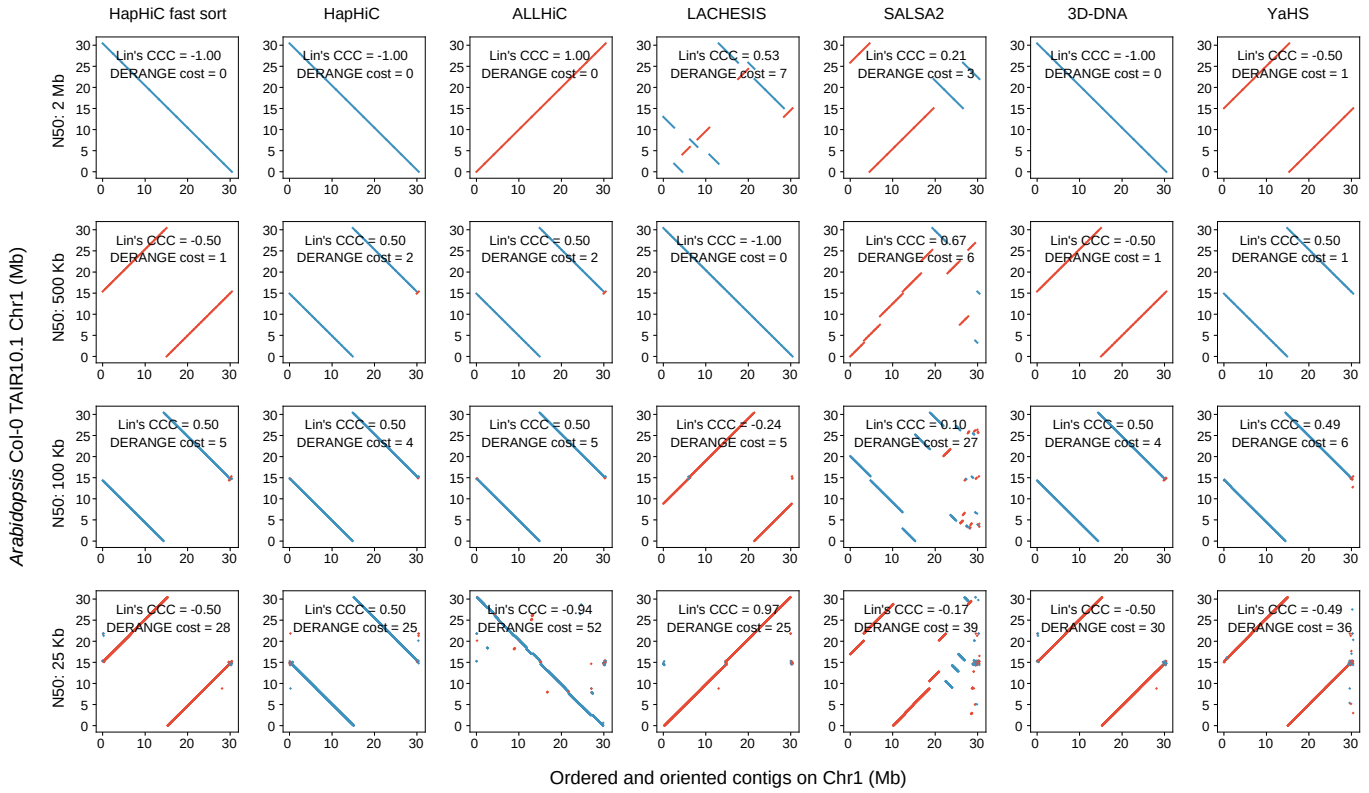
Supplementary Fig. 36 | Comparison of the two assemblies as the reference genomes for ALLHiC in chromosome assignment. The evaluation metrics include contiguity (**a**), misassignment rate between homologous chromosomes (**b**), misassignment rate between non-homologous chromosomes (**c**), and Δ*k* of manual parameter tuning for ALLHiC (**d**).



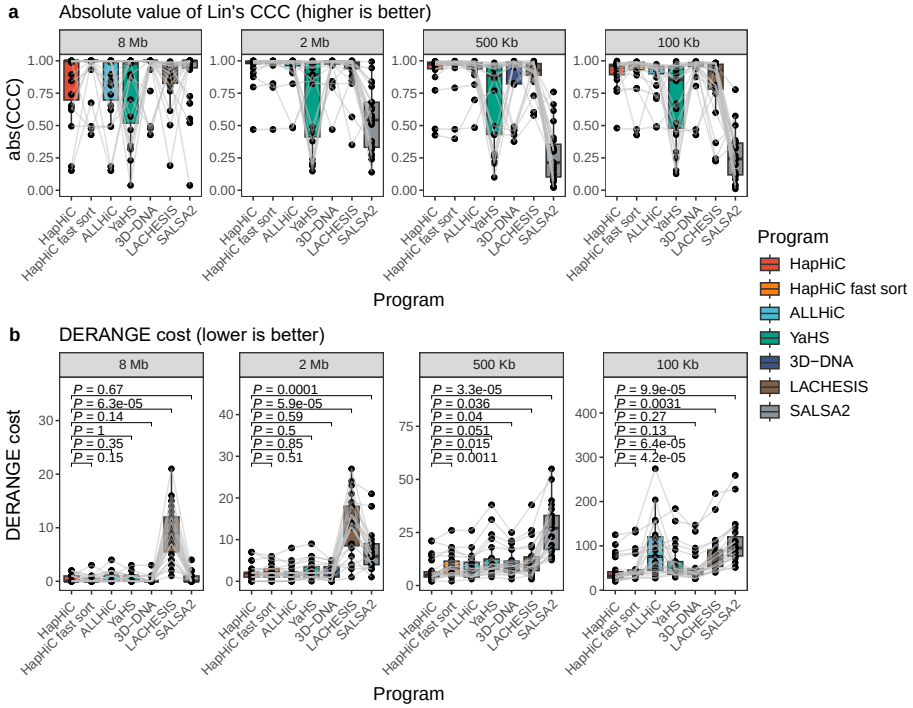
Supplementary Fig. 37 | Performance evaluation of HapHiC and ALLHiC in contig ordering and orientation using the built-in scoring system of ALLHiC. This evaluation was conducted on simulated assemblies of rice IRGSP-1.0, *Arabidopsis* TAIR10.1, and human CHM13v2.0_noY genomes with varying contig N50 values. **a**, The scores increase with the number of generations, with HapHiC demonstrating a significant decrease in the iterations needed for the genetic algorithm to converge. **b**, A comparison of the final scores between HapHiC and ALLHiC. When only fast sorting was employed, the initial scores of HapHiC were already comparable to or even higher than the final scores achieved by ALLHiC. In assemblies with low contig contiguity, these scores were further improved during the subsequent optimization process. P values were derived from two-sided Wilcoxon signed-rank tests (n = 12, 5, and 23 chromosomes for IRGSP-1.0, TAIR10.1, and CHM13v2.0_noY, respectively; "NA" indicates the statistical test was not applicable due to identical values). Box plots: the box depicts the interquartile range; the line represents the median value; and the whiskers represent the minimum and maximum values (excluding outliers).



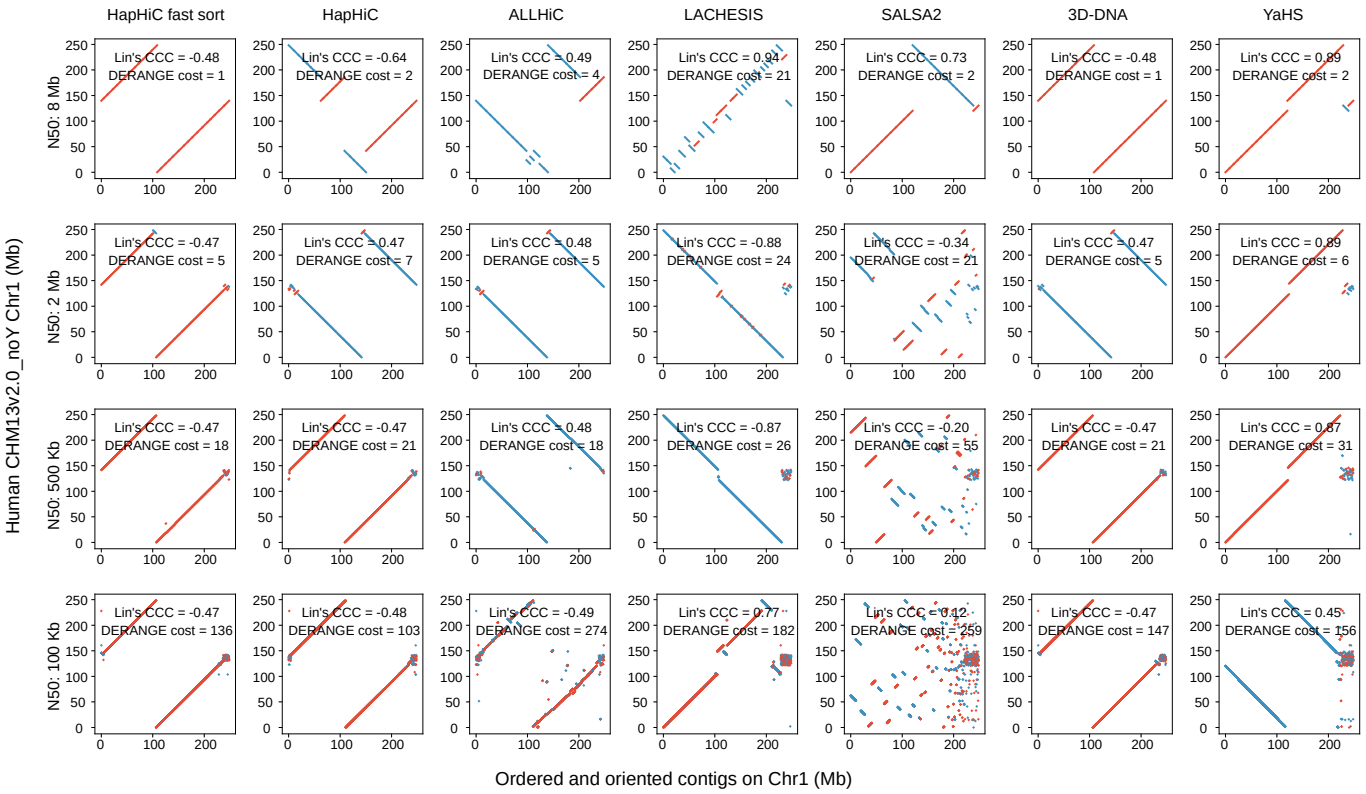
Supplementary Fig. 38 | Performance of each Hi-C-based scaffolding tools in ordering and orienting the contigs of the *Arabidopsis* TAIR10.1 chromosomes with varying contig N50 values. **a**, The absolute values of Lin's concordance correlation coefficients (CCCs) for each Hi-C-based scaffolding tool (n = 5 chromosomes). **b**, The DERANGE costs for each Hi-C-based scaffolding tool (P values from two-sided Wilcoxon signed-rank tests, n = 5 chromosomes; "NA" indicates the statistical test was not applicable due to identical values). Box plots: the box depicts the interquartile range; the line represents the median value; and the whiskers represent the minimum and maximum values (excluding outliers).



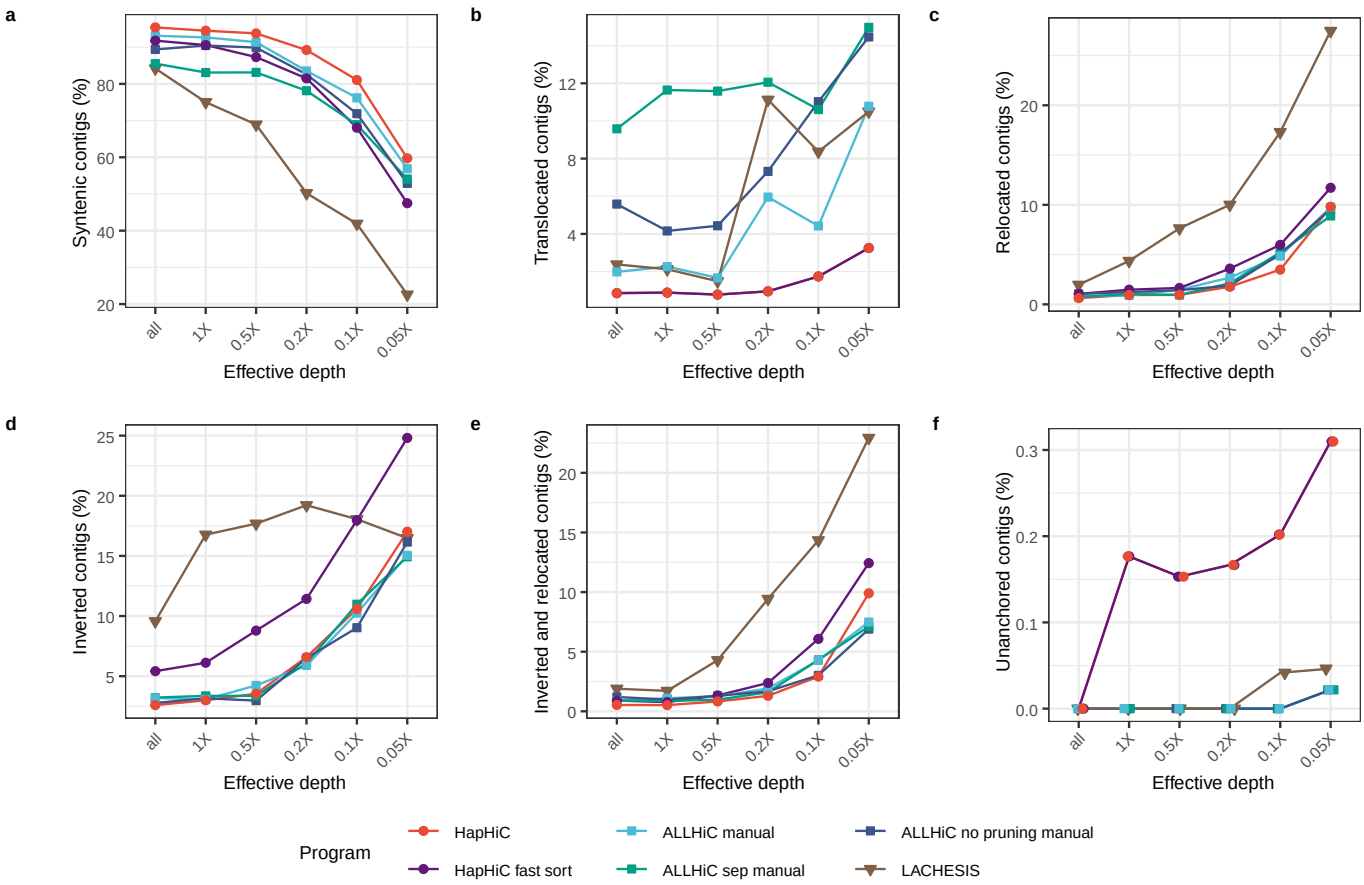
Supplementary Fig. 39 | Dot plots illustrate the concordance between chromosome 1 of the *Arabidopsis* TAIR10.1 genome and the contig ordering and orientation result produced by each Hi-C-based scaffolding tool. Red and blue dots represent forward and reverse complementary alignments, respectively. The corresponding Lin's concordance correlation coefficients (CCCs) and DERANGE costs are also displayed in these dot plots.



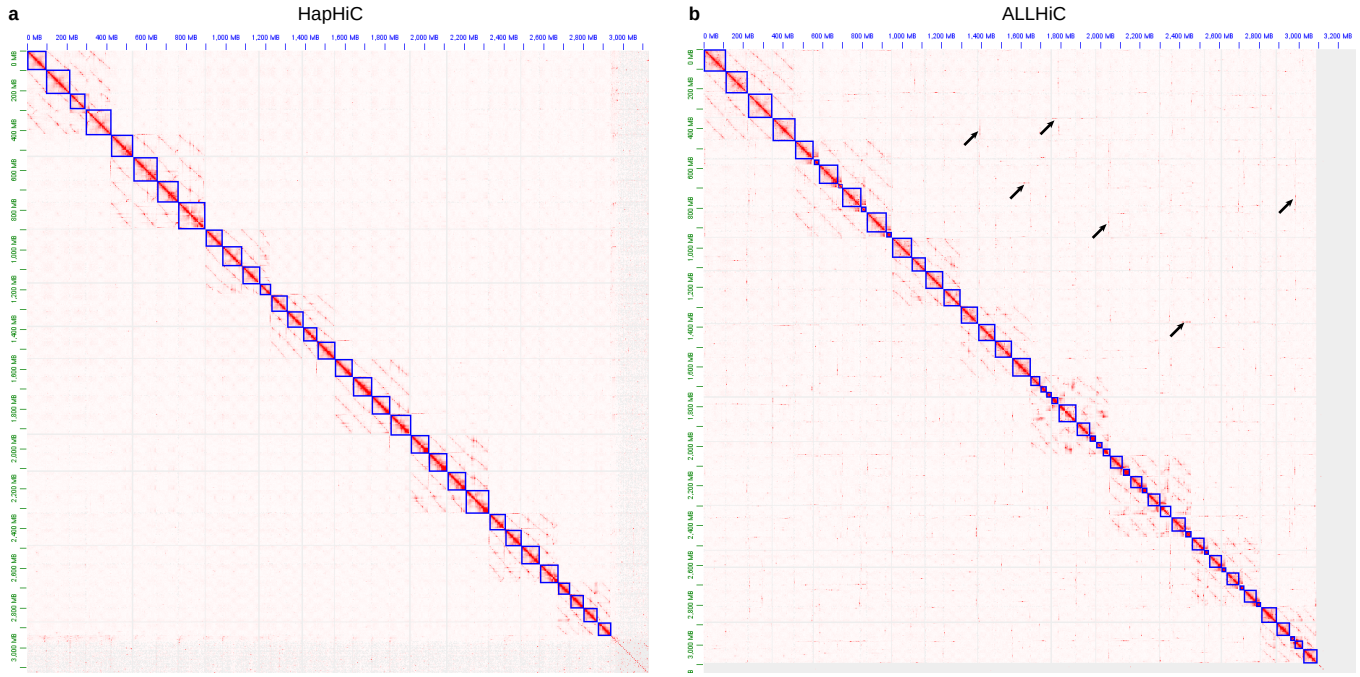
Supplementary Fig. 40 | Performance of each Hi-C-based scaffolding tools in ordering and orienting the contigs of the human CHM13v2.0_noY chromosomes with varying contig N50 values. **a**, The absolute values of Lin's concordance correlation coefficients (CCCs) for each Hi-C-based scaffolding tool (n = 23 chromosomes). **b**, The DERANGE costs for each Hi-C-based scaffolding tool (P values from two-sided Wilcoxon signed-rank tests, n = 23 chromosomes). Box plots: the box depicts the interquartile range; the line represents the median value; and the whiskers represent the minimum and maximum values (excluding outliers).



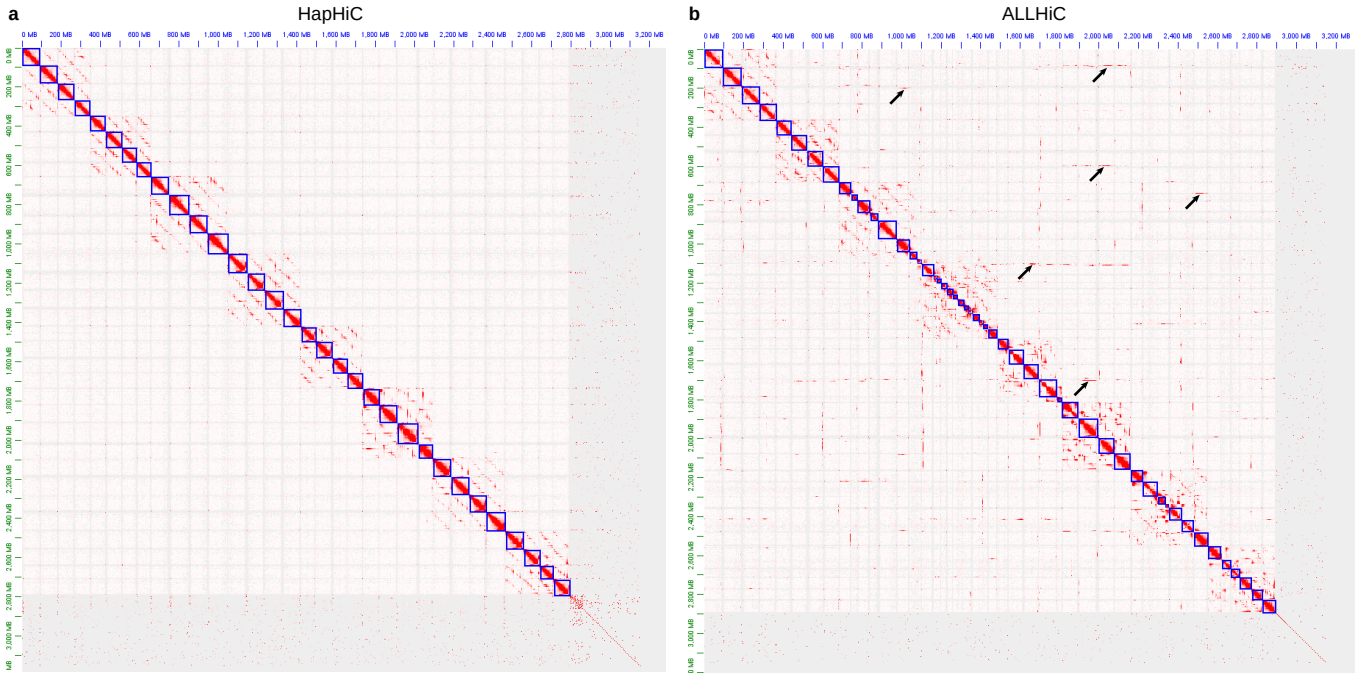
Supplementary Fig. 41 | Dot plots illustrate the concordance between chromosome 1 of the human CHM13v2.0_noY genome and the contig ordering and orientation result produced by each Hi-C-based scaffolding tool. Red and blue dots represent forward and reverse complementary alignments, respectively. The corresponding Lin's concordance correlation coefficients (CCCs) and DERANGE costs are also displayed in these dot plots.



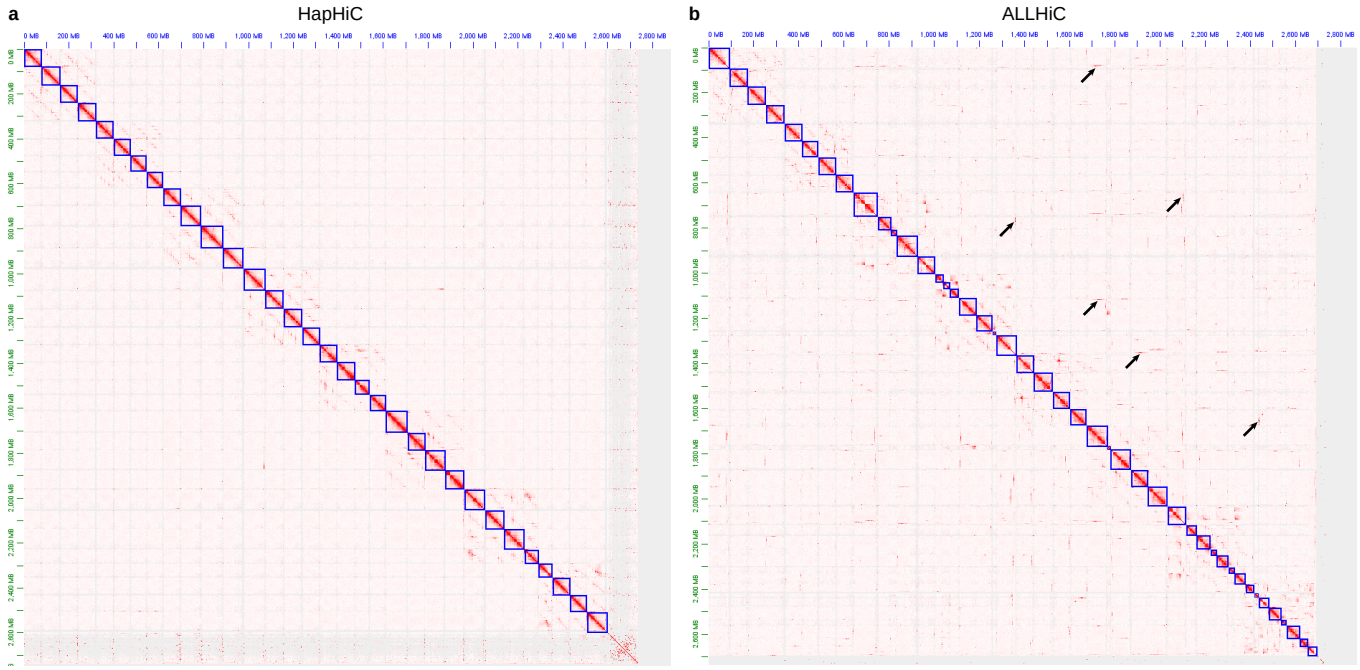
Supplementary Fig. 42 | Performance evaluation of three clustering-based scaffolding tools in contig ordering and orientation under varying effective depths. Metrics used for this evaluation include (**a**) syntenic contigs, (**b**) translocated contigs, (**c**) relocated contigs, (**d**) inverted contigs, (**e**) inverted and relocated contigs, (**f**) unanchored contigs. The details of these metrics are provided in **Supplementary Methods**.



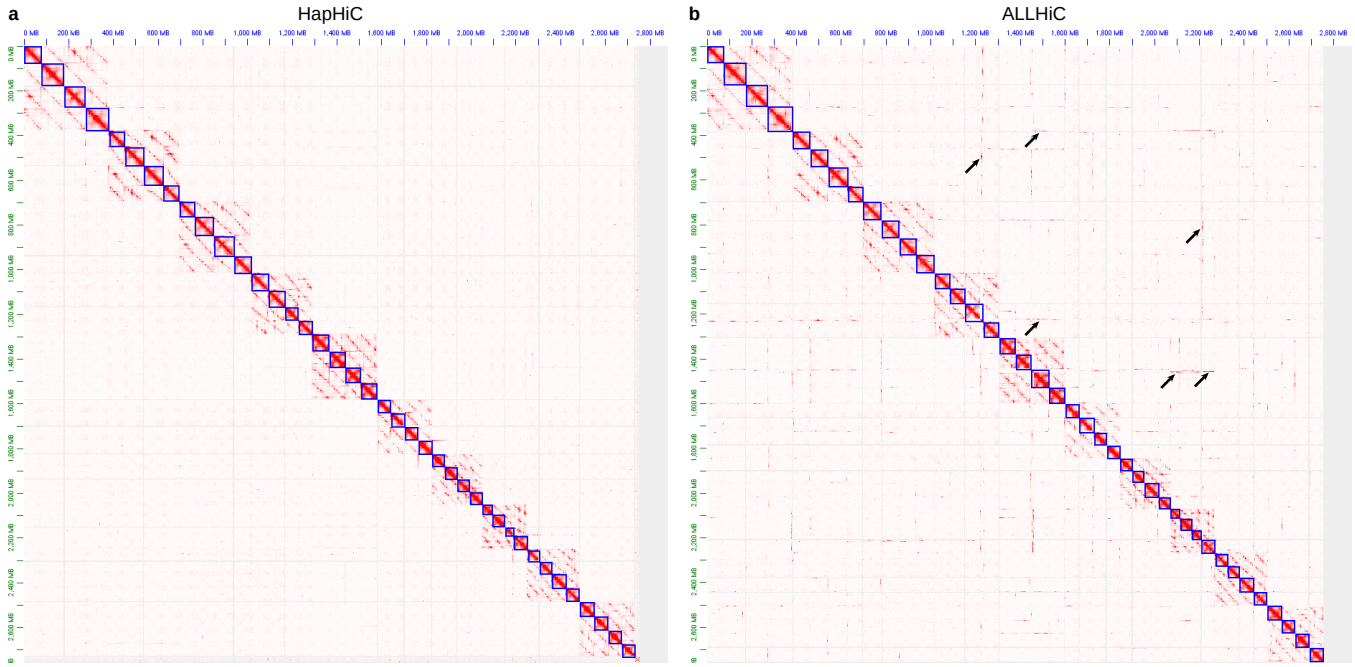
Supplementary Fig. 43 | Hi-C contact maps of HapHiC (**a**) and ALLHiC (**b**) scaffolds of the *S. spontaneum* AP85-441 genome. Black arrows indicate examples of misassignments between non-homologous chromosomes in the ALLHiC scaffolds.



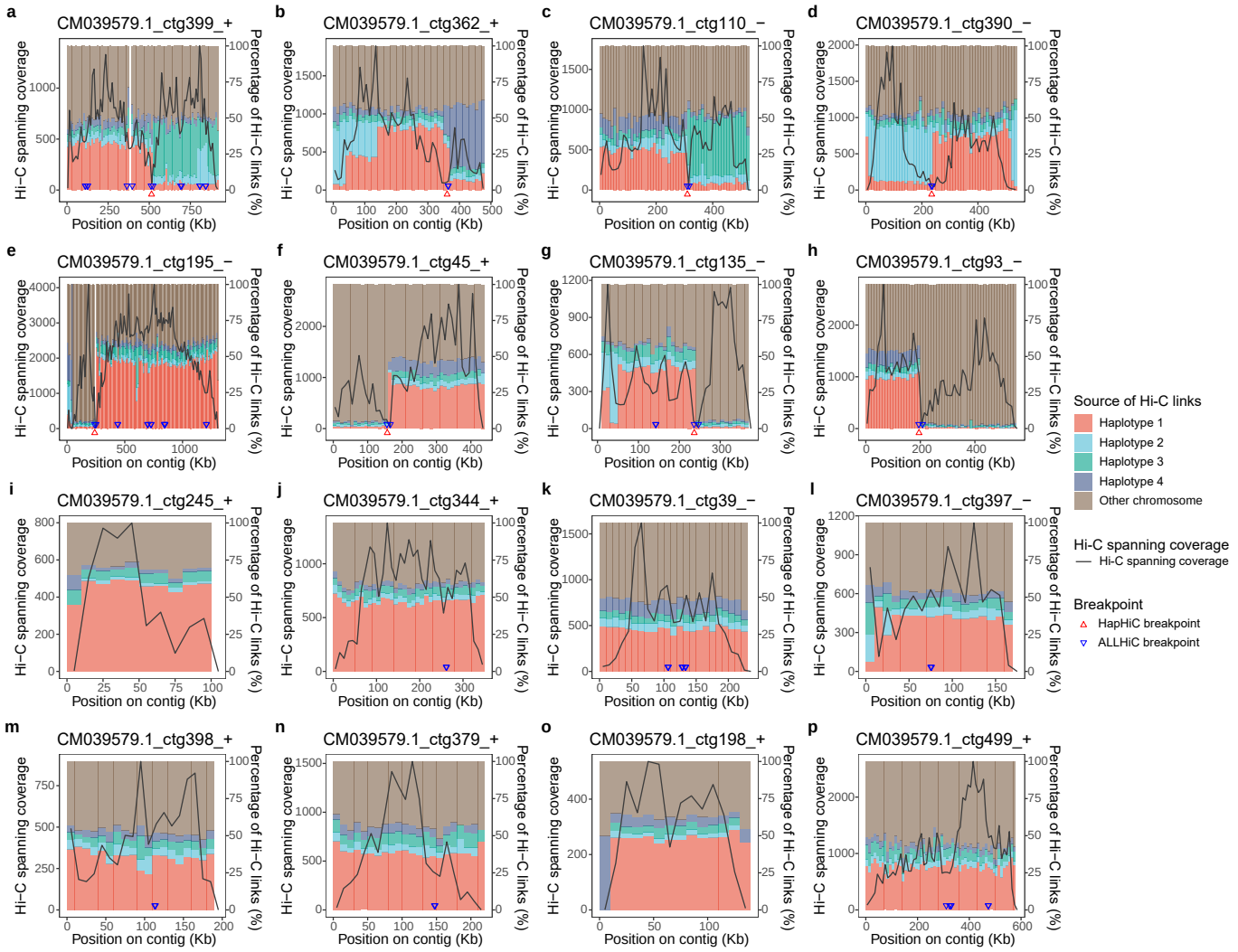
Supplementary Fig. 44 | Hi-C contact maps of HapHiC (**a**) and ALLHiC (**b**) scaffolds of the *M. sativa* XinJiangDaYe genome. Black arrows indicate examples of misassignments between non-homologous chromosomes in the ALLHiC scaffolds.



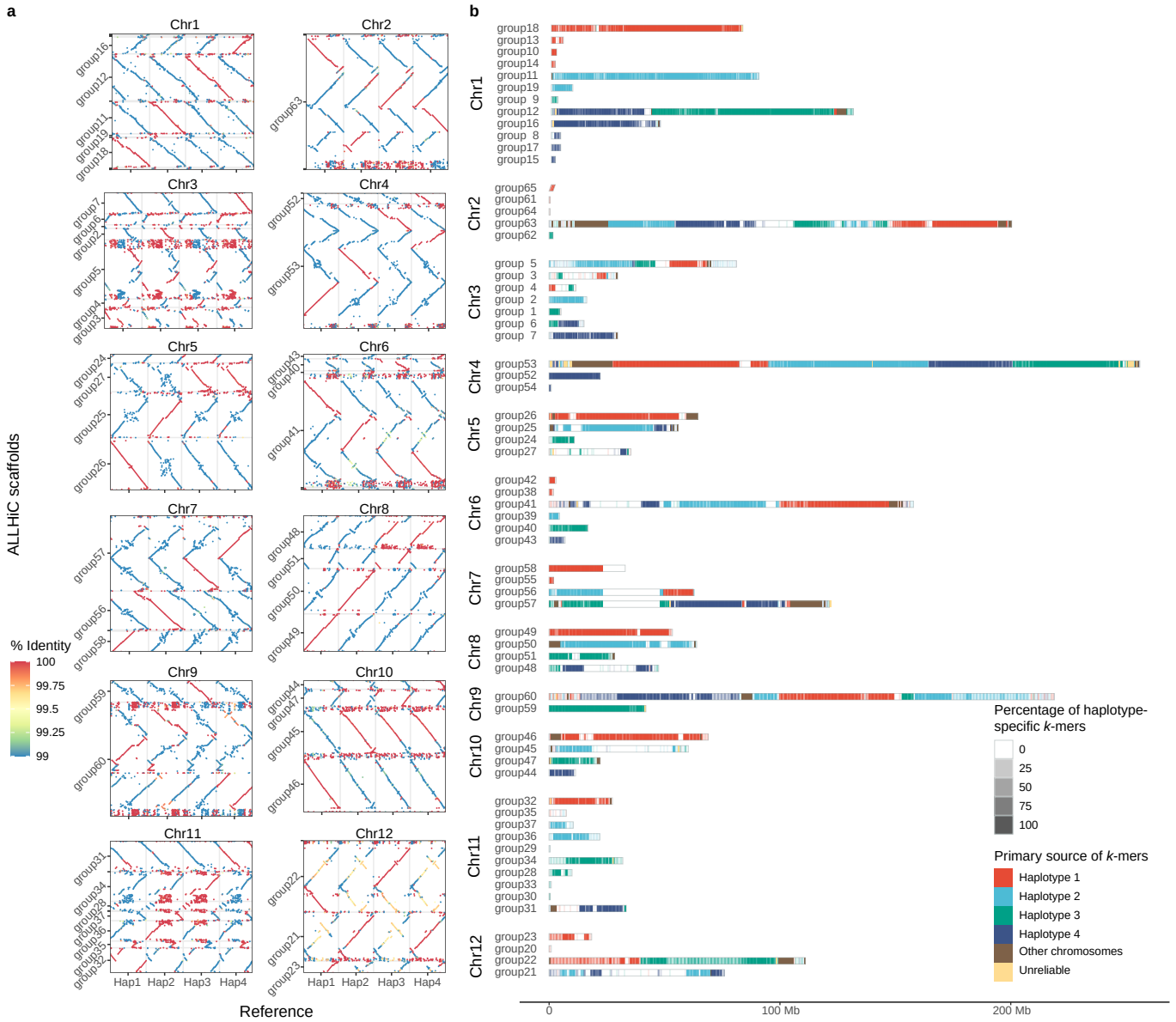
Supplementary Fig. 45 | Hi-C contact maps of HapHiC (**a**) and ALLHiC (**b**) scaffolds of the *M. sativa* Zhongmu-4 genome. Black arrows indicate examples of misassignments between non-homologous chromosomes in the ALLHiC scaffolds.



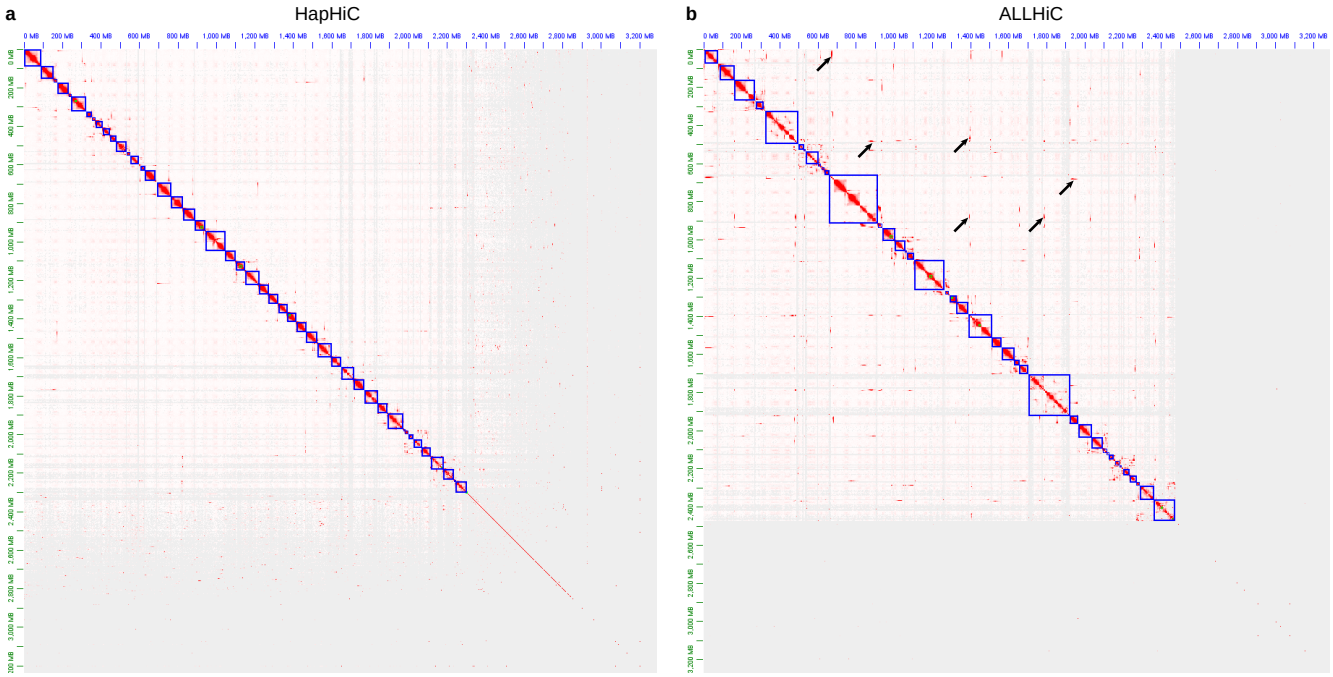
Supplementary Fig. 46 | Hi-C contact maps of HapHiC (**a**) and ALLHiC (**b**) scaffolds of the *S. spontaneum* Np-X genome. Black arrows indicate examples of misassignments between non-homologous chromosomes in the ALLHiC scaffolds.



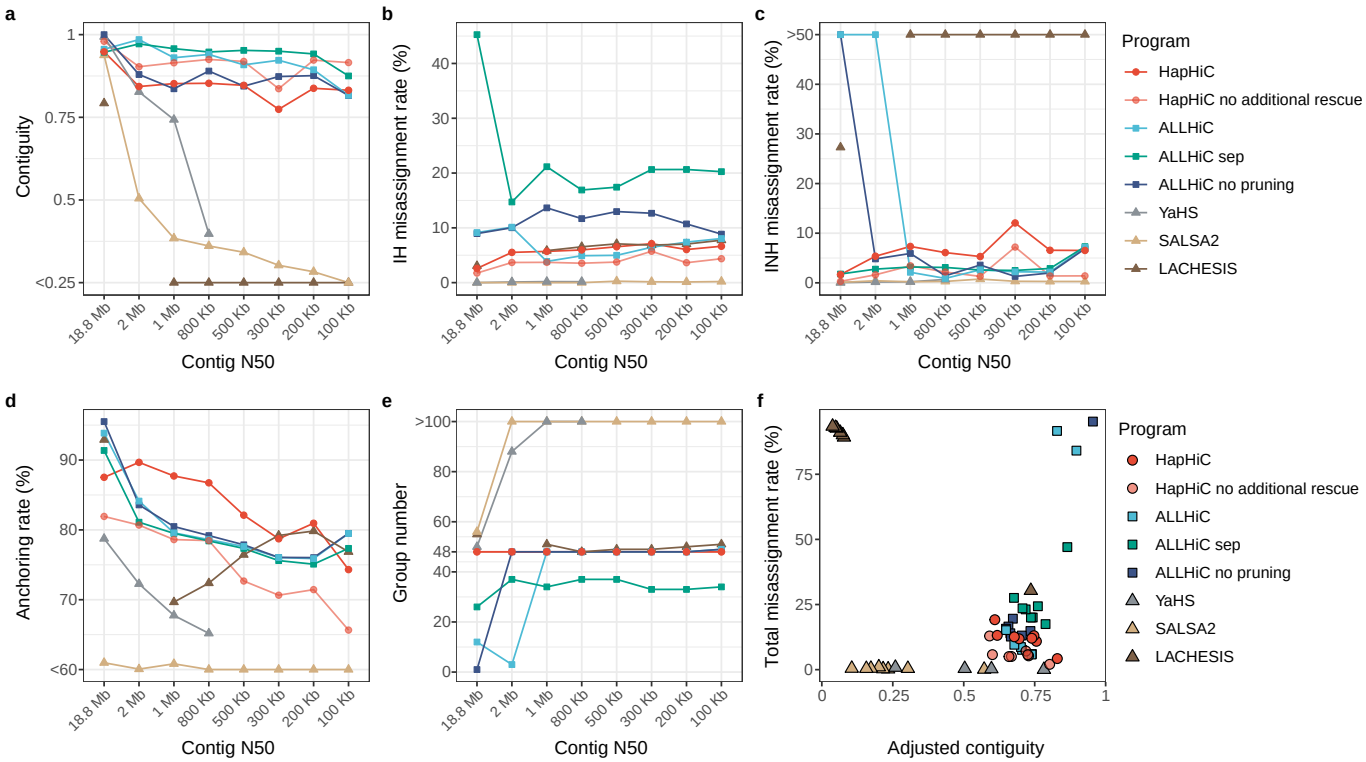
Supplementary Fig. 47 | Examples of assembly correction made by HapHiC and ALLHiC. These examples include four chimeric contigs formed between homologous chromosomes (**a**-**d**), four chimeric contigs formed between non-homologous chromosomes (**e**-**h**), and eight non-chimeric contigs (**i**-**p**). The line charts represent the Hi-C spanning coverages along contigs (left axes), while histograms depict the percentages of Hi-C links based on their sources along contigs (right axes). The source of each Hi-C link is determined by the mapping position of the other end of the read pair. Red and blue triangles indicate the breakpoints determined by HapHiC and ALLHiC, respectively.



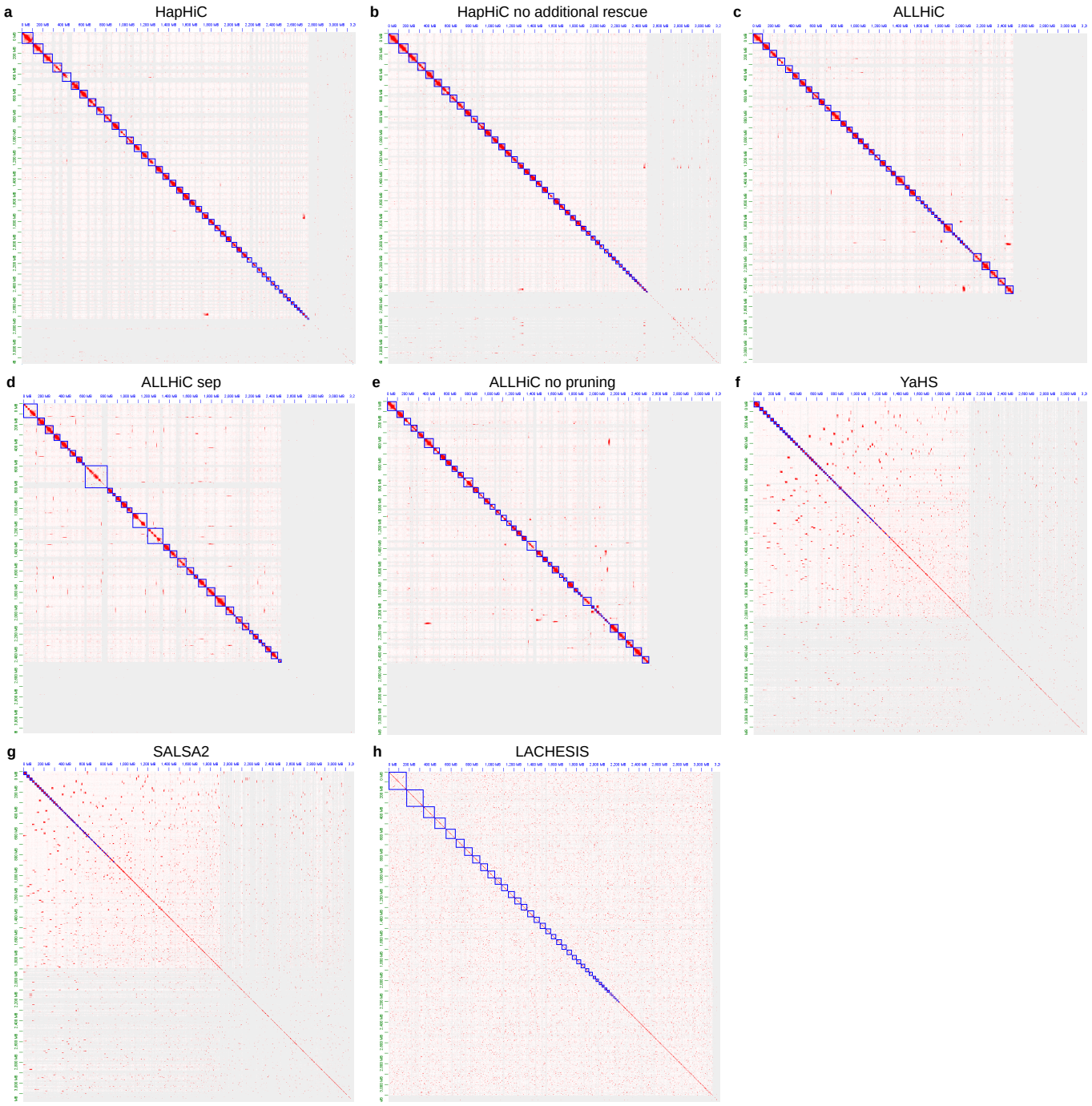
Supplementary Fig. 48 | Performance evaluation of ALLHiC in scaffolding the haplotypes of potato C88 genome. **a**, The dot plots illustrate the alignments between the ALLHiC scaffolds and the haplotypes of potato C88 genome, with dot colors indicating the sequence identities of alignments. **b**, A *k*-mer-based analysis reveals the primary source of each position along the contigs from the potato C88 haplotypes. The color indicates the primary source of *k*-mers, while the degree of transparency represents the percentage of haplotype-specific *k*-mers.



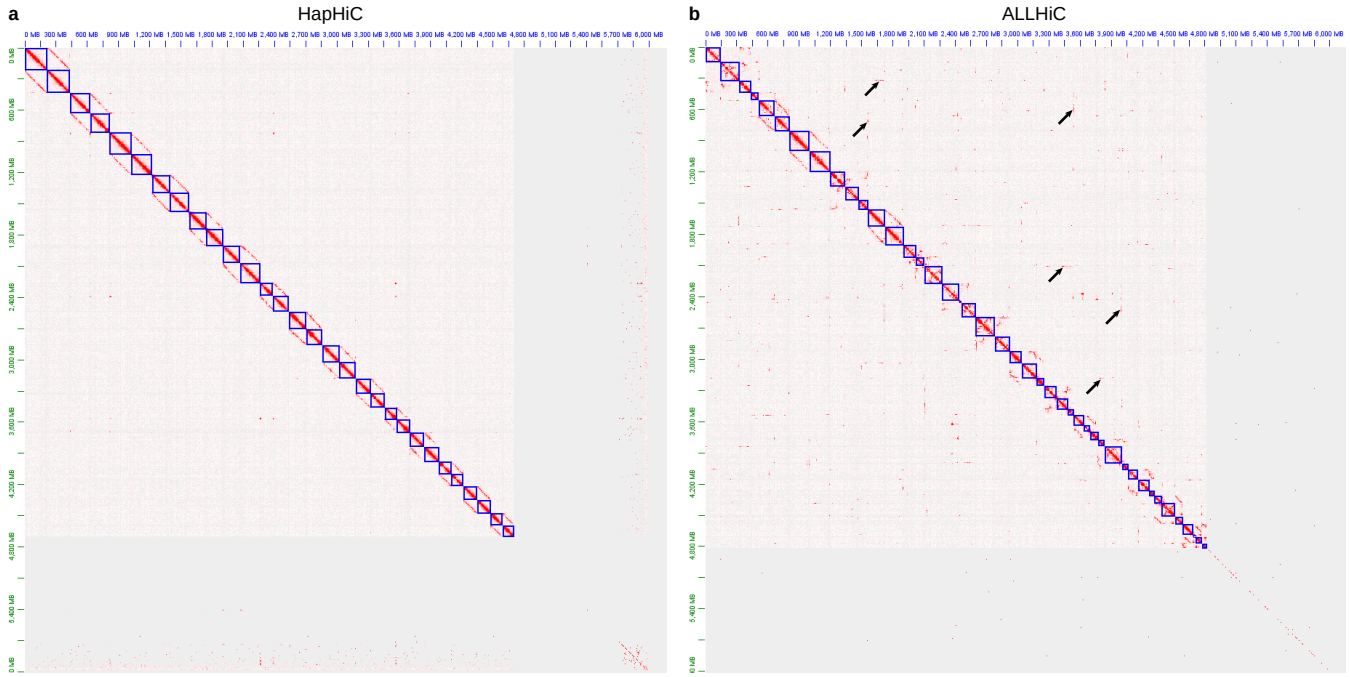
Supplementary Fig. 49 | Hi-C contact maps of HapHiC (**a**) and ALLHiC (**b**) scaffolds of the potato C88 genome. Black arrows indicate examples of misassignments between non-homologous chromosomes in the ALLHiC scaffolds.



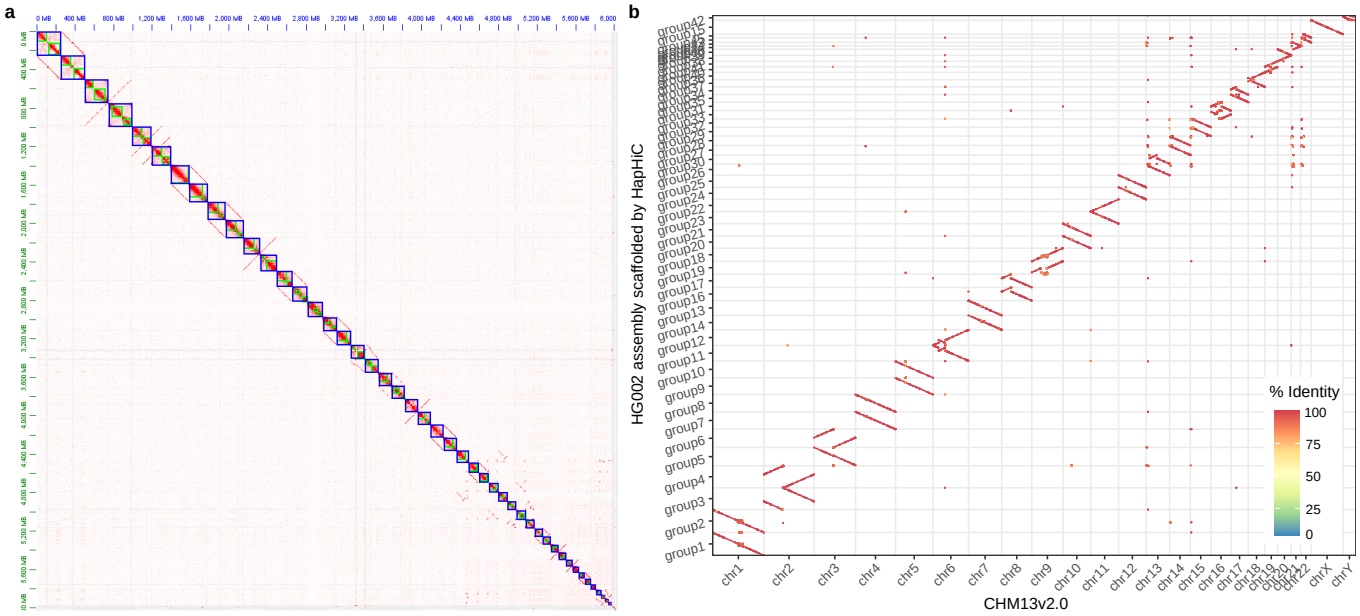
Supplementary Fig. 50 | Performance evaluation of Hi-C-based scaffolding tools in chromosome assignment on C88.v1 assemblies with varying contig N50 values. The evaluation metrics include contiguity (**a**), misassignment rate between homologous chromosomes (**b**), misassignment rate between non-homologous chromosomes (**c**), anchoring rate (**d**), number of scaffolds (**e**). **f** A summary plot illustrating the performance of each scaffolding tool. HapHiC was executed in default mode (HapHiC), or without additional rescue round (HapHiC no additional rescue). The total misassignment rate includes misassignment rate between both homologous and non-homologous chromosomes. The adjusted contiguity is calculated by multiplying the contiguity by the anchoring rate. The parameters for these scaffolding tools are provided in Section **3 Hi-C-based scaffolding** of **Software Commands**.



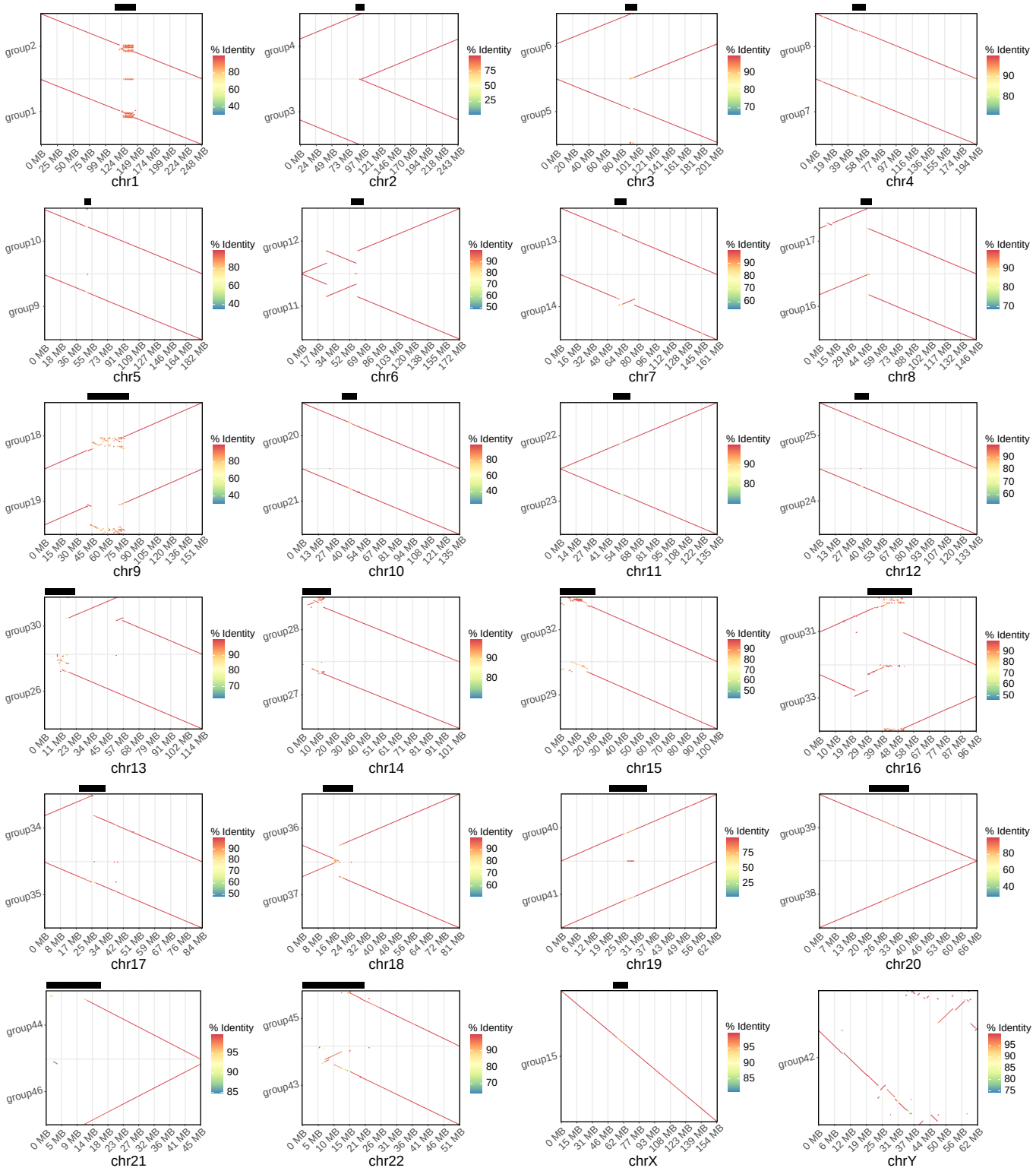
Supplementary Fig. 51 | Contact heatmap of potato C88 scaffolds generated by each scaffolding tool at a contig N50 of 800 Kb.



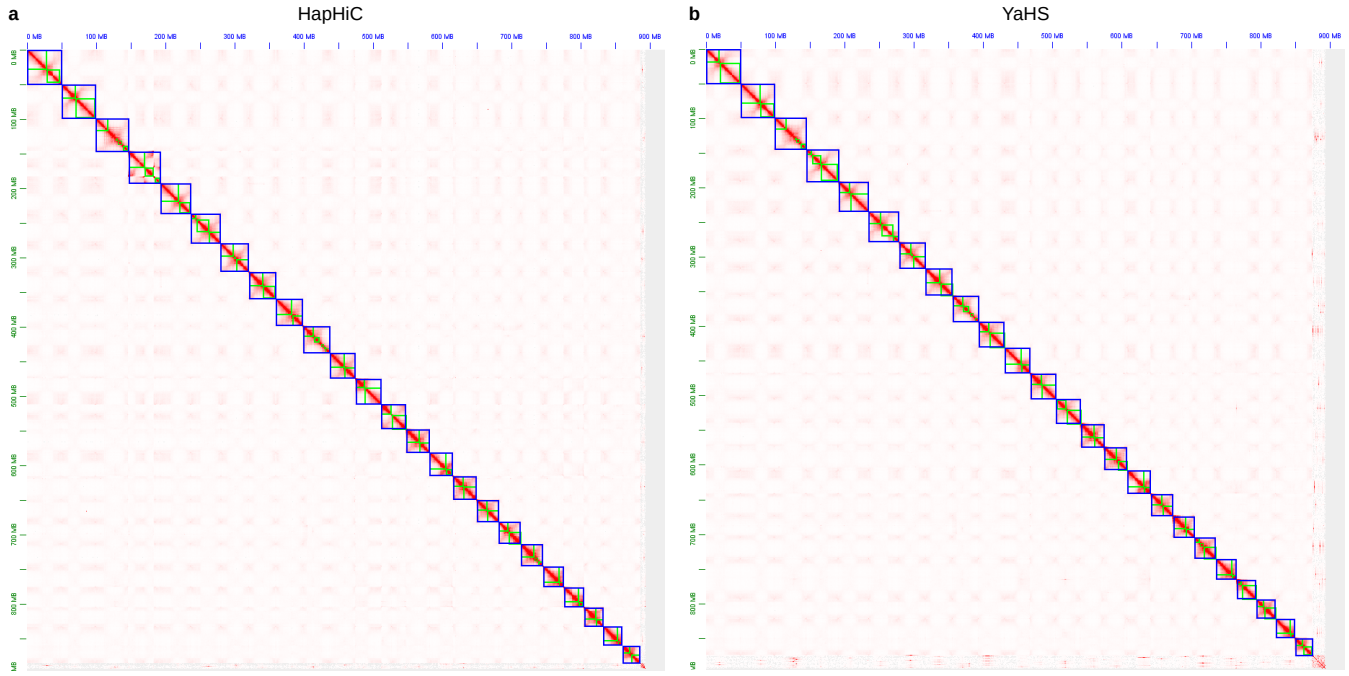
Supplementary Fig. 52 | Hi-C contact maps of HapHiC (**a**) and ALLHiC (**b**) scaffolds of the *C. sinensis* Tieguanyin genome. Black arrows indicate examples of misassignments between non-homologous chromosomes in the ALLHiC scaffolds. The reference genome used for ALLHiC is the chromosome-level monoploid assembly of *C. sinensis* Tieguanyin which was constructed from primary contigs.



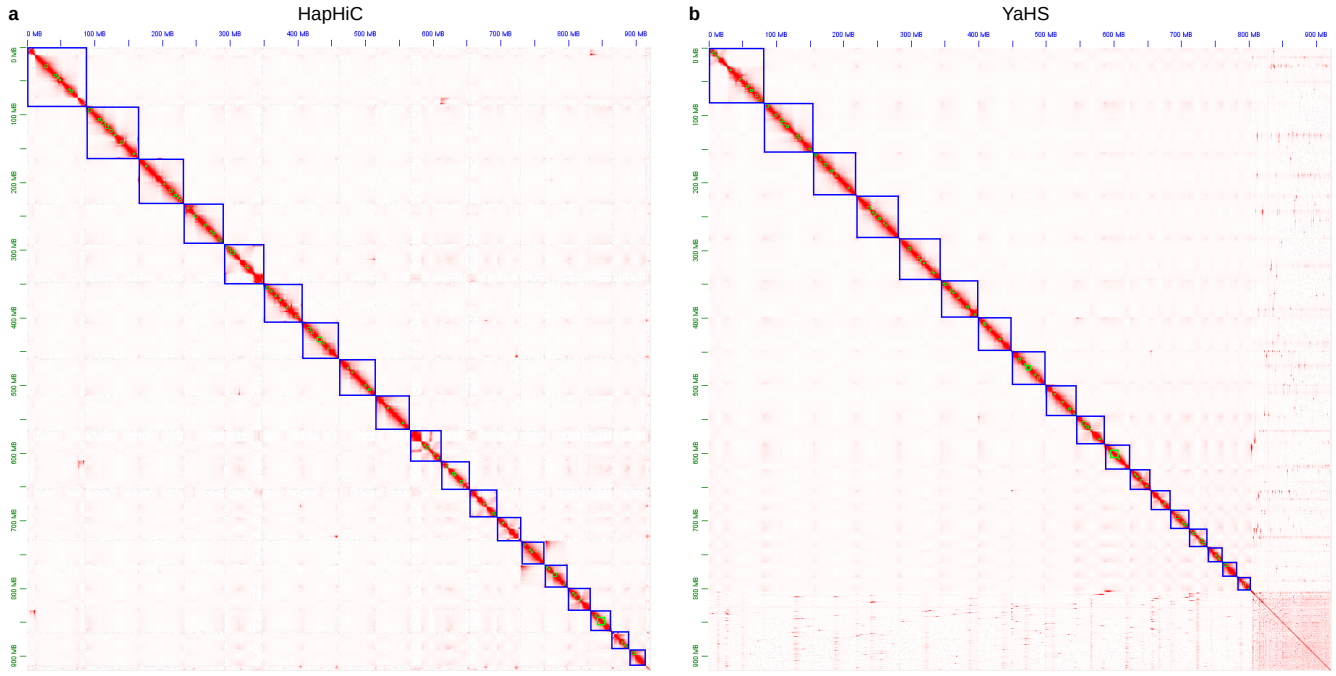
Supplementary Fig. 53 | Scaffolding of the diploid human genome, HG002_v2.1, using HapHiC. **a** Contact heatmap of the scaffolds generated by HapHiC. **b** A dot plot illustrating the whole-genome alignments between HG002 scaffolds and CHM13 T2T chromosomes.



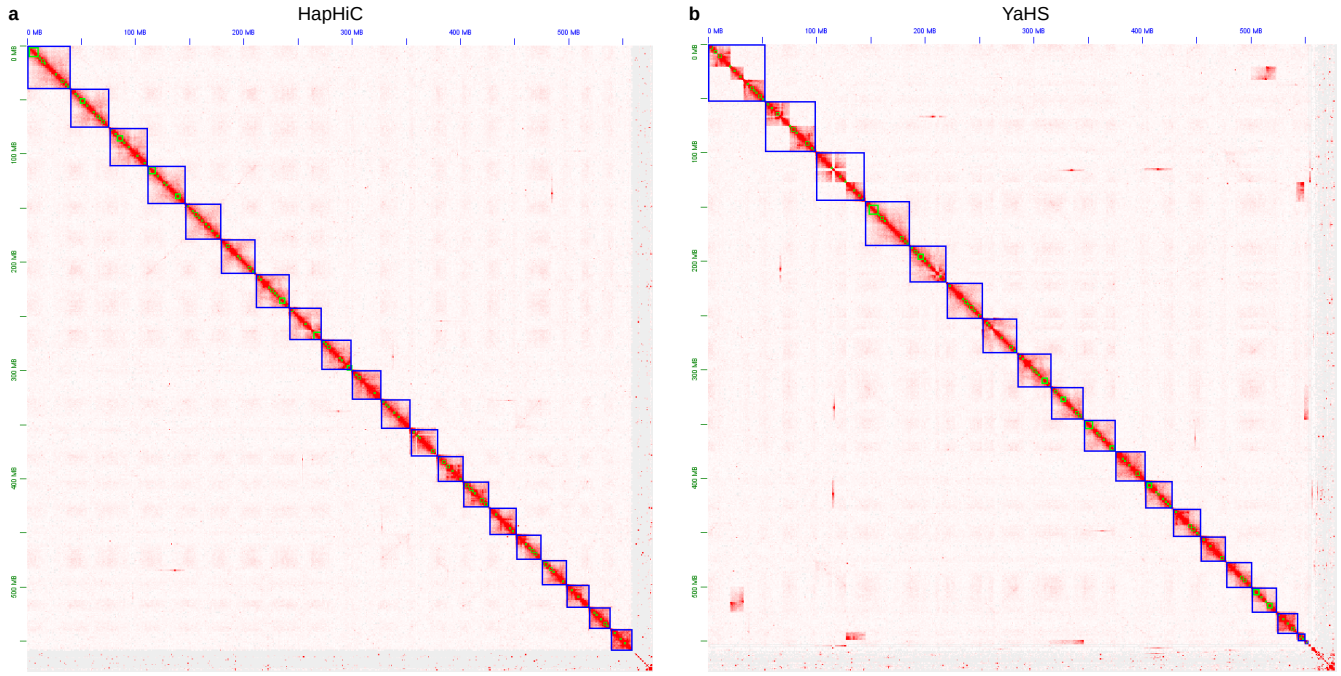
Supplementary Fig. 54 | Dot plots illustrating the alignments between each CHM13 T2T chromosome and HG002 scaffold constructed by HapHiC. Black bars indicate the repeat-rich regions in the centromeres.



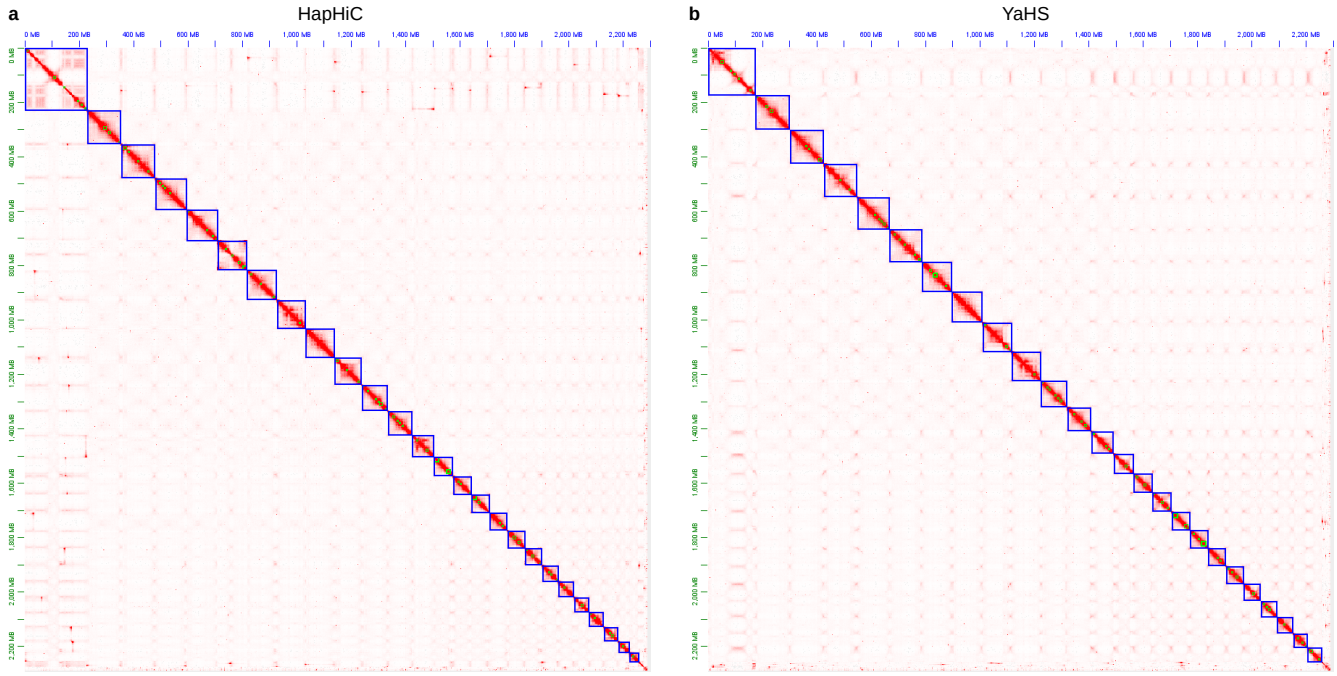
Supplementary Fig. 55 | Hi-C contact maps of HapHiC (**a**) and YaHS (**b**) scaffolds of the *Oryza alta* genome.



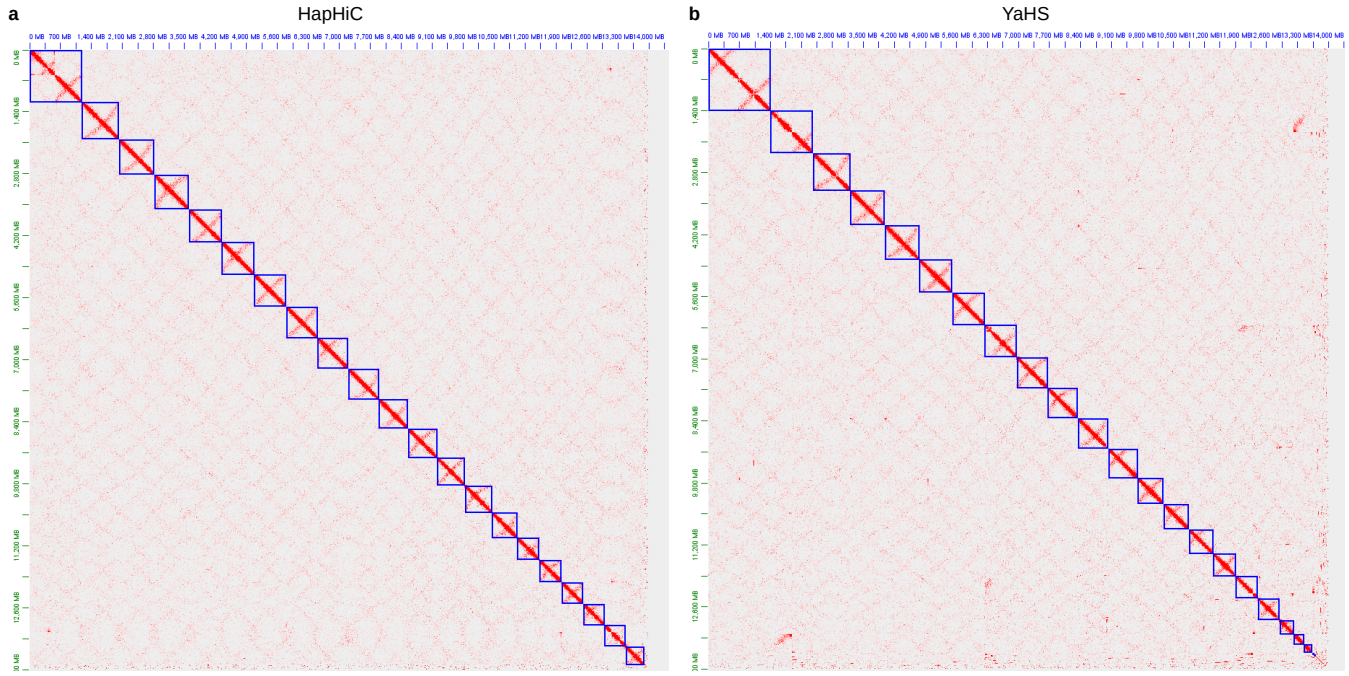
Supplementary Fig. 56 | Hi-C contact maps of HapHiC (**a**) and YaHS (**b**) scaffolds of the *Brassica napus* genome.



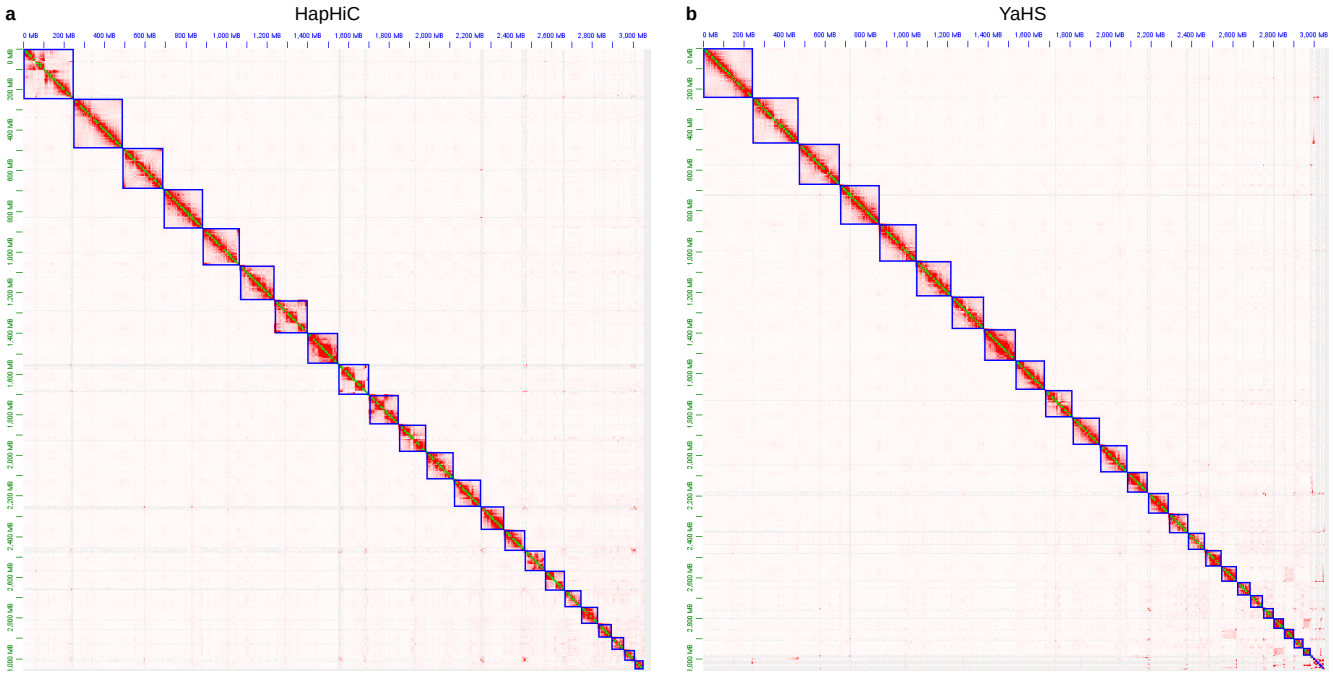
Supplementary Fig. 57 | Hi-C contact maps of HapHiC (**a**) and YaHS (**b**) scaffolds of the *Eragrostis tef* genome.



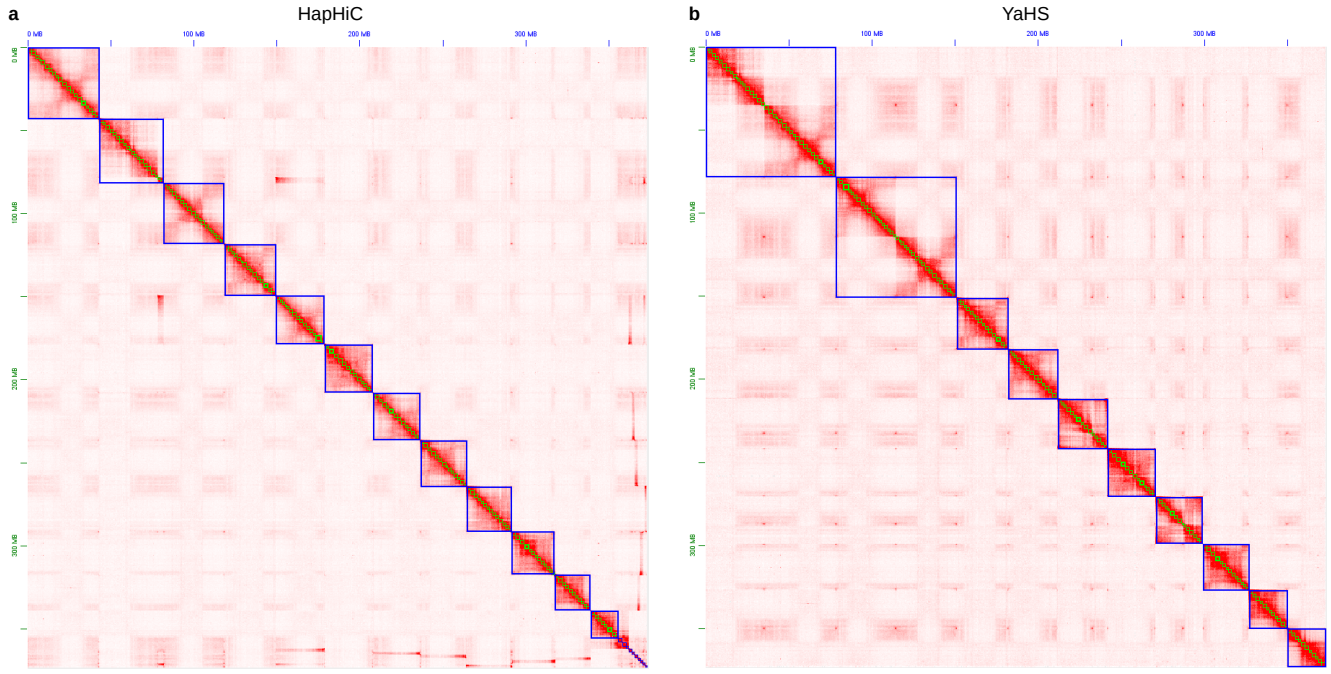
Supplementary Fig. 58 | Hi-C contact maps of HapHiC (**a**) and YaHS (**b**) scaffolds of the *Gossypium hirsutum* genome.



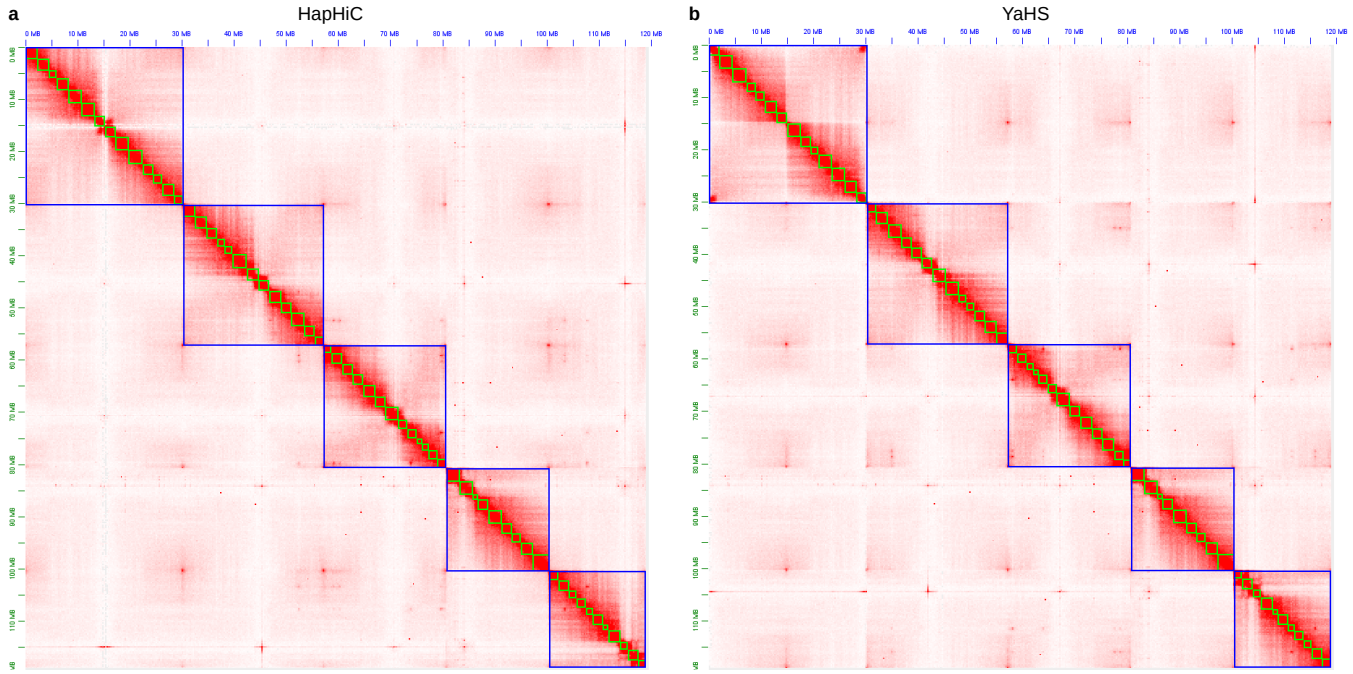
Supplementary Fig. 59 | Hi-C contact maps of HapHiC (**a**) and YaHS (**b**) scaffolds of the *Triticum aestivum* genome.



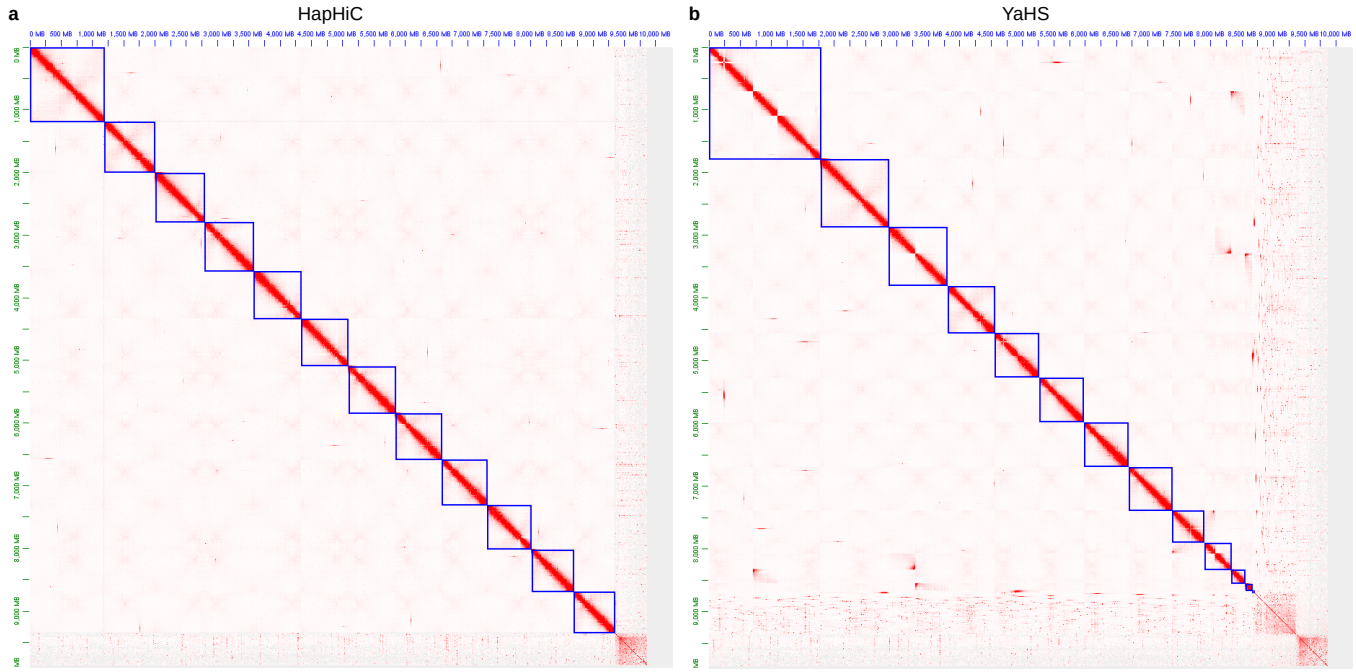
Supplementary Fig. 60 | Hi-C contact maps of HapHiC (**a**) and YaHS (**b**) scaffolds of the human CHM13v2.0_noY genome.



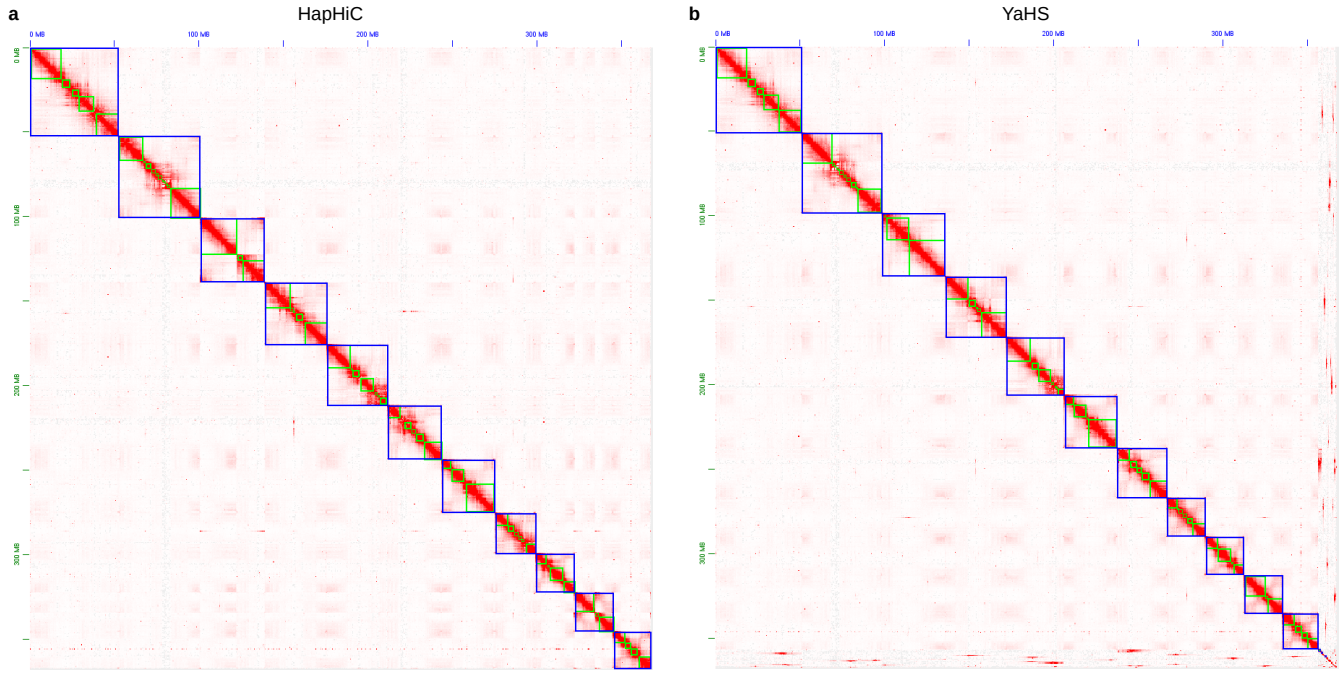
Supplementary Fig. 61 | Hi-C contact maps of HapHiC (**a**) and YaHS (**b**) scaffolds of the rice IRGSP-1.0 genome.



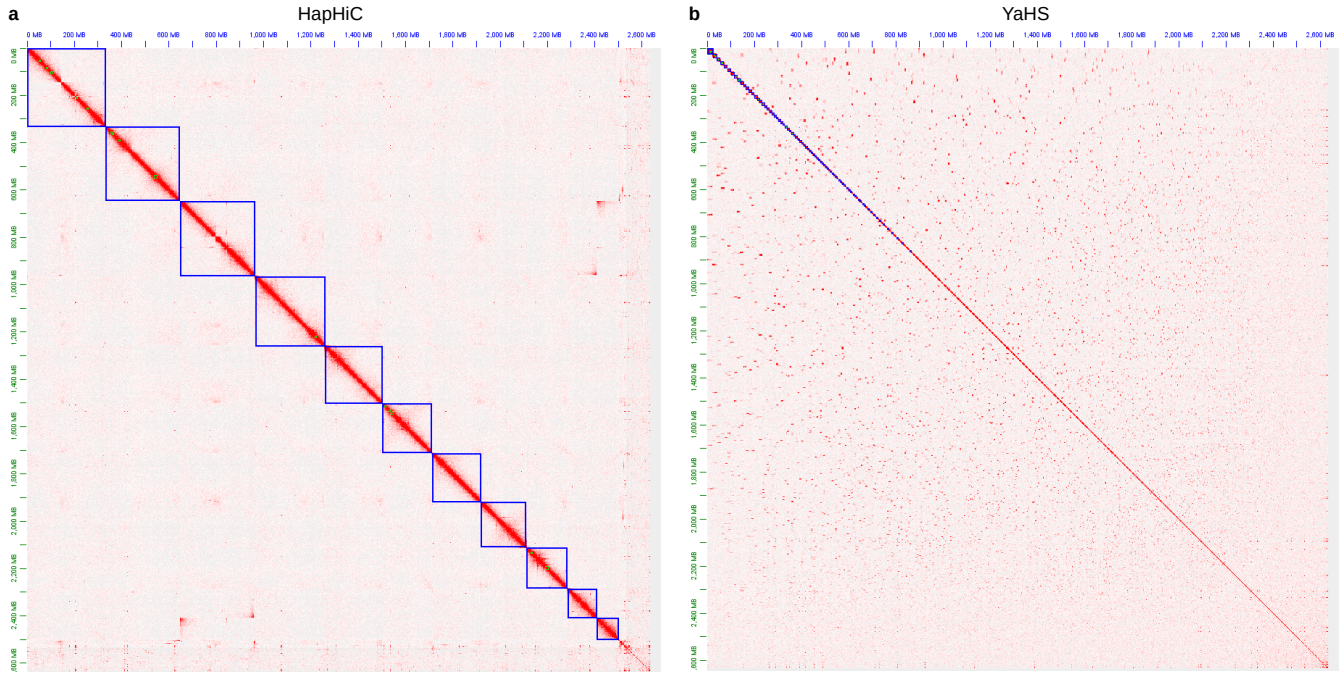
Supplementary Fig. 62 | Hi-C contact maps of HapHiC (**a**) and YaHS (**b**) scaffolds of the *Arabidopsis* TAIR10.1 genome.



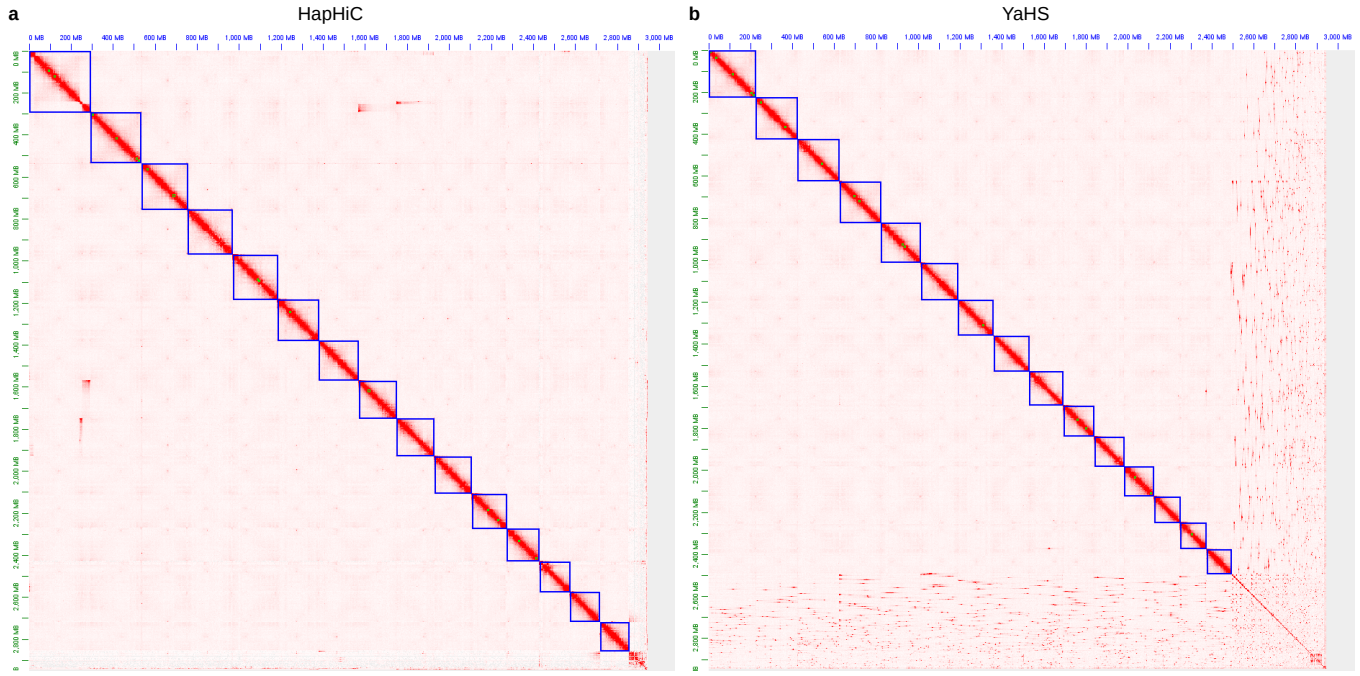
Supplementary Fig. 63 | Hi-C contact maps of HapHiC (**a**) and YaHS (**b**) scaffolds of the *Ginkgo biloba* genome.



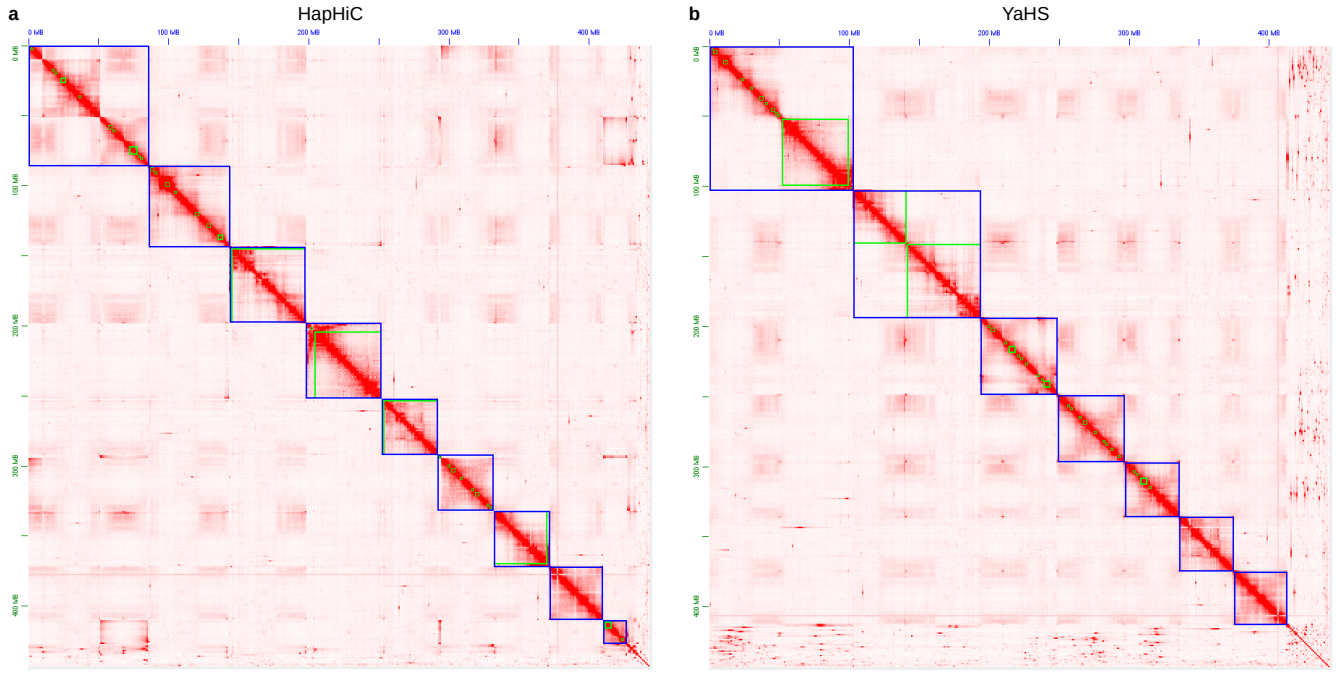
Supplementary Fig. 64 | Hi-C contact maps of HapHiC (**a**) and YaHS (**b**) scaffolds of the *Corylus mandshurica* genome.



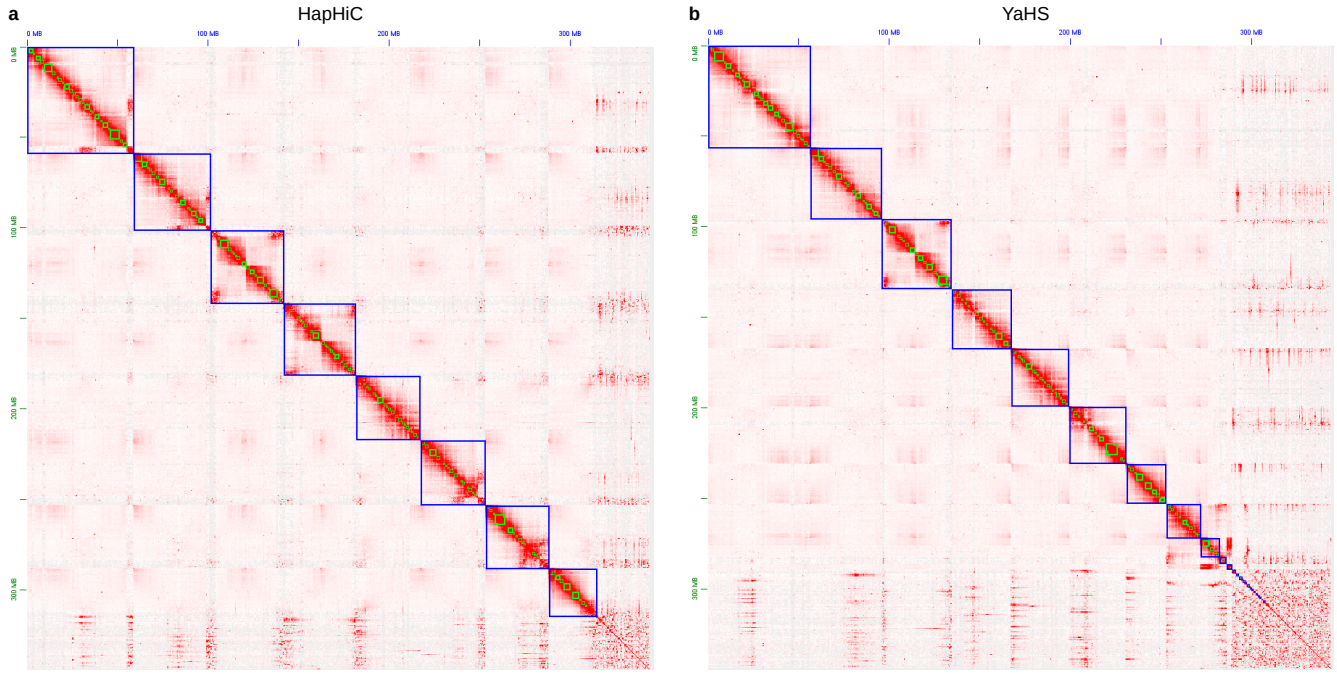
Supplementary Fig. 65 | Hi-C contact maps of HapHiC (**a**) and YaHS (**b**) scaffolds of the *Papaver somniferum* genome.



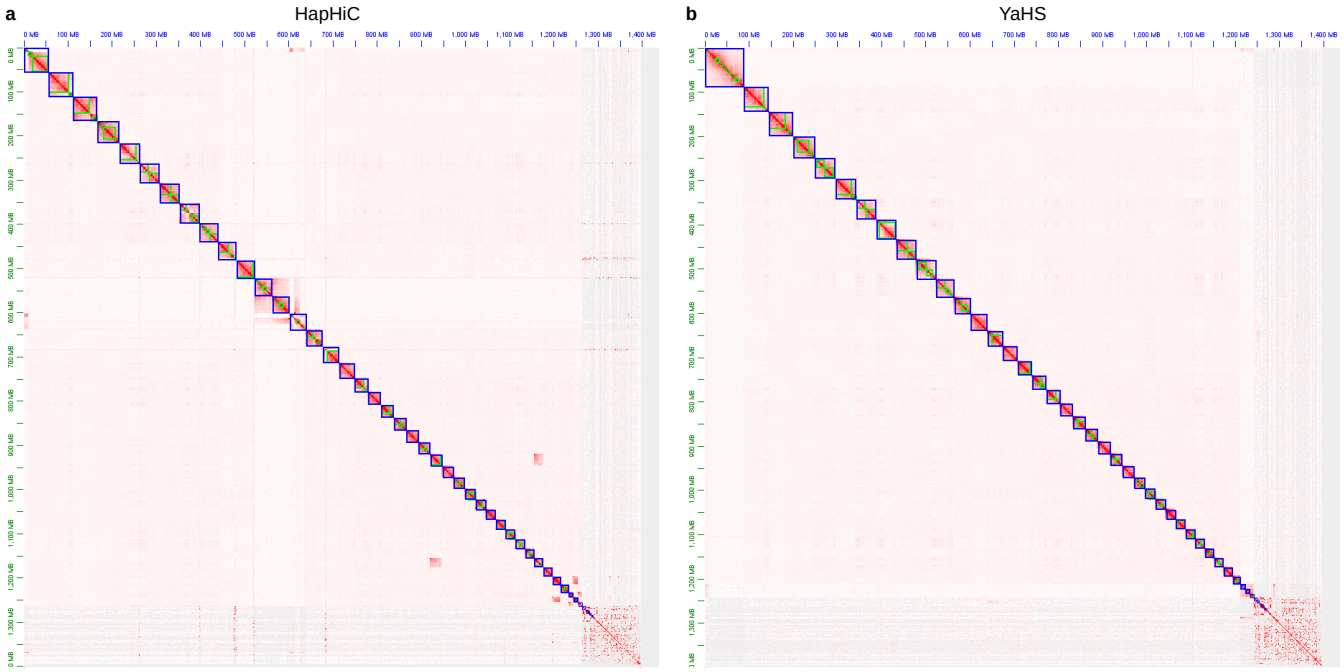
Supplementary Fig. 66 | Hi-C contact maps of HapHiC (**a**) and YaHS (**b**) scaffolds of the *Camellia sinensis* Huangdan genome.



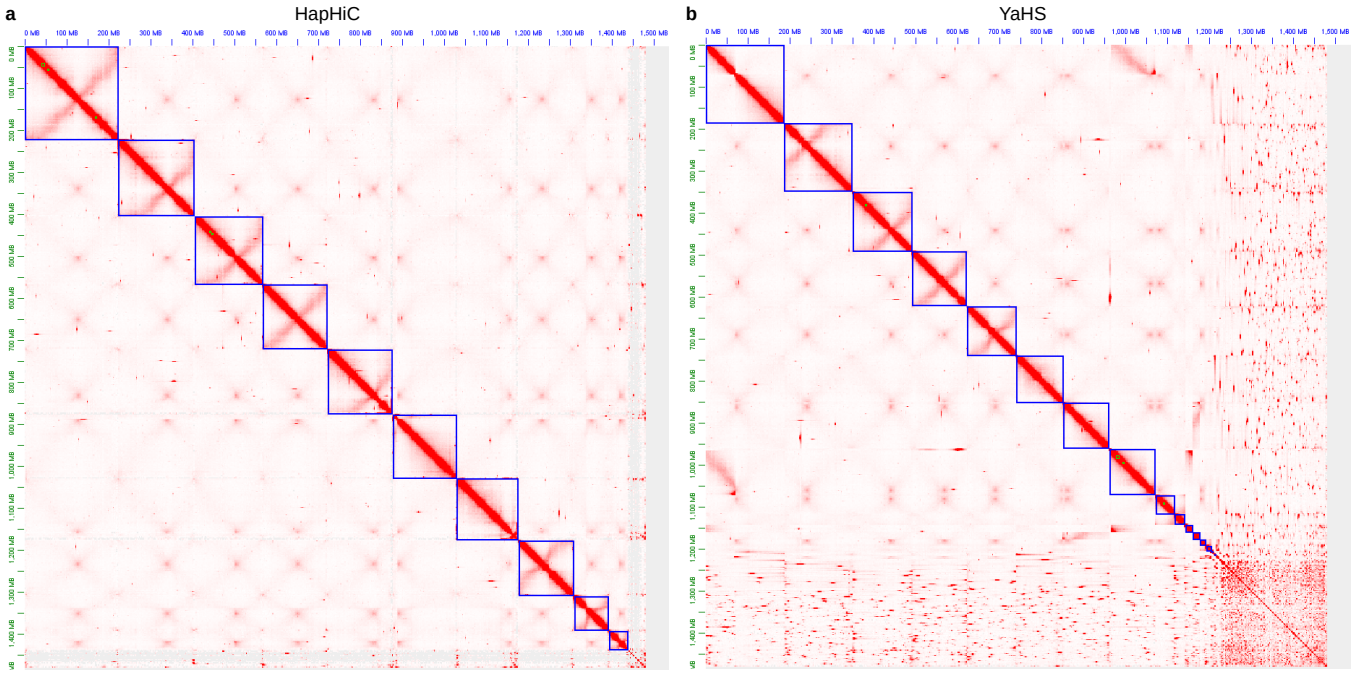
Supplementary Fig. 67 | Hi-C contact maps of HapHiC (**a**) and YaHS (**b**) scaffolds of the *Echinochloa haploclada* genome.



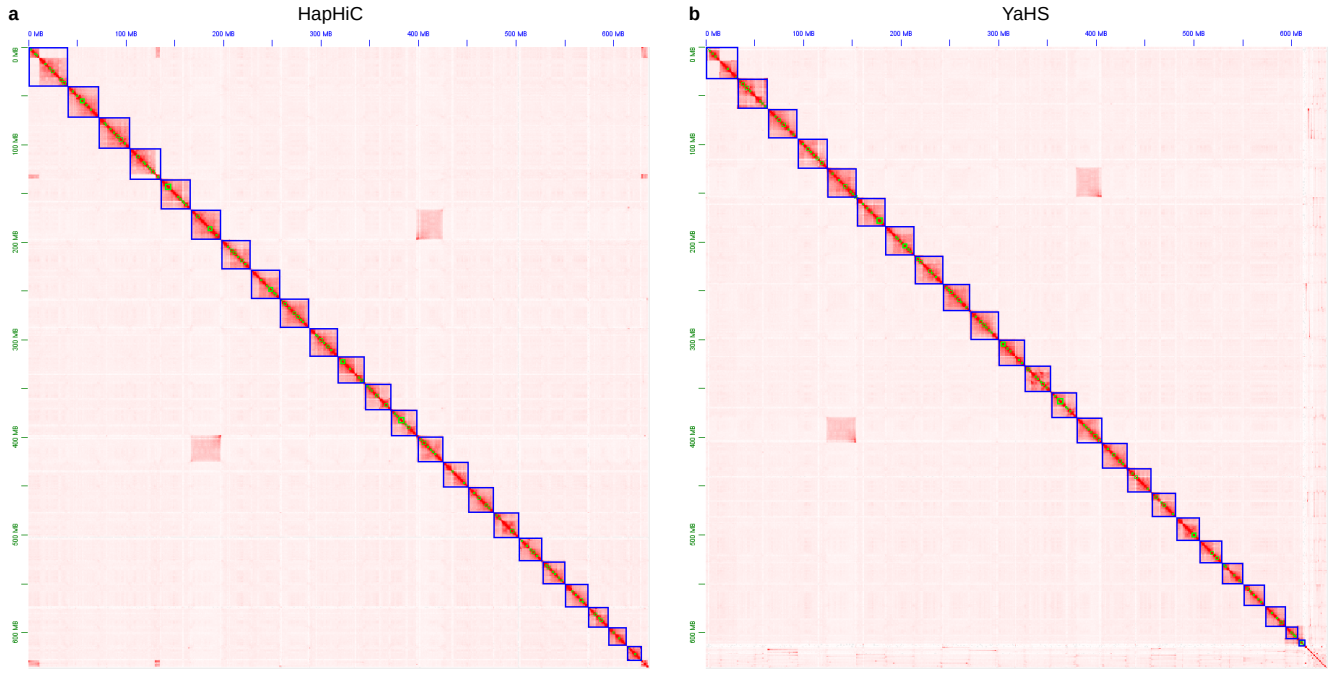
Supplementary Fig. 68 | Hi-C contact maps of HapHiC (**a**) and YaHS (**b**) scaffolds of the *Prunus avium* genome.



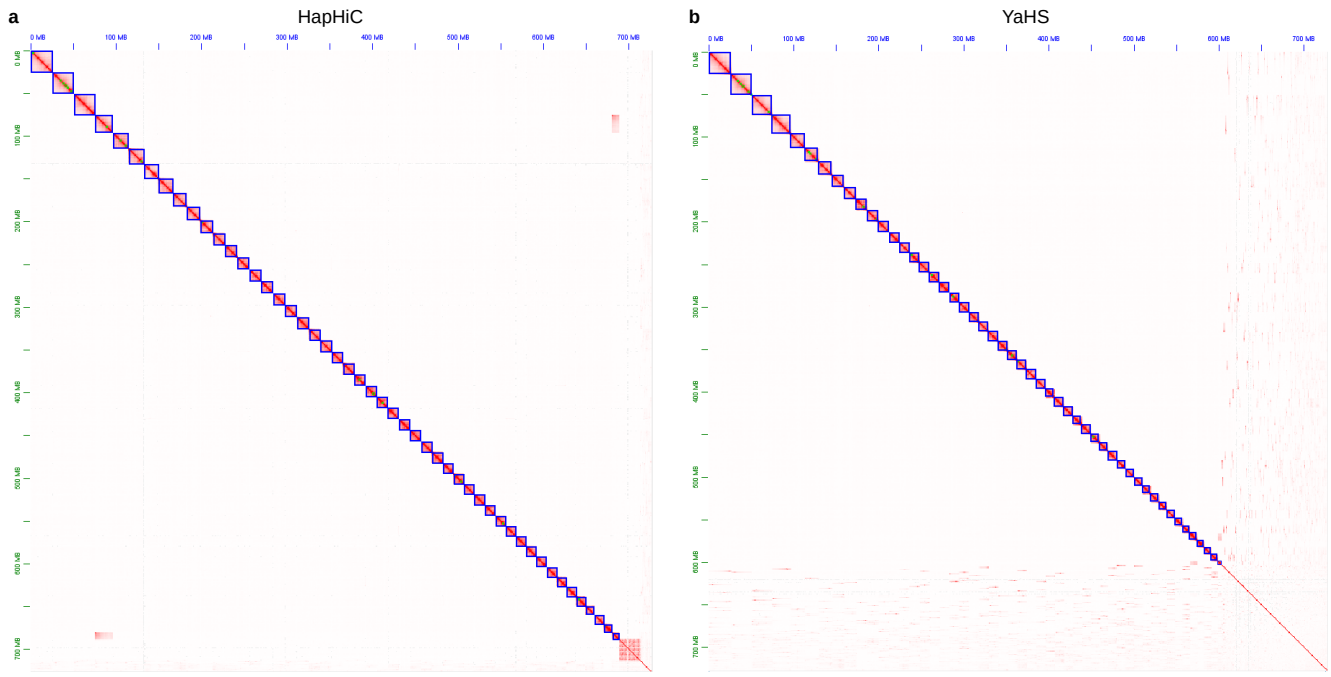
Supplementary Fig. 69 | Hi-C contact maps of HapHiC (**a**) and YaHS (**b**) scaffolds of the *Accipiter gentilis* genome.



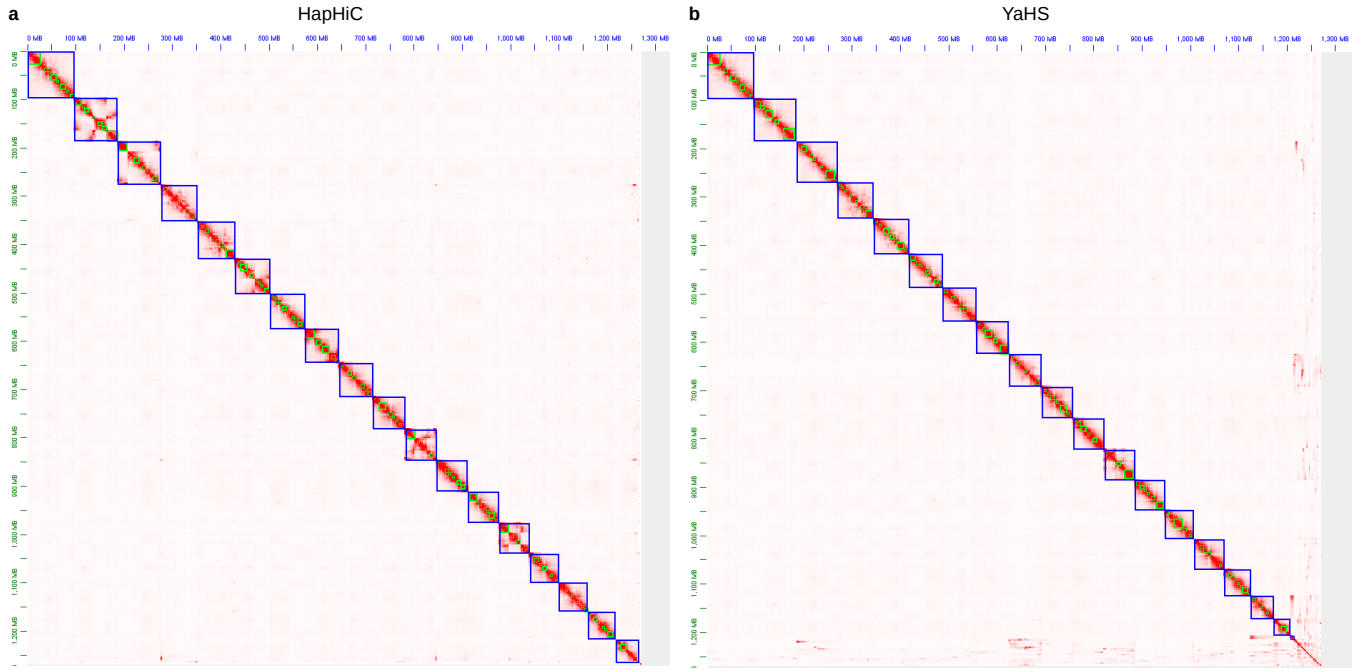
Supplementary Fig. 70 | Hi-C contact maps of HapHiC (**a**) and YaHS (**b**) scaffolds of the *Xenopus tropicalis* genome.



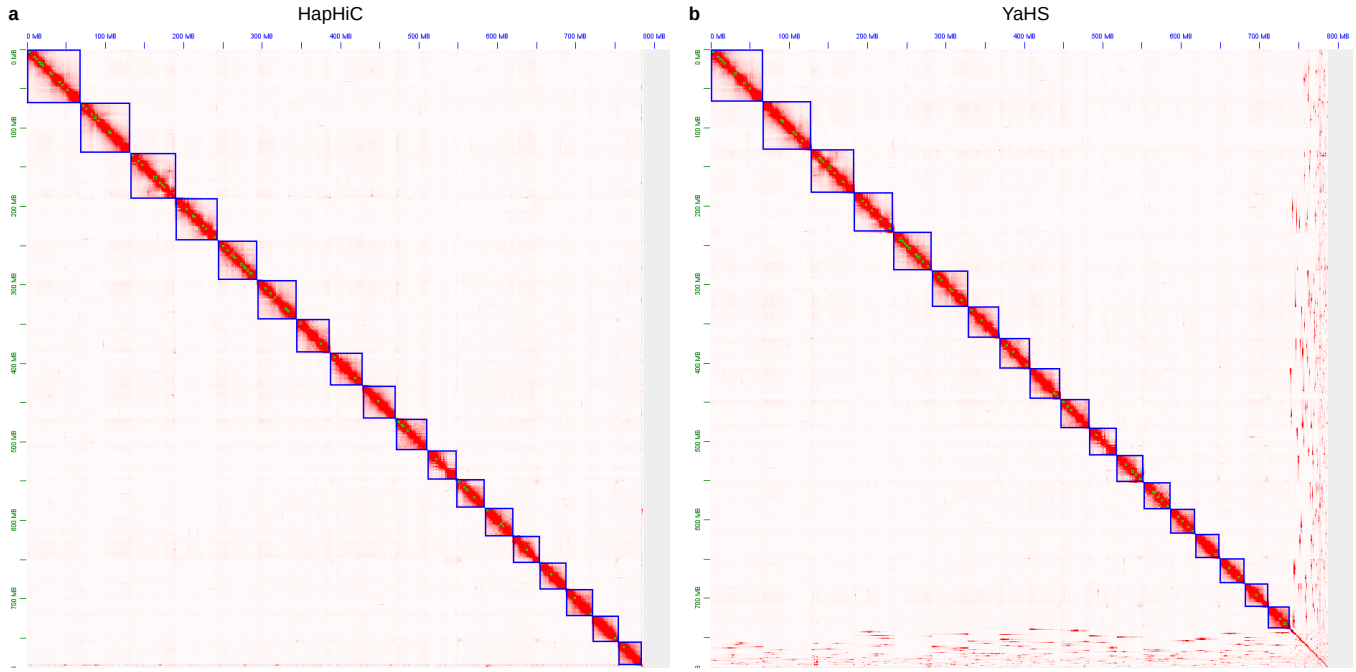
Supplementary Fig. 71 | Hi-C contact maps of HapHiC (**a**) and YaHS (**b**) scaffolds of the *Symphodus melops* genome.



Supplementary Fig. 72 | Hi-C contact maps of HapHiC (**a**) and YaHS (**b**) scaffolds of the *Antheraea pernyi* genome.



Supplementary Fig. 73 | Hi-C contact maps of HapHiC (**a**) and YaHS (**b**) scaffolds of the *Steromphala cineraria* genome.



Supplementary Fig. 74 | Hi-C contact maps of HapHiC (**a**) and YaHS (**b**) scaffolds of the *Lumbricus rubellus* genome.

Descriptions of Supplementary Data

Supplementary Data 1-6 are provided online as separated excel files. Titles for each file and sheet are listed as follows:

Supplementary Data 1: Performance evaluation of Hi-C-based scaffolding tools in chromosome assignment under various adverse factors **(14 sheets in total)**

- **Sheet 1:** Performance evaluation of Hi-C-based scaffolding tools in chromosome assignment on assemblies with varying contig N50 values
- **Sheet 2:** Performance evaluation of Hi-C-based scaffolding tools in chromosome assignment on assemblies with varying coefficients of variation of contig length
- **Sheet 3:** Performance evaluation of Hi-C-based scaffolding tools in chromosome assignment with varying effective Hi-C sequencing depths
- **Sheet 4:** Performance evaluation of Hi-C-based scaffolding tools in chromosome assignment on assemblies with varying proportions of chimeric contigs
- **Sheet 5:** Performance evaluation of ALLHiC with and without assembly correction in chromosome assignment on assemblies with varying proportions of chimeric contigs
- **Sheet 6:** Performance evaluation of Hi-C-based scaffolding tools in chromosome assignment on assemblies with varying proportions of collapsed contigs
- **Sheet 7:** Performance evaluation of Hi-C-based scaffolding tools in chromosome assignment on assemblies with varying sequence divergence levels
- **Sheet 8:** Performance evaluation of Hi-C-based scaffolding tools in chromosome assignment on assemblies with varying switch error rates
- **Sheet 9:** Performance evaluation of Hi-C-based scaffolding tools in chromosome assignment on assemblies with identical switch error rate (20%) but varying contig N50 values
- **Sheet 10:** Performance evaluation of Hi-C-based scaffolding tools in chromosome assignment on assemblies with varying ploidies
- **Sheet 11:** Performance evaluation of Hi-C-based scaffolding tools in chromosome assignment on assemblies with varying ploidies and 5% each of chimeric, collapsed, and switch errors
- **Sheet 12:** Performance evaluation of ALLHiC in chromosome assignment

using the first haplotype of *M. sativa* XinJiangDaYe as the reference genome

- **Sheet 13:** Performance evaluation of Hi-C-based scaffolding tools in chromosome assignment using potato C88 assemblies
- **Sheet 14:** Minimum effective depth for ordering and orientation

Supplementary Data 2: Performance evaluation of Hi-C-based scaffolding tools in contig ordering and orientation on assemblies with varying contig N50 values **(two sheets in total)**

- **Sheet 1:** Performance evaluation of HapHiC and ALLHiC in contig ordering and orientation using the built-in scoring system of ALLHiC
- **Sheet 2:** Performance evaluation of all Hi-C-based scaffolding tools in contig ordering and orientation using Lin's concordance correlation coefficient and DERANGE cost

Supplementary Data 3: Time and memory usage of Hi-C-based scaffolding tools **(five sheets in total)**

- **Sheet 1:** Time and memory usage of HapHiC and ALLHiC in contig ordering and orientation
- **Sheet 2:** Time and memory usage of the entire pipeline for each Hi-C-based scaffolding tool
- **Sheet 3:** Time and memory usage of each Hi-C-based scaffolding tool in assembly correction
- **Sheet 4:** Time and memory usage of each Hi-C-based scaffolding tool in processing Hi-C data at varying depths
- **Sheet 5:** Time and memory usage of HapHiC, ALLHiC, and YaHS pipelines in scaffolding published assemblies

Supplementary Data 4: Validation of HapHiC in published genome assemblies **(two sheets in total)**

- **Sheet 1:** Details of published genome assemblies and Hi-C data used for HapHiC validation
- **Sheet 2:** Quantified performance of HapHiC, ALLHiC, and YaHS in scaffolding published assemblies

Supplementary Data 5: Comparison of assembly correction results between HapHiC

and ALLHiC on manually annotated contigs in the first haplotype of *S. spontaneum* Np-X chromosome 1 **(one sheet in total)**

Supplementary Data 6: Correlation analysis of genetic maps with the A subgenome of *M.* × *giganteus* (MgiA) and the genome of *M. sinensis* (MsiA) **(one sheet in total)**